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Mining Scientific Data for the Study of the Heterostructural Design and the Optoelectronic Properties of Quantum Cascade Lasers

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*“To my parents Patrick and Winrose, my siblings, my daughter Cecilia, my lovely wife
Celestine Nanjala and my late Grandmother Cecilia Nanyama Kerre...”*

Résumé

Les lasers à cascade quantique (QCL) sont des dispositifs laser à semi-conducteurs qui fonctionnent dans la gamme de l'infrarouge moyen (de 10 THz à 100 THz) et de l'infrarouge lointain (de 100 GHz à 10 THz). Les propriétés des lasers à cascade quantique (QCL) peuvent être classées en deux catégories : les propriétés de conception/hétérostructure (propriétés d'empilement des matériaux) et les propriétés optoélectroniques (comportement des performances du laser, par exemple puissance, température de fonctionnement, fréquence laser, etc.). Des informations structurées sur les différents designs de QCL et leurs propriétés de fonctionnement jouent un rôle majeur dans le déchiffrement de la structure complexe de ces lasers, ouvrant la perspective d'améliorer ainsi la compréhension du lien entre les performances du dispositif et les paramètres de conception du design. Une représentation structurée des propriétés des QCL pouvant faciliter l'exploration des propriétés afin d'en comprendre leurs relations inhérentes est donc souhaitable. Les connaissances acquises grâce à ce travail pourraient être utilisées pour développer de nouveaux design de QCL avec des propriétés cibles telles que la température de fonctionnement par exemple.

La plupart des propriétés des lasers QCL sont décrites dans des textes scientifiques (articles, thèses,...), ce qui pose un défi pour extraire efficacement les propriétés et les organiser formellement en vue d'une exploration structurée plus approfondie. Les bases de connaissances existantes ne capturent pas les informations sur les propriétés des QCL et les relations entre elles. Il n'existe donc pas de normes/vocabulaires communs pour accéder, partager et analyser les données sur les propriétés des QCL dans diverses sources textuelles hétérogènes de manière interopérable. À notre connaissance, l'extraction des propriétés des QCL à partir de textes scientifiques n'a pas non plus été explorée à ce jour. Pour remédier à ces problèmes, cette thèse explore les techniques d'extraction d'informations (IE) et de représentation des connaissances (KR) dans le but de développer de nouvelles méthodes d'extraction des propriétés des QCL à partir de textes et de proposer des normes pour la représentation et l'analyse des propriétés des QCL de manière interopérable afin d'obtenir des informations.

Cette thèse est structurée comme suit : (i) une revue complète de l'état de l'art en matière d'extraction d'informations et de représentation des connaissances dans le domaine de la science des matériaux en relation avec le domaine QCL, (ii) un ensemble de données d'instructions original sur les propriétés QCL extraites de divers travaux scientifiques, (iii) deux méthodes d'exploration des propriétés QCL à partir de textes (l'une basée sur une approche fondée sur des règles et l'autre sur des approches d'apprentissage automatique (ML)), (iv) une représentation formelle des concepts de propriétés QCL sous la forme d'un modèle ontologique afin de formaliser les connaissances sur les concepts de propriétés QCL et les relations entre les propriétés de conception et de fonctionnement, et enfin (v) sur la base de l'approche d'apprentissage automatique pour l'extraction de propriétés QCL à partir de textes, nous présentons un workflow original pour l'enrichissement sémantique des propriétés QCL dans les textes en générant un graphe de connaissances des propriétés QCL. Les résultats expérimentaux indiquent la capacité des pipelines d'extraction d'informations proposés à extraire les propriétés QCL d'intérêt à partir de textes. Nous validons également la capacité des modèles d'ontologie et de graphe de connaissances proposés à capturer les exigences du domaine des propriétés QCL.

mots-clés: Extraction d'informations, Représentation des connaissances, Graphes de connaissances, Science des matériaux, Apprentissage automatique, Ontologies, Lasers à cascade quantique

Abstract

Quantum Cascade Lasers (QCLs) are semiconductor laser devices whose operation is within the mid infrared range (from 10 THz to 100 THz) and far infrared range (from 100 GHz to 10 THz). Quantum Cascade Laser (QCL) properties can be classified into the design/heterostructure (material stacking properties) and the optoelectronic properties (laser performance behaviour e.g power, working temperature, lasing frequency etc). Structured information on the various QCL designs and their working properties plays a major role in deciphering the complex structure of the laser designs hence enhancing the understanding of performance of the laser device in relation to its design properties. A structured representation of the QCL properties that can support exploration of the properties to understand the inherent relationships is therefore highly desired. The insights obtained from this work can be utilized in developing new QCL designs with target properties such as the working temperature.

Most of the QCL laser properties are described in scientific texts, which poses a challenge in efficiently extracting the properties and formally organizing them for further structured exploration. The existing Knowledge Bases do not capture information on QCL properties and the relationships between them. There are therefore no standards/shared vocabularies for accessing, sharing and analyzing QCL properties data in various heterogeneous textual sources in an interoperable manner. To the best of our knowledge, extraction of QCL properties from scientific texts has also not been explored. To address these problems, the thesis explores Information Extraction (IE) and Knowledge Representation (KR) techniques with an objective of developing novel methods for extraction of QCL properties from text and proposing standards for representation and analysis of QCL properties in an interoperable manner for insights.

The thesis is developed in the following parts : (i) A comprehensive review of the state of the art in Information Extraction and Knowledge Representation in the materials science domain in relation to the QCL domain, (ii) An original instruction dataset for extracting QCL properties from various scientific texts, (iii) Two methods for mining QCL properties from scientific texts (one based on a rule-based approach and the other one based on machine learning (ML) approaches), (iv) A formal representation of QCL properties concepts in the form of an ontology model in order to formalize the knowledge of QCL properties concepts and the relationships between the design and working properties and lastly (v) Based on the machine learning approach for QCL properties extraction from text, we present an original workflow for the Semantic enrichment of QCL properties in text by generating a QCL properties Knowledge Graph. Experimental results indicate the ability of the proposed information extraction pipelines in extracting the QCL properties of interest from scientific texts. We also validate the ability of the proposed ontology and knowledge graph models in capturing the QCL properties domain requirements.

Keywords: Information Extraction, Knowledge Representation, Knowledge Graphs, Materials Science, Machine Learning, Ontologies, Quantum Cascade Lasers

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Contents

Title	i
Résumé	v
Abstract	vii
Acknowledgments	ix
Table of Contents	xvi
List of Figures	xix
List of Tables	xxii
Acronyms	xxiii
1 Introduction	1
1.1 Background of the Study	2
1.2 FAIR Data Principles	6
1.3 Motivation of the Thesis	8
1.4 Problem Statement	9
1.5 Hypotheses and Research Directions	10

1.6	Research Questions	11
1.7	Contributions	12
1.7.1	Generation of an Original Dataset for Extraction of QCL properties from Scientific Texts	12
1.7.2	Extraction of QCL Properties from Scientific Texts	13
1.7.3	Formal Representation of Properties in the QCL Domain	13
1.7.4	Semantic Enrichment of the QCL Properties Descriptors in Scientific Texts	14
1.8	Publications and Data	14
1.8.1	International Scientific Journals	14
1.8.2	International Scientific Conferences/Workshops	15
1.8.3	Data	15
1.9	Thesis Document Outline	15
1.10	Conclusion	17
2	Knowledge Representation and FAIR Data Principles: Introduction to Concepts, Methods and Related Works	19
2.1	Introduction	20
2.2	Knowledge Representation	20
2.2.1	Introduction	20
2.2.2	Semantic Web	21
2.2.3	Linked Data	21
2.2.4	Ontologies	25
2.2.5	RDF Graphs and Property Graphs	29
2.2.6	RDF Graph Databases	32
2.2.7	SPARQL Query	32
2.2.8	Related Work	34
2.3	Knowledge Representation and FAIR Data Principles	36
2.4	Conclusion	37

3	Information Extraction: Definitions, Methods and Related Works	39
3.1	Introduction	40
3.2	Information Extraction	40
3.2.1	Rule-Based Methods	41
3.2.2	Machine Learning Based Methods	43
3.3	Evaluation of Information Extraction Methods in the Materials Science Domain .	49
3.4	Related Works	51
3.4.1	Information Extraction in Materials Science Domain	51
3.4.2	Existing Databases of Materials Science Data	53
3.4.3	Challenges and Thesis Positioning	53
3.5	Conclusion	54
4	A Dataset for Extraction of Quantum Cascade Laser Properties from Scientific Texts	57
4.1	Introduction	58
4.2	Motivation for the Dataset Generation	58
4.3	Materials and Methods	59
4.3.1	Sentence Sampling	59
4.3.2	Dataset Generation	59
4.3.3	Data Annotation	60
4.4	Value of the Dataset and Valorization	61
4.4.1	Value of the Dataset	61
4.4.2	Dataset Valorization	62
4.4.3	Limitations of the Dataset	62
4.5	Conclusion	62
5	Extraction of Quantum Cascade Laser Properties From Scientific Texts	65
5.1	Introduction	66

5.2	Rule Based Approach	66
5.2.1	Introduction	66
5.2.2	Methodology	67
5.2.3	Experiments and Results	72
5.3	Machine Learning Approach	75
5.3.1	Introduction	75
5.3.2	Methodology	75
5.3.3	Experiments and Results	79
5.4	Conclusion	82
6	Formal Representation of the Quantum Cascade Laser Properties: An Onto-logical Approach	85
6.1	Introduction	86
6.2	Ontology Modeling	86
6.2.1	Ontology Engineering Methodology	86
6.2.2	Requirements Analysis	87
6.2.3	Modeling the Quantum Cascade Laser Properties	89
6.3	Ontology Evaluation	95
6.3.1	Source Code	96
6.3.2	Evaluation Metrics	96
6.3.3	Results and Discussions	98
6.3.4	The QCL Ontology Model and the FAIR Data Principles	102
6.4	Ontology Limitations	103
6.5	Conclusion	103
7	Semantic Enrichment of the Quantum Cascade Laser Properties	105
7.1	Introduction	106
7.2	Knowledge Graph Generation	106

7.2.1	Data Extraction from Scientific Texts	106
7.2.2	Knowledge Graph Generation	107
7.3	Knowledge Graph Evaluation	110
7.3.1	Source Code	111
7.3.2	Evaluation Metrics	111
7.3.3	Evaluation Results	112
7.4	Conclusion	117
8	Conclusion and Perspectives	119
8.1	Summary of the Thesis	120
8.1.1	Comprehensive Literature Review of Knowledge Representation and Information Extraction in Materials Science	121
8.1.2	Generation of Datasets for Extraction of QCL properties from Text	122
8.1.3	Extraction of Quantum Cascade Laser Properties from Text	122
8.1.4	Formal Representation of Quantum Cascade Laser Properties	122
8.1.5	Semantic Enrichment of Quantum Cascade Laser Properties in Text	123
8.2	Significance and Limitations	123
8.3	Perspectives	124
8.3.1	Further Improvement of Efficiency in Mining QCL properties from Text	124
8.3.2	Extension of Knowledge Representation for QCL Properties	124
8.3.3	Learning Methods for QCL Properties Prediction	124
	Bibliography	126
	Appendices	143
A	Links to Resources	143
A.1	Source Code Links	143
A.1.1	Instruction Dataset Source Code	143

A.1.2	Rule-Based IE Source Code	143
A.1.3	RAG-Based IE Source Code	143
A.1.4	The QCL Ontology Source Code	143
A.1.5	The QCL KG Generation Source Code	143
A.2	Dataset Links	144
A.2.1	Instruction Dataset Link	144

List of Figures

1.1	Schematic Representation of the Quantum Cascade Laser (from [181]).	2
1.2	Sample Heterostructure Description (image extracted from [96]). Shows the calculated conduction band diagrams with the barriers (Injection barrier and collection barrier) for both the design bias and parasitic bias.	4
1.3	Modeling of Polymer Nanocomposites (Figure from [117]). The modeling captures knowledge on the Polymer Nanocomposite components and the roles of the different components such as the treatment roles.	5
1.4	Workflow for supervised learning of molecular properties (image from [134]). A model is optimized with labeled data and subsequently used to estimate molecular properties for an unknown (unlabeled) data set.	6
1.5	Outline of the Thesis Document	16
2.1	Sample Ontology for Material Properties	26
2.2	Sample RDF Graph Representing Personal Information	30
2.3	Sample RDF Graph for Material Properties	30
2.4	Sample Data Modeling with a Property Graph	31
2.5	Query Results	33
2.6	Query Graph Pattern	34
3.1	The chemDataExtractor pipeline (Image from [41]).	42
3.2	The LSTM Model Architecture (Image from [140]).	44
3.3	General Architecture of a Bidirectional LSTM (Image from [11]).	46

3.4	Pre-training of BERT (Image from [39]). $E_1, E_2 \dots E_N$ are the embeddings, $T_1, T_2, \dots T_N$ the token layers while T_{rm} are the individual tokens in the token layers.	47
3.5	RAG Approach (Image from [104]). x denotes the query which is encoded by the process q . The documents provides context that is passed to the generator p that generates contextual information used in getting the response.	48
3.6	RAG Paradigms: Naive RAG, Advanced RAG and Modular RAG (Image from [52]).	50
4.1	Dataset Generation	60
5.1	Sample Extracted Heterostructure Material Record (example data extracted from [190]).	72
5.2	The RAG Pipeline for Mining QCL Properties	76
5.3	Sample Regenerated Prompt by GPT-4 Turbo Model	78
5.4	Model Execution Time	82
6.1	Upper Concepts in the Ontology.	90
6.2	Ontology Section for Quantum Cascade Laser Optoelectronic Characteristics. . .	91
6.3	Heterostructure Design Properties Concepts and Relations	92
6.4	A Visualization of the QCL Heterostructure Class from the Ontology.	93
6.5	Concepts for the Quantum Cascade Laser Working Mode.	93
6.6	Concepts for Provenance Information.	94
6.7	Concepts in the QCL Ontology.	94
6.8	A Neo4j Visualization of the Ontology model.	95
6.9	Result for the Query for Competency Question 4.2	99
6.10	Result for the Query for Competency Question 6.1	100
6.11	Result for the Query for Competency Question 7.1	101
7.1	The Knowledge Graph Schema Model	108
7.2	Modeling the Relationship Between Frequency and the Heterostructure.	109
7.3	A Visualization of an Instance of the Heterostructure	110
7.4	The Knowledge Graph Generation Workflow	110

7.5	SPARQL Query Results for CQ 5.2	114
7.6	Results for CQ 5.1	115
7.7	Results for CQ 6.2	116
8.1	Summary of the Contributions	121

List of Tables

2.1	Ontologies for Specific Sub-domains in Material Science	35
3.1	The Parser Elements (from [165]).	43
4.1	Summary Statistics for the Instruction Dataset	61
5.1	The Parser Elements (elements extracted from [165]).	70
5.2	Quantum Cascade Laser Properties, Units and the Keywords.	70
5.3	Performance Evaluation Metrics for the Default Autosentence Parser.	74
5.4	Performance Evaluation Metrics for the Defined Parsers.	74
5.5	Test Data Statistics	79
5.6	Performance Metrics for the Laser Working Temperature.	80
5.7	Performance Metrics for Laser Output Power.	80
5.8	Performance Metrics for the Laser Frequency.	80
5.9	Performance Metrics for the Laser Design Type.	81
5.10	Performance Metrics for the Laser Heterostructure Materials.	81
5.11	Average Performance of the Models for all the Properties	81
6.1	Competence Questions (CQs)	97
6.2	Competence Questions (CQs)	99

7.1	Namespace URIs	108
7.2	Classes of Competency Question	112
7.3	The Competency Questions	113
7.4	Results for Schema Completeness Analysis	116
7.5	Results for Property Completeness Analysis	116

Acronyms

BIBO	B ibliographic O ntology
CQ	C ompetency Q uestion
RD	R esearch D irection
DL	D escription L ogic
FOL	F irst O rders L ogic
LD	L inked D ata
LOD	L inked O pen D ata
IR	I nheritance R ichness
EMMO	E uropean M aterials and M odeling O ntology
QCL	Q uantum C ascade L aser
KG	K nowledge G raph
MDO	M aterials D esign O ntology
RDF	R esource D escription F ramework
QUDT	Q uantities U nits D imension and D atatype O ntologies
KR	K nowledge R epresentation
IE	I nformation E xtraction
ML	M achine L earning
RAG	R etrieval A ugmented G eneration
LLM	L arge L anguage M odel
CNER	C hemical N amed E ntity R ecognition
NLP	N atural L anguage P rocessing
CNN	C onvolutional N eural N etwork
BERT	B idirectional E ncoder R epresentations from T ransformers
PG	P roperty G raph

Chapter 1

Introduction

1.1	Background of the Study	2
1.2	FAIR Data Principles	6
1.3	Motivation of the Thesis	8
1.4	Problem Statement	9
1.5	Hypotheses and Research Directions	10
1.6	Research Questions	11
1.7	Contributions	12
1.7.1	Generation of an Original Dataset for Extraction of QCL properties from Scientific Texts	12
1.7.2	Extraction of QCL Properties from Scientific Texts	13
1.7.3	Formal Representation of Properties in the QCL Domain	13
1.7.4	Semantic Enrichment of the QCL Properties Descriptors in Scientific Texts	14
1.8	Publications and Data	14
1.8.1	International Scientific Journals	14
1.8.2	International Scientific Conferences/Workshops	15
1.8.3	Data	15
1.9	Thesis Document Outline	15
1.10	Conclusion	17

“There must be a beginning of any great matter, but the continuing unto the end until it be thoroughly finished yields the true glory.”

– Francis Drake, *Brainy Quote*

1.1 Background of the Study

Quantum Cascade Lasers (QCL) are semiconductor lasers that exhibit an emission in mid infrared range (from 10 THz to 100 THz) and far infrared (from 100 GHz to 10 THz) portion of the electromagnetic spectrum. This laser technology was first proposed in 1994 at Bell Laboratories [45]. Unlike other typical inter-band semiconductor lasers whose laser radiation emission is based on the recombination of electron-hole pairs across the material band gap, QCLs are unipolar in nature and the electromagnetic radiation is emitted through the use of intersub-band transitions in a repeated stack of semiconductor multiple quantum well heterostructures [86].

A QCL device is made up of an active region consisting of a nanometric stack of different semiconductor materials. The interaction between these different layers of materials gives the stack its optoelectronic properties, i.e. its radiation emission behavior according to the electric current injected in the structure. These stacks are called heterostructures. The heterostructures are complex but in general, weakly structured post growth. This implies that most of the properties of the laser are defined by its growth sheet, i.e. the description of the different stacked layers: their thickness, the nature of the material, the order, etc. Figure 1.1 shows the schematic representation of a QCL device.

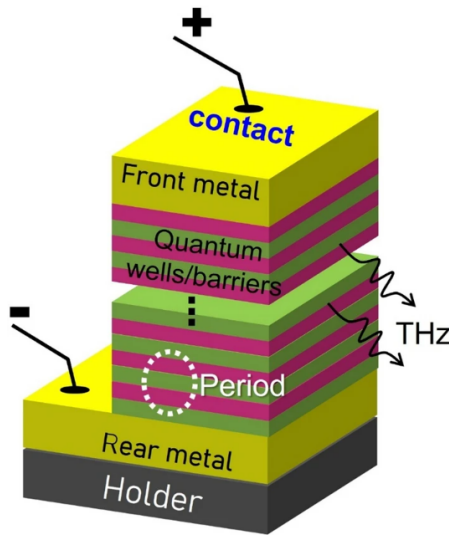


Figure 1.1: Schematic Representation of the Quantum Cascade Laser (from [181]).

The QCL laser properties can therefore be broadly classified into two categories: the *design properties*, which captures the laser design characteristics and the *optoelectronic/working characteristics* which detail the laser emission/performance behaviour as a result of injection of current

into the laser device. Examples of optoelectronic characteristics include emitted power, emission frequency, working temperature and the threshold current density. The design properties on the other hand include metallization geometry (metal-to-metal, plasmonic waveguide), materials used, laser geometry (the ridge length and width). One may simulate the structure of the QCL, but it is difficult to catch the effect of fabrication imperfections.

Since their inception, QCLs have gained interest and a lot of effort has been put in place to make them more robust and efficient. QCLs are capable of various industrial applications in various domains. The potential applications vary from the medical domain to the other industrial applications such as spectroscopy, fundamental physics among others. For instance, in the biomedical field, the radiation can be used in detection of abnormal tissues, including cancers [173]. The waves also have potential application in the pharmaceutical industry, for instance in detecting organic compounds in drugs and identification of two- or three dimensional distributions of molecules in pharmaceuticals based on the terahertz absorption spectrum [156]. The radiation also has a potential application in electronics: the THz domain waves prefigure extremely high speed telecommunication networks (high speed WiFi, 6G) [84]. Other potential applications also include gas sensing in the mid-infrared range (MIR) [168] and non destructive testing [71].

The laser design features influence the emission behaviour of a QCL device. Some of the properties are dependent on the working of mode the laser, for instance the working temperature or the value of the injected current. Understanding how the various QCL properties influence each other enables optimal QCL laser fabrication. An optimal design process would therefore lead to an optimal emission behaviour of a particular QCL device. Uncovering the hidden relationships between the QCL design and performance features therefore improves the design of QCL devices with optimal performance, for instance QCL devices operating at suitable working temperatures and emitting laser beams within the desired frequency ranges.

The analysis of QCL properties to understand the relationships between design and working characteristics requires access to quality and structured QCL properties data. The QCL properties are usually mentioned in scientific articles proposing various QCL device designs together with their corresponding working/optoelectronic properties. Currently, numerical simulation of the electronic band structure of the device and human expertise are utilized in the analysis of the properties for QCL device optimization. Access to structured data will therefore enable a data driven discovery of the non-trivial correlations between the QCL properties. Data driven discovery methods entail the use of data mining methods to uncover hidden relations in data properties. These approaches can help to catch technical imperfection from the QCL fabrication process.

A sample description of a QCL device with its corresponding properties from scientific texts is given in the sentence below.

"A Terahertz quantum cascade laser with a rather high injection coupling strength based on an indirectly pumped scheme is designed and experimentally implemented. To effectively suppress leakage current, the chosen quantum cascade module of the device is based on a five-well GaAs/Al_{0.25}Ga_{0.75}As structure. The device lases up to 151 K with a lasing frequency of 2.67 THz." (text extracted from [139]).

This text description has several QCL properties which includes: heterostructure materials (GaAs/Al_{0.25}Ga_{0.75}As), working temperature (151 K) and the lasing frequency (2.67 THz). In the heterostructure material *GaAs/Al_{0.25}Ga_{0.75}As*, there is two information : the type of material used *GaAs/Al_xGa(1 - x)As* and the concentration of Aluminium Al ($x = 0.25$). Some of the QCL properties contain special characters for instance the layer sequence which includes forwarded slashes (/) and commas (,). Figure 1.2 shows the description of a heterostructure from a sample article [96]. The figure contains information on the structure layer thicknesses. This information is not mentioned in text in some cases. This requires image analysis techniques in order to extract this information.

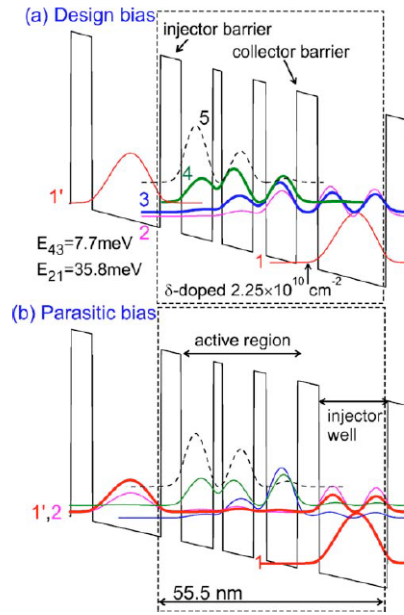


FIG. 1. (Color online) Calculated conduction band diagrams at the design bias (8.4 kV/cm, 46.5 mV/module) and the parasitic bias (6.65 kV/cm, 37 mV/module) with the four-well modules outlined in boxes. Starting from the injector barrier on the left, the layer thicknesses in Å are **49/78/23/76/32/76/52/168** with barriers indicated by bold fonts, and the 52 Å collector barrier is delta-doped at $2.25 \times 10^{10} \text{ cm}^{-2}$ in the center.

Figure 1.2: Sample Heterostructure Description (image extracted from [96]). Shows the calculated conduction band diagrams with the barriers (Injection barrier and collection barrier) for both the design bias and parasitic bias.

The heterogeneous and unstructured nature of the texts possess difficulties in efficiently mining the QCL properties from scientific texts to generate structured data that can be used to analyze these properties for insights regarding proposals of QCL design features with optimal performance behaviour. Some of the laser properties such as the working temperature poses challenges to extract as there may be several temperatures mentioned in text (such as the temperature used for the fabrication process) hence making it difficult to extract the temperature of interest. Development of efficient methods for extraction of QCL properties from text is therefore highly desired in order to generate data that can be used for various tasks in order to understand the implications of various design properties on the laser performance. We thus explore the field

of *Information Extraction (IE)* for the purpose of semi-automatic extraction of QCL properties. We propose rule-based and machine learning techniques that utilizes the patterns and the context in text to identify the QCL properties of interest.

Indeed, information Extraction (IE) methods have gained a lot of interest in the recent past in mining of materials science data from text [41, 133]. IE refers to the process of identifying and extracting specific entities, relationships, events and other information from natural language texts to generate structured output [30]. The adoption of the Text and Data Mining (TDM) policy¹ has proposed a legal framework for the use of intelligent algorithms in the accelerated acquisition and analysis of materials science data [147]. This has led to increased efforts in the adoption of Artificial Intelligence (AI) methods in IE tasks in the materials science domain. Other efforts in materials science data analysis include prediction of the various QCL properties based on other properties using Machine Learning (ML) algorithms [28, 166, 69].

The field of Knowledge Representation (KR) has also been explored to represent complex relationships and concepts in various domains including the materials science domain. KR is a subfield of AI that entails development of systems for representing information about the world in a form that can be understood by computers to enable reasoning on it [21]. The use of the Semantic Web technologies such as Ontologies and Knowledge Graphs (KG) has enabled deployment of KR systems for representation and analysis of data in materials science domain. Examples include the ontology for materials design [105] and a Knowledge graph for materials science [123].

Semantic web technologies allows data access and interoperability across various files, web pages and applications [6, 35]. Ontologies provide a formal conception of the terms and relationships while KGs capture the data instances based on the semantics provided by various ontologies or vocabularies [48, 59]. The expressive nature of KGs allows modeling of complex scenarios [143]. This makes them suitable for modeling complex interactions between various materials science properties for instance between chemical compounds and their physical properties such as temperature. Figure 1.3 shows sample modeling of composition of polymer nanocomposites [117].

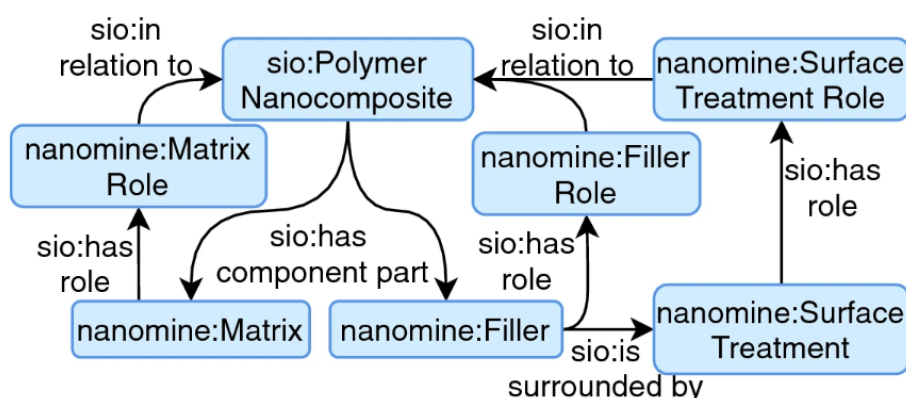


Figure 1.3: Modeling of Polymer Nanocomposites (Figure from [117]). The modeling captures knowledge on the Polymer Nanocomposite components and the roles of the different components such as the treatment roles.

¹[https://www.europarl.europa.eu/RegData/etudes/BRIE/2018/604942/IPOL_BRI\(2018\)604942_EN.pdf](https://www.europarl.europa.eu/RegData/etudes/BRIE/2018/604942/IPOL_BRI(2018)604942_EN.pdf)

Data driven approaches have been proposed for accelerated materials design. Examples of such works include database-driven Materials Selection for Semiconductor Heterojunction Design [155] and data-driven Strategies for Accelerated Materials Design [134]. Figure 1.4 shows a workflow for supervised learning of molecular properties as proposed in [134].

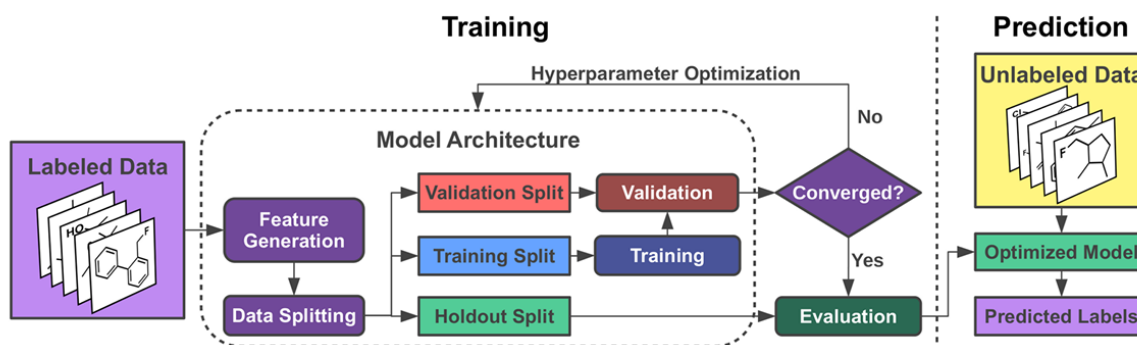


Figure 1.4: Workflow for supervised learning of molecular properties (image from [134]). A model is optimized with labeled data and subsequently used to estimate molecular properties for an unknown (unlabeled) data set.

Data driven approaches require structured data. The adoption of IE and KR in the materials science domain presents opportunities in efficient materials data extraction from unstructured text sources and formally representing data in a structured way. This will enable data driven materials design approaches. The structured data can be enriched with information from other vocabularies that define standards for units of measurements and other standards hence providing semantics for material modeling. The structured data will integrate information from various sources hence making their analysis fast and efficient as opposed to the manual analysis of this data. These methods can be extended to various materials science domains for instance the materials and the semiconductor physics.

1.2 FAIR Data Principles

The FAIR data principles were proposed in 2016. The sole aim of these principles is to provide guidelines to improve the **F**indability, **A**ccessibility, **I**nteroperability, and **R**euse of digital data assets [118]. These enable development of platforms that support machine readable data. The principles are divided into four major categories as outlined below.

Findable

This principle entails the first step of using data and ensures that data and metadata are easily found by both human and computers. This principle has the several guidelines that ensures it is achieved. They are as outlined below.

- *F1. (Meta)data are assigned a globally unique and persistent identifier. This guideline*

plays a crucial role in supporting other guidelines as it enables unique identification of data hence minimizing ambiguity. Persistent identifiers in this context refers to links that lead to a resource defining a particular concept or scenario.

- *F2. Data are described with rich metadata.* The metadata should give comprehensive information of the data being described. This includes giving information on the quality and the context of the data being described. This enables one to find the data and use it based on the information given in the metadata.
- *F3. Metadata clearly and explicitly include the identifier of the data they describe.* This guideline is critical to the FAIR data as it enables findability of the data using the metadata. This is important as in most cases the data and the metadata that describes it are usually separate files. These two files can be associated by including the global unique and persistent identifier for the data in the metadata.
- *F4. (Meta)data are registered or indexed in a searchable resource.* This principle ensures discoverability of data or metadata. The chances of data being useful increases with it being known to exist. Indexing this data enables search engines to discover the data hence making it more useful.

Accessible

This principle focuses on the accessibility of the data. Once the data is found by the user, it should be clear how the data will be accessed possibly including authentication and authorization where it is deemed necessary. The principle of accessibility is divided into major guidelines that support it:

- *A1. (Meta)data are retrievable via identifiers using standard communications protocols.* In order to achieve A1, further guidelines are proposed; A1.1 the protocol is open, free, and universally implementable and (A1.2) the protocol allows for an authentication and authorization procedure where it is deemed necessary. The use of standard protocols ensures free access to data in a standardized way irrespective of the devices trying to access data.
- *A2. Metadata are accessible, even when the data are no longer available.* Data may be unavailable due to various reasons such as maintenance costs hence rendering the links to the data invalid. This guideline enables tracking the data sources during the times of need when the data is no longer available.

Interoperable

This principle emphasizes the need for the data to be integrated with other data. The data should also be interoperable with other applications or workflows. This will enable several activities such as storage, analysis and processing. In order to meet the FAIR Interoperable principle, the following guidelines are proposed:

- *I1. (Meta)data use a formal, accessible, shared, and broadly applicable language for knowledge representation.* This implies that there should be a common way of understanding the data between computers and humans without specified algorithms or mappings.
- *I2. (Meta)data use vocabularies that follow FAIR principles.* The vocabularies used to represent the meta data should be FAIR. The vocabulary should be documented and using globally unique and persistent identifiers in order to enable controlled common use.
- *I3. (Meta)data should include qualified references to other (meta) data.* A qualified reference is the one that gives comprehensive information on its intent. The main aim of this guidelines is to ensure the existence of more meaningful links between the metadata. This plays a crucial role in enriching the contextual knowledge about the data being described which translates to enriching the entire data model.

Reusable

The goal of this principle is to optimize the reuse of data. This is captured in the Reusability guideline R1 which states that (Meta)data should be richly described with a plurality of accurate and relevant attributes. This guideline is divided further as described below.

- *R1.1. (Meta)data are released with a clear and accessible data usage license.* The data usage license indicates the conditions under which the data can be used and to what extent. This is to minimize ambiguity that could limit the usage of a particular data resource.
- *R1.2. (Meta)data are associated with detailed provenance.* Detailed provenance for the data enhances the ability to track the data sources and ensures appropriate credit is given to the data curators. This also gives credibility to the data hence improving its reuse.
- *R1.3. (Meta)data meet domain-relevant community standards.* The data should meet the domain standards for data sharing and representation. This will increase its reuse within the specific community as it adheres to the common standards within that domain.

1.3 Motivation of the Thesis

The main motivation of this thesis is based on the **need for structured access and analysis of QCL properties data in a FAIR manner**. There exists a lot of experimental data for QCL properties captured in various sources in form of scientific texts. Analysis of this data directly within the scientific texts is not technically feasible considering the volume of the texts and their unstructured nature.

Data driven approaches have shown promising results for novel materials design in the semiconductor domain. Existence of structured platforms capturing QCL properties data from various sources will enable a data driven approach in understanding the implication of the QCL design features on the laser working properties such as working temperature. The insights obtained from this analysis can be used to guide the optimization of the QCL designs in order to

achieve optimal performance. This can also accelerate research in the discovery of novel QCL design features that positively impact the QCL laser performance. Other potential use cases could be the prediction of the laser performance based on its design features. The platforms can also be used to maintain provenance information for these properties to for comparison and tracking conflicting QCL properties data.

While this thesis proposes to deal with semantic data integration to enable data driven approaches for property understanding on QCL properties data, the idea is not only for the users of the results, but the insights can also be used for semantic data integration from heterogeneous sources in other specific domain properties, especially from text sources for researchers in IE and KR.

1.4 Problem Statement

In this section, we present the problem of **providing an interoperable, semantic-aware access and analysis of QCL properties data** for exploration in order to understand the relationship between the various QCL properties. We present this in form of a scenario in an example as below.

Example 1.1. Problem Definition: *Consider a Scenario where a semiconductor engineer needs a structured access to QCL properties data for analysis for various insights in a FAIR manner.*

There are several cases in which such information is vital, for instance in cases where there is need to fabricate a QCL device with certain design features corresponding to certain desired optoelectronic properties such as working temperature and optical power for an optimized operation. In addition to this, for the existing QCL laser devices, there may also be the need to quickly understand the relationship between the various design and performance characteristics in order to get insights for future QCL fabrication processes.

There may also be the need to understand trends in QCL semiconductor laser fabrication over a given period of time and the other features on which the QCL devices are based on. This creates the need for valid references to various sources documenting the QCL laser device design and performance characteristics. In order to achieve a structured collection of QCL properties data that can be used for exploration, several issues need to be dealt with. These are as outlined below.

- The QCL design and optoelectronic/working properties data together with their metadata exists in heterogeneous sources in the form of scientific articles. The text is usually in the unstructured format (with QCL properties captured in diverse scientific texts) hence presenting challenges in their structured extraction. The domain specific nature of the QCL properties also poses additional challenges in uniquely them properties in text.
- Decisions regarding designing of QCL lasers with target working/optoelectronic properties are usually carried on the basis of manual experimental data analysis and experience of the experts which takes time hence delaying the design process. A gap also exists between

what is expected from numerical simulation and the experimental result as a consequences of imperfection in fabrication process.

To deal with these issues and enable QCL properties data exploration in order to understand relationships among them, there is therefore need for efficient methods to extract QCL properties data from the unstructured scientific texts to create structured QCL properties data and propose knowledge representation systems that can be used to formally organize this data for exploration. This should enable a standardized access to QCL properties data in a FAIR manner allowing exchange and analysis of the data in an interoperable way by the community. The information maintained should also include the metadata for the QCL properties in order to trace them back to the sources for provenance. This will enable a quicker exploration of these properties hence providing insights for accelerated fabrication of QCL designs with target working properties/characteristics.

Despite the existence of articles capturing QCL properties data (for instance 124 articles containing the term "quantum cascade laser" with a random search on ArXiv), another related challenge in achieving this structured QCL data access is also the lack of quality training datasets for training and evaluation of methods for QCL properties extraction from scientific texts.

1.5 Hypotheses and Research Directions

In order to enable structured access to QCL properties data from various heterogeneous sources for exploration, we argue that it is necessary to *explore the development of efficient methods for generating structured QCL properties data from scientific texts and proposing KR platforms to organize this data for analysis*. We therefore define the hypotheses to guide the study as below.

- **Hypothesis 1:** Information Extraction methods can be used to propose techniques for efficiently generating structured QCL properties data from unstructured scientific texts.
- **Hypothesis 2:** Formal representation of Knowledge on QCL design and working properties successfully supports structured access, analysis and sharing of QCL properties data in accordance to the FAIR data principles.
- **Hypothesis 3:** Semantic enrichment of QCL properties data via a KG enables the analysis of this data for insights regarding the relationship between the QCL design and working properties.

Based on the defined hypotheses, the thesis therefore proposes two research directions (RDs) as outlined below.

- **Research Direction 1 (RD1):** The development of **robust datasets** and **novel IE methods for extraction of structured QCL properties data** from scientific texts.
- **Research Direction 2 (RD2):** The development of robust KR platforms for **formalizing the knowledge for QCL properties** and for the **semantic enrichment** of the QCL properties in text.

Both RD1 and RD2 complement each other to ensure a structured exploration of QCL properties from diverse unstructured scientific texts.

1.6 Research Questions

In relation to the identified RDs, we formulate research questions in order to guide the research as below.

- **Research Question 1:** *How can Natural Language Processing (NLP) techniques be utilized together with QCL domain expert knowledge to curate an original dataset for extraction of QCL properties from scientific texts?*

This research question falls under RD1. In this research question, we argue that NLP techniques can be adopted together with QCL expert knowledge in order to curate original datasets for the purpose of developing and evaluating IE methods for extracting QCL properties from scientific texts. These datasets can be used as benchmarks in advancing the IE methods for QCL properties extraction from text.

- **Research Question 2:** *How can information extraction techniques be utilized for the task of generating structured QCL properties data from scientific texts?*

This research question also falls under RD1. In this research question, we argue that IE methods can be adopted in order to design algorithms for generating structured QCL properties data from scientific texts. This involves tackling the initial challenge of accessing this data that is spread in wide heterogeneous sources in form of unstructured text and generating structured QCL experimental data. This data can be integrated in platforms that enable analytics on the various QCL properties

- **Research Question 3:** *How can knowledge representation systems such as Ontologies be utilized for representing knowledge on QCL design and working properties?*

This research question falls under RD2. In this research question, we hypothesize that KR techniques such as Ontologies can be adopted to formalize knowledge for the QCL properties. This will provide a platform for organizing QCL properties data and allow automated reasoning and inference on it by both machine and humans. This will also act as a community standard for accessing and analyzing QCL data in an interoperable way.

- **Research Question 4:** *How can knowledge representation systems such as Knowledge Graphs be utilized for semantic enrichment of QCL properties in scientific texts?*

This research question also falls under RD2. In this research question, we argue that Knowledge Graphs can be adopted to capture and store QCL properties data and be used as a platform for

data mining tasks on the relationships between QCL properties. This can also provide provenance information in order to track the properties to their sources.

In recent years, KR technologies such as Ontologies and Knowledge Graphs have gained a lot of interest in representation of materials science data [105, 24, 123]. Information Extraction methods have also been proposed for generation of databases of materials properties from scientific texts [41, 93, 77, 133].

Despite these great achievements, the existing KR tools cannot be readily used to capture information on QCL properties data due to reasons such as lack of formal definitions for the properties concepts and the relationships between them. No standards have been proposed to formalize knowledge in this domain. For the IE techniques, there is need to develop methods that generalize well on the QCL properties data extraction task. The machine learning based IE techniques also need quality training data in order to efficiently perform on the task of QCL property extraction from text.

One of the main challenges of enabling interoperable access to QCL properties data is the inter-disciplinary nature of this work. This requires the ability to enable machines (computers) to integrate, reason and make sense of domain specific data with a lot of ambiguity, not forgetting the heterogeneous nature of the data sources. This calls for collaboration with QCL experts in this nature of work in order to deal with this complex devices. Based on this, QCL experts were involved in several stages, from requirements elicitation for ontology and KG development, problem specification, dataset generation and validation of the solutions.

We investigate each of these aspects in the coming chapters in order to design methods to extract QCL properties from scientific texts to generate structured QCL properties data and propose semantically enriched/structured platforms that can be used to analyze this data for insights regarding the optimal design of QCL devices. We experiment and validate the proposals with QCL data and present the results.

1.7 Contributions

Based on the problems elaborated in the previous section, the objective of this study is to address four main aspects of generating knowledge representation platforms for exploring QCL properties in scientific texts and we briefly discuss them below.

1.7.1 Generation of an Original Dataset for Extraction of QCL properties from Scientific Texts

In this contribution, we address the problem of scarce quality training datasets for extracting QCL properties from text as presented in Research Question 1. We propose a novel instruction dataset for training and evaluation of LLMs for QCL property extraction from scientific texts as presented in [90]. We use sample sentences from scientific articles to generate new sentences using the GPT-3.5 instruct model (through paraphrasing and generation of new sentences). The generated sentences are labeled with the help of experts in order to form the final dataset.

The proposed instruction dataset provides a background in understanding the relationship between the various QCL properties. The dataset enables development of ML algorithms to extract QCL properties from text to generate data that can be used for analysis of the relationships between these properties. The dataset also provides a benchmark dataset for evaluating large language models’ performance for the task of information extraction of properties in the QCL domain. The dataset has been made available to the public for research purposes under the CC-BY license².

1.7.2 Extraction of QCL Properties from Scientific Texts

In this contribution, we address Research Question 2. We explore rule-based and machine learning methods in order to develop methods for QCL properties extraction from text. This is important for generating structured data from scientific literature that can be used to analyze the relationships between the various QCL properties.

In the rule based method, we propose an approach for adapting the chemDataExtractor toolkit for QCL property extraction from text [165]. The toolkit is based on the autosentence parser that does not generalize well on properties in the QCL domain. We define new syntactic rules for identification and extraction of QCL properties from text. This method is suitable in case of lack of quality training data for developing machine learning models for QCL property extraction from text.

The machine learning approach is based on LLMs. We propose a Retrieval Augmented Generation (RAG) approach for QCL properties extraction from text [104, 52]. LLMs are trained on general knowledge data and often struggle on domain specific tasks. This implies that they cannot be readily used for extraction of QCL properties without adaptation. We hypothesize that exposing the model to labeled data consisting of sample text describing QCL properties and the corresponding extracted properties improves its ability in recognizing the QCL domain specific concepts for their extraction from text. In order to achieve this, we adopt the dataset proposed in 1.7.1 to provide context to the model during QCL properties extraction in real time. We investigate a dynamic prompting strategy via RAG where every prompt is automatically updated with quality training examples from a labeled instruction dataset. We show that this improves the model learning capabilities as opposed to static prompting where few shot training examples have to be specified in every prompt for every property being extracted. This method does not also require intensive computational resources as compared to LLM finetuning approaches.

1.7.3 Formal Representation of Properties in the QCL Domain

In this work, we address Research Question 3 by proposing a formal representation of properties in the QCL domain. The formal representation is in form of an Ontology that gives formal definitions of the concepts for the QCL design and working/optoelectronic characteristics and the relationships between them [89]. It also provides formal definitions for provenance information in order to trace the QCL characteristics information to the source. The Ontology is developed

²<https://doi.org/10.57745/U3U7XR>

based on user requirements specified in collaboration with the QCL device experts. Expert defined use cases are also adopted for the ontology validation.

The proposed ontology adopts terms from various Ontologies such as the Materials Design Ontology [105], European Materials and Modelling Ontology (EMMO) [54], Provenance Ontology [100], cheBI Ontology [36] and the QUDT Ontology [62]. We define new terms for the concepts of interest that are not captured by the existing ontologies and vocabularies. The proposed ontology is important in providing a platform for FAIR access to information for the QCL properties, the relationships between them and their provenance information by both machines and human. The ontology model is also made available to the public for research purposes under the CC BY license³.

1.7.4 Semantic Enrichment of the QCL Properties Descriptors in Scientific Texts

In this contribution, we address Research Question 4 by proposing a workflow for generating a Knowledge Graph for the QCL properties from scientific texts. The workflow is based on the machine learning based information extraction approach proposed in section 1.7.2. The approach is used to extract QCL properties from text to generate structured data for QCL properties that is used to generate the KG.

We define a KG schema for organizing the data extracted from text. The schema is based on the QCL properties Ontology proposed in section 1.7.3 and other vocabularies such as the Bibliographic Ontology (BIBO)⁴. We generate a QCL properties KG for 36 QCL devices containing 2979 triples capturing the design, working properties, provenance information for the properties and the links to the various references for the QCL properties.

The generated KG provides a semantically enriched and interlinked data platform for analyzing QCL properties from various heterogeneous and unstructured scientific texts while providing links to the provenance information and references on which the properties are based. The KG can be extended with more properties for more QCL devices using the proposed KG generation pipeline.

1.8 Publications and Data

The contributions of this thesis have led to several publications in various international venues. We present them as below.

1.8.1 International Scientific Journals

- **Contribution 1 (Presented in Section 1.7.1):** Kerre, D., Laurent, A., Maussang, K., & Owuor, D. (2025). An instruction dataset for extracting quantum cascade laser properties

³https://github.com/DeperiasKerre/qcl_Onto/tree/main/qclontology/version-1.0

⁴<https://dcmi.github.io/bibo/>

from scientific texts. Data in Brief, 58, 111255 [90]. <https://doi.org/10.1016/j.dib.2024.111255>

- **Contribution 3 (Presented in Section 1.7.3):** Kerre, D., Laurent, A., Maussang, K., & Owuor, D. (2025). A Concise Ontological Model of the Design and Optoelectronic Properties in the Quantum Cascade Laser Domain. Semantic Web, 16 (4), 22104968251359870 [89]. <https://doi.org/10.1177/22104968251359870>

1.8.2 International Scientific Conferences/Workshops

- **Contribution 2 (Presented in Section 1.7.2):** Kerre, D., Laurent, A., Maussang, K., & Owuor, D. (2023, August). A text mining pipeline for mining the quantum cascade laser properties. In European Conference on Advances in Databases and Information Systems (pp. 393-406). Cham: Springer Nature Switzerland [88]. https://doi.org/10.1007/978-3-031-42941-5_34
- **Contribution 4 (Presented in Section 1.7.4):** Kerre, D., Laurent, A., Maussang, K., & Owuor, D. (2025). Semantic Enrichment of the Quantum Cascade Laser Properties in Text-A Knowledge Graph Generation Approach. In LLM-TEXT2KG 2025: 4th International Workshop on LLM-Integrated Knowledge Graph Generation from Text (Text2KG) Co-located with the Extended Semantic Web Conference (ESWC 2025), June 1 - June 5, 2025, Portoroz, Slovenia [87]. https://ceur-ws.org/Vol-4020/Paper_ID_4.pdf

1.8.3 Data

The contributions have also led to the publication of an original dataset. The dataset is publicly shared for research purposes. The details are as outlined below.

- Kerre, Deperias; Laurent, Anne; Maussang, Kenneth; Owuor, Dickson, 2025, "An Instruction Dataset for Extracting Quantum Cascade Laser Properties from Scientific Text", Recherche Data Gouv, V1. <https://doi.org/10.57745/U3U7XR>

1.9 Thesis Document Outline

Figure 1.5 shows an outline of the remaining sections of this thesis. Each chapter corresponds to the contributions made by this thesis as outlined in section 1.7 and the publications in section 1.8.

A description of each of the chapters is as outlined below.

In Chapter 2, we introduce preliminary concepts in Knowledge Representation (KR) and discuss how KR can be utilised to achieve the FAIR (Findability, Accessibility, Interoperability, and Reuse) data principles. We discuss the different technologies and the different standards used in their development. We also present a comprehensive review of related works in KR for

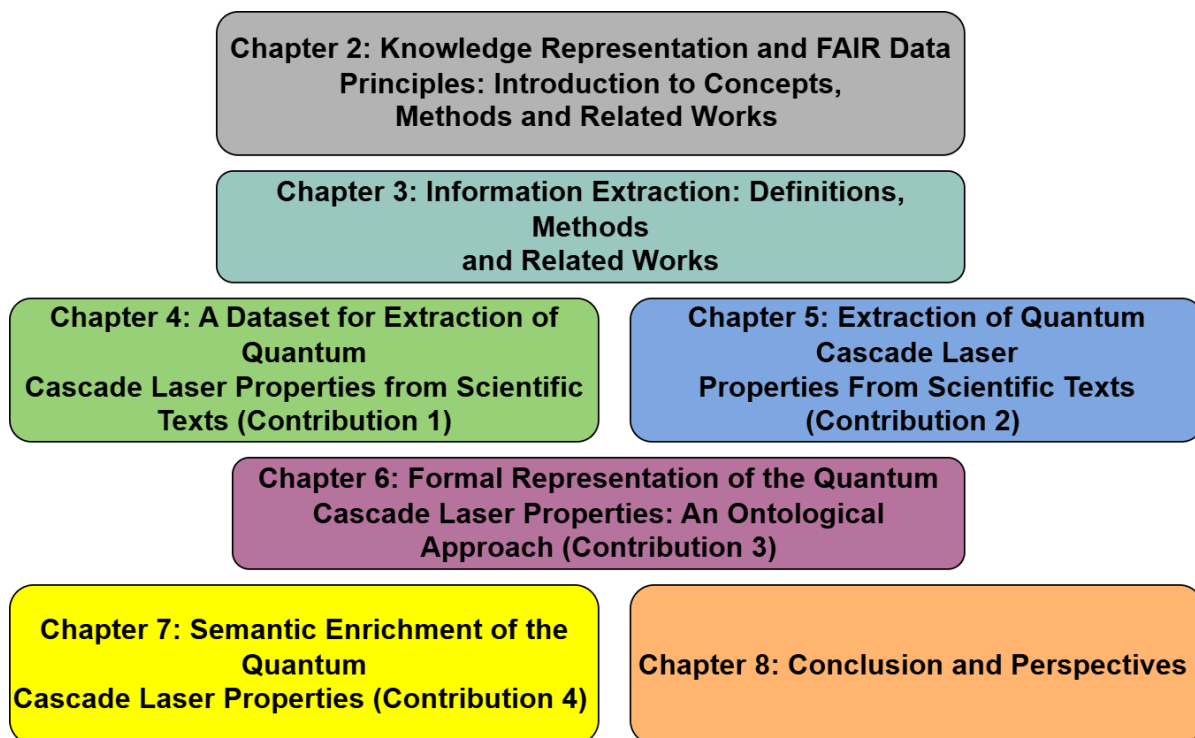


Figure 1.5: Outline of the Thesis Document

the materials science domain in relation to our domain of interest. We conclude this section by giving a summary of the discussed concepts and the open issues that need to be addressed. The concepts discussed in this chapter provides a foundation for the work in chapters 6 and 7.

In Chapter 3, we introduce the concepts of open Information Extraction (IE) in the natural language processing (NLP) perspective. We discuss the methods used and also review the related works in the materials science domain. Lastly, we present open challenges that need to be addressed in the perspective of our work. The concepts discussed provide a foundation for the work in chapter 5.

In Chapter 4, we propose an original instruction dataset for extracting QCL properties from scientific texts. We explore the use of LLMs and domain expert knowledge in generating a labeled dataset using sample sentences from scientific articles. We detail the the dataset generation workflow and the dataset statistics. We also provide more information on the dataset such as the availability.

In Chapter 5, we explore the possibility of developing methods for generating structured QCL properties data from scientific texts using both rule-based and machine learning approaches. This step enables the generation of structured QCL data that can be used for analysis in various platforms. We propose two methods for QCL property extraction from scientific texts using both rule based and mchine learning approaches. We evaluate them and discuss the results.

In Chapter 6, we propose a formal representation of the QCL properties and the relationship between them in form of an ontology. This is based on foundational ontologies in materials science

and other vocabularies. The proposed ontology formalizes the knowledge on QCL properties by providing semantics for the various concepts and relationships for QCL properties. We evaluate the ontology model's consistency and domain applicability and discuss the results.

In Chapter 7, we adopt the machine learning-based approach for QCL properties extraction from text in Chapter 5 and propose a workflow for generating a Knowledge Graph for QCL properties from text. We validate the consistency of the generated Knowledge Graph and evaluate it on domain expert competency questions and discuss the results.

In Chapter 8, we give a conclusion of this study and emphasize the applicability of the proposed approaches in formal representation and structured exploration of QCL properties for insights regarding the development of QCL devices with target properties. For future perspectives, we propose several research directions that can be explored as an extension to the contributions made by this study.

1.10 Conclusion

In this chapter, we have given the background to the study, an introduction to the FAIR data principles, the problem statement, the main research questions and hypotheses, the main contributions of this thesis, a summary of the publications and the thesis document outline.

In the next chapter, we introduce the concept of Knowledge Representation. We give an overview of the tools and technologies used such as Knowledge Graphs, Ontologies and the standard specifications such as Resource Description Framework (RDF), RDF Schema (RDFS), Web Ontology Language (OWL) and SPARQL Protocol and RDF Query Language (SPARQL). We demonstrate the role of Knowledge Representation in ensuring FAIR data. We also present a review of the state of the art for Knowledge Representation in materials science and conclude with open research challenges that remain.

Knowledge Representation and FAIR Data Principles: Introduction to Concepts, Methods and Related Works

2.1	Introduction	20
2.2	Knowledge Representation	20
2.2.1	Introduction	20
2.2.2	Semantic Web	21
2.2.3	Linked Data	21
2.2.4	Ontologies	25
2.2.5	RDF Graphs and Property Graphs	29
2.2.6	RDF Graph Databases	32
2.2.7	SPARQL Query	32
2.2.8	Related Work	34
2.3	Knowledge Representation and FAIR Data Principles	36
2.4	Conclusion	37

“The Semantic Web is not ‘merely’ the tool for conducting individual tasks that we have discussed so far. In addition, if properly designed, the Semantic Web can assist the evolution of human knowledge as a whole.”

– Tim Berners-Lee, James Hendler and Ora Lassila (2001)

2.1 Introduction

In this chapter, we briefly introduce the concepts of Knowledge Representation (KR), the Semantic Web, the various standards used in KR, and the various technologies used in KR such as Ontologies and Knowledge Graphs. We also discuss the role of KR in achieving the FAIR Data principles. Lastly, we review related works for KR in the the materials science domain and present the open issues with regard to our domain of interest.

2.2 Knowledge Representation

2.2.1 Introduction

Knowledge representation is subfield of artificial intelligence (AI) that is dedicated to the development of frameworks for representing information about the world entities in a form that can be understood by computers to enable reasoning on it [21]. These formal frameworks enables computers to reason and solve complex tasks such as answering questions or making inferences for data mining tasks. Some of the earliest works on computer based KR were General Problem Solver [124] and Advice Taker both proposed in 1959 [115].

The intuition of KR is borrowed from the concepts of psychology on how humans represent knowledge and solve problems [152]. These concepts are used in designing formal frameworks that enable representation of complex systems of knowledge. The various kinds of automated reasoning are made possible by adopting the concepts of logic [74, 10].

One of the key issues in the design of knowledge representation frameworks is balancing the trade-off between expressivity¹ and tractability² [103]. This enables the design of knowledge representation formalisms that allow expressivity in the concepts captured while at the same time having control over them. One of the ways to assess this is by use of First Order Logic (FOL)³ [116]. FOL however has limitations in ease of use and implementation due to the numerous ways of expressing the same thing. This can be handled by adopting Description Logic, which has more set of symbols for expressivity with limited use cases [10].

¹Expressivity refers to the knowledge representation language’s ability to represent a wide range of knowledge and concepts.

²Tractability refers to the efficiency of a knowledge representation language in reasoning and inference tasks in a given amount of time and resources.

³FOL is the formalization of the predicate logic that is used for reasoning about objects, their properties, and relationships between them.

Semantic networks are one of the popular formalisms of knowledge representation. A semantic network captures semantic relations between concepts within the network [102]. Semantic networks can either be directed or undirected and can be instantiated with data to explore the relations. One of the most interesting types of such semantic network within the World Wide Web (WWW) community is the Semantic Web [17]. The Semantic Web enables sharing of machine readable data on the web. In the following sections, we introduce the semantic web and discuss the principles adopted in order to achieve its goal.

2.2.2 Semantic Web

In 2001, Tim Berners-Lee, James Hendler, and Ora Lassila proposed the concept of the Semantic Web, which is an extension of the Web that enables reuse and sharing of data across various applications [17]. This implies that the shared data is machine readable. This enables the computers understand and reason even on complex data representations. The semantic web is set up by the World Wide Web Consortium (W3C) through a set of standards that formalize the data sharing process on the Web⁴. In this context, the data shared ranges from schemas, vocabularies, datasets and the data semantics (meaning of this data) [68]. The interconnection of data on the web to enable structured exploration is termed as Linked Data (LD) [19]. The Semantic Web principles provides a foundation for achieving a web of linked data.

In order to provide semantics within the web of data, several standards have been proposed and adopted by the W3C. The standards facilitate common protocols and data formats for data sharing and reuse on the web. We discuss these standards under linked data in the following section.

2.2.3 Linked Data

Linked Data constitutes a set of best practices for publishing and establishing a connection between structured data on the web [19]. LD can be also be termed as Linked Open Data (LOD) if interconnected resources are open. The web of linked data is based on a set of rules for publishing data on the web proposed by Berners-Lee in 2006 [16]. These forms the basis of the principles of linked data and therefore the standards adopted in publishing linked data [66]. We discuss these standards in the following sections:

Uniform Resource Identifier (URI)

URIs are used to uniquely identify resources on the web of data. They constitute of a compact sequence of characters that identifies abstract or physical resources and their specification is guided by internet standard of the WWW⁵. URIs also provide information on how to act on a resource or obtaining a representation of the resource, for instance the format, the protocol to be used, homepage etc.

⁴<https://www.w3.org/standards/semanticweb>

⁵<https://web.archive.org/web/20171230123648/http://abcdrfc.free.fr/rfc-vf/rfc3986.htm>, accessed on 14 November 2024

For illustration purposes, consider the following URI: “<https://www.strathmore.edu>”. This URI identifies the resource (Strathmore Homepage) and implies that the resource representation can be obtained via the (Hypertext Transfer Protocol Secure) HTTPS from a host named www.strathmore.edu.

Resource Description Framework (RDF)

RDF⁶ is a Semantic Web standard that is used for representation and data interchange on the web [98, 2]. It has features that enables description and exchange of metadata hence enabling a standardized exchange of data on the web, irrespective of the data format or data schema structure.

RDF adopts URIs in naming the relationship between resources. The resources (which range from people, organizations, places etc) are in form of named entities. This connection forms a directed labeled graph. In this graph, the edges represent the named link between two resources which are represented by the graph nodes. Attributes are associated with the entities through literals.

RDF has a set of concepts that builds the foundation of an RDF triple, the basic element of knowledge representation in RDF. The concepts includes Resources, Literals, Properties and Statements and we present them below.

Resource: A *resource* is an object or a thing, which can be real or abstract. Resources can range from objects, people, documents, concepts ect.

Literal: A *literal* refers to a value with an associated data type such as a string or an integer. Literals can be used to capture values for instance age of people or names of items etc.

Node: A *Node* refers to an entity (resource) representing information about anything in the world. A blank node is a node that represents a resource with a missing URI or Literal.

Statement: An RDF statement is an object-attribute-value triple. This triples consists of a resource, a property and a value where the value can either be a resource or a literals.

Based on these RDF concepts, an RDF triple can be defined as illustrated below.

Definition 2.2.1 (RDF Triple). Let B be the set of all blank nodes, U the set of URIs and L the set of Literals. An RDF triple t is given as $t = (s, p, o)$, where $s \in (U \cup B)$, $p \in U$ and $o \in (U \cup B \cup L)$

In Definition 2.2.1, s stands for the subject which is being described in the triple, p the property which represents the relationship between the subject s and o the object value. We show an example of a simple triple that captures personal information below.

Example 2.2.1 RDF Triple Generation: Consider the following description: *John Fabian*

⁶<http://www.w3.org/TR/2014/REC-rdf11-concepts-20140225/>

and Mark Strong are friends. They have known each other for more than 10 years.

Using vocabularies from RDF and FOAF ⁷, we can generate the following triples:

triple 1: <foaf:#JohnFabian foaf:knows foaf:#MarkStrong>

triple 2: <foaf:#JohnFabian RDF.type foaf:Person>

triple 3: <foaf:#MarkStrong RDF.type foaf:Person>

In this example, triple 1 captures the relationship ‘know’ i.e p between John Fabian (subject) and Mark Strong (o). The triple gives a description of the person that the subject John Fabian knows. The triples 2 and 3 indicates that the subjects John Fabian and Mark Strong are persons. The semantics of the relations are provided by the vocabularies in the prefixes. i.e foaf and RDF.

Resource Description Framework Schema (RDFS)

RDFS⁸ is a W3C standard that extends the RDF data model by providing a data-modeling vocabulary for RDF data [22, 27]. This provides a structure to the RDF resources by describing basic formalism for representing these resources.

The core concept of the RDFS vocabulary is the *rdfs:Class*. An *rdfs:Class* consists of a group of resources. The members of an individual class are referred to as the class instances and the *rdf:type* property is used to assert that a class is an instance of another class. The concept of *subclass* is also used as follows: if a class A is a *subclass* of A’ then all instances of A will also be instances of A’. The concept of subclass is indicated using the *rdfs:subClassOf* property which asserts that a particular property is a subclass of another property.

The *rdfs:Resource* class is the class of all RDF resources i.e all things described by RDF. This class is an instance of the *rdfs:Class*. There is also the *rdfs:Literal* class which is the class of literal values such as strings and integers. This class is an instance of *rdfs:Class* and a subclass of *rdfs:Resource*. The *rdfs:Datatype* class refers to the class of datatypes. This class is both an instance of and a subclass of *rdfs:Class* and each instance of this class is a subclass of *rdfs:Literal*.

The other important class in RDFS is the *rdfs:Property*. This is the class of all properties in RDFS, both data and object properties and it is an instance of *rdfs:Class*. The instances of *rdfs:Property* class includes: *rdfs:subClassOf*, *rdf:type*, *rdfs:range*, *rdfs:domain*, *rdfs:label*, *rdfs:subPropertyOf* and *rdfs:comment*.

The *rdfs:range* property is used to state that the values of a property are instances of one or more classes while the property *rdfs:domain* is used to state that any resource that has a given property is an instance of one or more classes. The *rdfs:subPropertyOf* property is used to state that all resources related by one property are also related by another. The *rdfs:label* property can be used to provide a human-readable version of a resource’s name. For instance, a triple <X *rdfs:label* Y> states that Y is a human-readable version of the name for the resource X. The *rdfs:domain* of *rdfs:label* property is *rdfs:Resource* and the *rdfs:range* of *rdfs:label* is *rdfs:Literal*.

⁷FOAF (an acronym for friend of a friend), a machine-readable ontology describing persons, their activities and their relations to other people and objects: <http://xmlns.com/foaf/spec/>

⁸<https://www.w3.org/TR/rdf-schema/>

Lastly, the *rdfs:comment* property is usually used to provide a human-readable description of a resource. For instance the triple $\langle U \text{ rdfs:comment } V \rangle$ states the *V* is a human-readable description of the resource *U*. Likewise, the *rdfs:domain* of *rdfs:comment* property is *rdfs:Resource* and the *rdfs:range* of *rdfs:comment* is *rdfs:Literal*.

The comprehensive description of the whole vocabulary can be accessed in the RDFS documentation. RDF has concrete syntaxes for describing the various RDF resources. We present a brief overview of the syntaxes with relevant examples in the following section.

Concrete Syntaxes for RDF

RDF has several abstract syntaxes for specifying the RDF documents. These captures the grammatical rules for defining RDF triples. The syntaxes include N-Triples, Turtle, RDF/XML, RDFa, JSON-LD, TriG and N-Quads. In this section, we present RDF/XML⁹ and Turtle¹⁰ syntaxes that we adopt in this work.

RDF/XML: The RDF/XML syntax for RDF documents constitutes the historical syntax for serializing an RDF document as an XML document. This enables exchange of RDF documents by different devices with varied operating systems. We present an example of a serialization in RDF/XML syntax as described below.

Example 2.2.2 RDF/XML Serilization: *Consider a scenario describing a document at an organization (in this case Inria), which has properties author (to capture its author) and topic (to capture the document topic).*

Listing 2.1 shows a sample serialization of an RDF document in the RDF/XML format.

Listing 2.1: RDF Document Serialization in RDF/XML syntax

```
<?xml version="1.0"?>
<rdf:RDF xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#"
xmlns:inria="http://inria.fr/rr/doc.html">
<inria:author rdf:resource="http://ns.inria.fr/pierre.pro#me"/>
<inria:topic>Semantic Web</inria:topic>
</rdf:Description>
</rdf:RDF>
```

The first line in the document is the XML declaration. The is followed by the root element of RDF documents: `<rdf:RDF>`. The `xmlns:rdf` and `xmlns:inria` namespaces specifies namespaces for the elements in the RDF document. The `<rdf:Description>` element captures the description of the resource which is identified by the `rdf:about` attribute. The elements `<inria:author>` and `<inria:topic>` are the properties of the resource being described.

Turtle: The Turtle syntax is widely used and more readable for the human eye. Listing 2.2 shows the serialization of the RDF document in example 2.2 into the Turtle syntax.

⁹<https://www.w3.org/TR/rdf12-xml/>

¹⁰<https://www.w3.org/TR/rdf12-turtle/>

Listing 2.2: RDF Document Serialization in Turtle syntax

```
@prefix rdf:<http://www.w3.org/1999/02/22-rdf-syntax-ns#>.
@prefix inria:<http://inria.fr/schema#>.
<http://inria.fr/rr/doc.html>
inria:author<http://ns.inria.fr/pierre.pero#me>;
inria:topic "Semantic Web"
```

A Turtle statement begins with prefix declaration associating prefix names to the namespaces used within the RDF document.

URIs are captured within angle brackets and the literals within quotation marks. Declaration of triples with the same subject end with a full stop with the declaration of the individual properties being separated by semicolons. If there exists more than one literal for the same property, this can be separated by commas.

Web Ontology Language (OWL)

Both RDF and RDFS are limited in terms of expressivity. The expressivity of RDF is limited to binary ground predicates while that of RDF schema is limited to subclass hierarchy and property hierarchy within the definitions of domain and range. The Web Ontology Language (OWL)¹¹ [5] provides more expressivity as compared to RDF and RDFS.

The main aim of OWL is to formalize descriptions of a domain by specifying the classes, properties and individuals. The language also enables assertion of properties and reasoning over the classes and the individuals within the classes. OWL comes in three sub-languages¹². This includes OWL-Full, OWL-DL and OWL-Lite.

OWL Lite is suitable for users who basically need a classification hierarchy with simple constraints. OWL DL is has DL in its name due to its correspondence with description logic. OWL DL is suitable for users who need maximum expressiveness while retaining computational completeness (i.e all conclusions are guaranteed to be computable) and decidability (all computations will finish within a finite time frame). Lastly, OWL-Full is suitable for users who need maximum expressiveness and the syntactic freedom of RDF. There is however no computational guarantees.

2.2.4 Ontologies

Ontologies are one of the most popular KR frameworks that are used to represent knowledge in various domains. They provide formal definitions or semantics for KR. An ontology, from a philosophical context, involves the study of the nature of being and existence. The term ontology is derived from two Greek words “onto”, which means being, and “logia”, which means written or spoken discourse. A review on the metaphysical aspect of ontology summarized the essence of ontology as follows: “provide a definitive and exhaustive classification of entities in all spheres of being” [161].

¹¹<https://www.w3.org/TR/owl-guide/>

¹²<https://www.w3.org/TR/owl-features/>

In the computer science domain, an ontology is interpreted as a “a specification of a conceptualization” [59]. It provides a formal definition of concepts and their relationships in a particular context. Based on these prepositions, an ontology can therefore be viewed as a description of of things, relationships, and their characteristics. In most cases, the description is usually bounded within a particular domain.

Ontologies in essence also constitutes a taxonomy, as a a graph structure that describes the hierarchical relationship of a group of things. Members within the taxonomy that further down the hierarchy inherit characteristics from one or more parents that occur above them [136]. This taxonomy is extended to enable the logical expression of relationships among things, membership in groups, multiple inheritance relationships, the symmetry of relationships, exclusiveness, and various characteristics of a given thing/real world entity.

The Semantic web based Ontologies are focused on the logical consistency and expressiveness of the ontology taxonomies (facts, relationships, properties) on the web. This is in order to allow for machine understanding and reasoning [40]. These Ontologies are developed using OWL. This allows for reasoning on the represented knowledge. Apart from enabling reasoning, they provide formal descriptions/semantics of concepts on the web that can be reused for various tasks [75].

An example of an ontology is shown in Figure 2.1. The ontology represents knowledge on the various properties of a material.

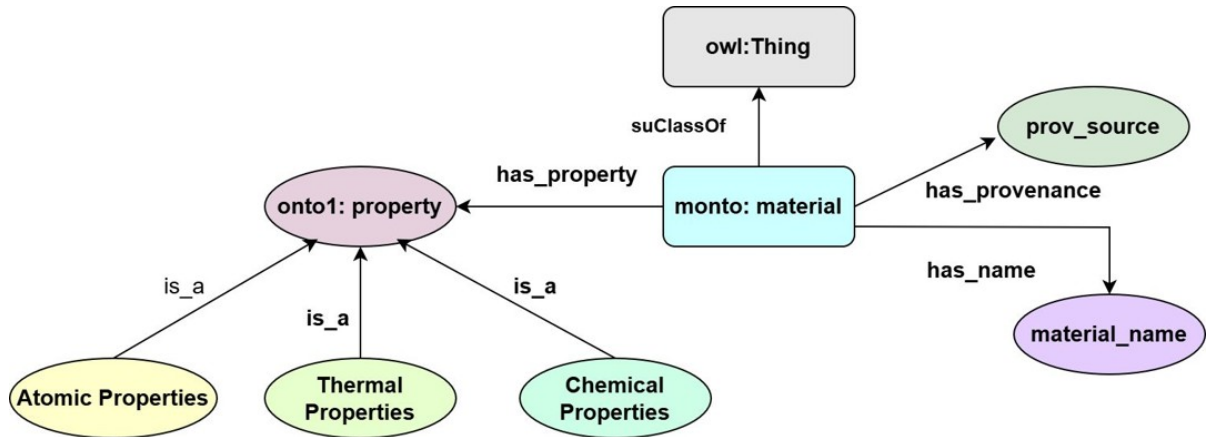


Figure 2.1: Sample Ontology for Material Properties

Ontology Engineering Methodologies

Ontology engineering constitutes the ontology development process, ontology life cycle and methodologies for building Ontologies [121, 120]. There are several options for building Ontologies, for instance, building from scratch, from an existing global or local ontology, from a corpus of information sources only, or from a combination of latter two approaches [34].

Different researchers have proposed several sophisticated and elaborate ontology engineering methods. An analysis of some of these methodologies is presented in [79]. One of the most

common ontology engineering methodology is the MENTHONTOLOGY [49]. This method was proposed for building domain ontologies from scratch. This ontology methodology does not support interoperability which is an important aspect to be considered during ontology development [79].

The Neon ontology engineering methodology is a comprehensive ontology engineering methodology that considers ontology development while taking the surrounding context into consideration [164]. This methodology has a series of 9 ontology development scenarios that correspond to the ontology development activities at the various stages of the ontology development life cycle. For instance, *scenario 1: From specification to implementation*, entails the first step in the ontology development life cycle which involves building an ontology from scratch. At this level, the ontology requirements and use cases are elicited in collaboration with the intended ontology users and the domain experts. Their role is crucial in making sure the ontology requirements are fully and correctly captured. Other interesting development scenarios include:

- *Scenario 2: Reusing and re-engineering non-ontological resources.* This scenario is used in cases where ontology developers need to assess non-ontological resources and determine whether they can be adopted in the ontology building process. The assessment is based on the requirements of the ontology. The task of re-engineering the selected resources into ontologies is also captured in this scenario.
- *Scenario 3: Reusing ontological resources.* In this scenario, ontology engineers reuse ontological resources in developing the new ontology. The resources reused may include reusing the whole ontology or some sections/terms in the ontology. This is also based on the requirements of the ontology being developed.
- *Scenario 4: Reusing and re-engineering ontological resources.* In this scenario, the ontology engineers both reuse and re-engineer ontological resources. The re-engineering may involve modification or extension of concepts in the ontologies being re-used.
- *Scenario 5: Reusing and merging ontological resources.* This scenario is adopted in cases where the ontology developers need to create a new ontology from two or more ontologies from the same domain.
- *Scenario 6: Reusing, merging, and re-engineering ontological resources.* This scenario is the same as scenario 5. The difference is instead of using the set of merged resources as it is, they are re-engineered before being used.
- *Scenario 7: Reusing ontology design patterns (ODPs).* This scenario entails ontology engineers accessing ODPs and reusing them in the ontology development life cycle.
- *Scenario 8: Restructuring ontological resources.* In this scenario, ontology developers perform a restructuring process before re-using ontological resources in the ontology being built. The restructuring may include activities such as modularizing, pruning, extending, and/or specializing.
- *Scenario 9: Localizing ontological resources.* This scenario entails Ontology developers adapting an ontology to other languages and culture communities hence producing a multilingual ontology.

A detailed description on these scenarios can be found in [164]. In most cases, not all the scenarios are adopted in the ontology development life cycle. Specific scenarios are adopted based on the ontology development activities and requirements of a particular ontology being developed. For instance, scenario 9 may not be adopted in cases where there is no need to localize/develop the ontology model in a particular local language.

The Neon ontology engineering methodology is more comprehensive as it captures most of the ontology development activities. The methodology also support the reuse, re-engineering and merging of existing ontological and non ontological resources hence enabling interoperability. This enables development of domain ontologies that are interoperable across various standard ontologies/vocabularies integrated into it. With regard to these mentioned benefits, the Neon methodology is adopted in this work.

Provenance Modeling in Ontologies

Provenance in KR is an important aspect as it entails maintaining the history of entities/data objects [122]. This captures several aspects such as the creators, the time, versions, changes made etc.

In the semantic web community, the prov-ontology serves as the W3C standard for representing provenance information [100]. This standard vocabulary has several classes such as the Agent class. This class can be reused to represent knowledge on provenance. In the materials science domain, it has been extended to represent provenance information on materials properties data [105].

Provenance modeling in KR plays a crucial role in several cases. For instance, it can be used to track conflicting information for further review. Several methods can be used to rank this information and establish correct facts among the conflicting information based on the sources.

Ontology Evaluation

Ontology evaluation is still an open research area and several ontology evaluation techniques have been proposed in literature. These methods can be categorized as follows: those based on the assessment of logical consistency of the ontology [61], metric based evaluation [167] and ontology validation [141].

Logical consistency implies lack of contradictions in the ontology hierarchy definition. This can be assessed by ontology checking tools and logical reasoners such as HermiT Reasoner [55], Pitfall Scanner [135] and the Pellet Reasoner [159].

Ontology validation entails assessing the ability of the proposed ontology in meeting the intended requirements. This is achieved through designed test cases in form of competency questions which are evaluated on the ontology. In ontology validation, the role of domain experts in which the ontology is to be applied is very crucial in the evaluation process. Ontology validation is used in domain specific ontology evaluation, for instance in the materials science domain [105].

The metric based evaluation entails evaluation of ontology schema and instance metrics. The schema metrics address the design of the ontology. Some of the schema metrics include *Attribute Richness*, *Inheritance Richness*, and *Relationship Richness*.

The instance quality metrics examines the way data is placed in the ontology. This plays a crucial role in indicating the effectiveness of the ontology design and the amount of knowledge represented by the ontology. The instance metrics are further divided into knowledge base metrics and class metrics. More information on these categories can be found in [167].

In this thesis, we explore the *Inheritance Richness* as a metric based evaluation in order to evaluate the ability of the proposed ontology to detail the domain specific knowledge in the schema. This metric details the distribution of information across different levels of the ontology's inheritance tree. Based on this, an ontology can be categorized as either horizontal or vertical. In a formal way, the inheritance richness of the schema (IRs) can be defined as the average number of subclasses per class.

An ontology with a low Inheritance Richness metric is regarded of a horizontal nature, which reflects a very detailed type of knowledge that the ontology represents. On the other hand, an ontology with a high Inheritance Richness metric value is regarded to be of a vertical nature, which implies that the ontology represents a wide range of general knowledge [167]. We provide more details of this evaluation in chapter 6.

2.2.5 RDF Graphs and Property Graphs

RDF Graphs

An RDF graph is an RDF document with several RDF triples. It is an instantiation of a ontology providing guiding principles for organizing data. This can be formally defined as as illustrated in Definition 2.2.2 below.

Definition 2.2.2 (RDF Graph). Consider a finite set of triples G such that each triple $t = (s, p, o)$. An RDF Graph G is given by $G = \{(s, p, o) \mid s \in V, p \in E, o \in V \cup L\}$, where V is the set of nodes, E the set of properties/edges, L the set of literal values, s the subject, p the predicate and o the object.

We demonstrate an example of an RDF graph in the example below.

Example 2.2.3 *RDF Graph:* Consider a description of a person which contains of the full name Mark Strong, the email mark.strong@hotmail.com and their home page given by the URL https:example.com/markPage.

Figure 2.2 shows a sample RDF graph for representing information about this person using the FOAF¹³ vocabularies. The prefixes RDF and foaf represent the namespace prefixes for the vocabularies being used: foaf (FOAF vocabulary) and RDF (the RDF namespace). This shows the relationships between the various entities in the RDF graph which consists of 5 RDF triples.

¹³<http://xmlns.com/foaf/0.1/>

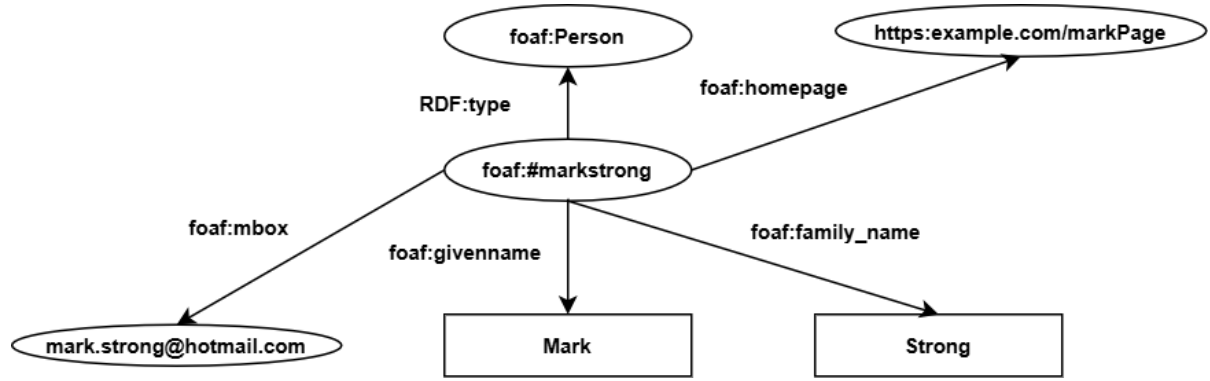


Figure 2.2: Sample RDF Graph Representing Personal Information

A sample RDF graph for representing data on materials properties is shown in Figure 2.3. The semantics are provided by the ontology sample shown in Figure 2.1.

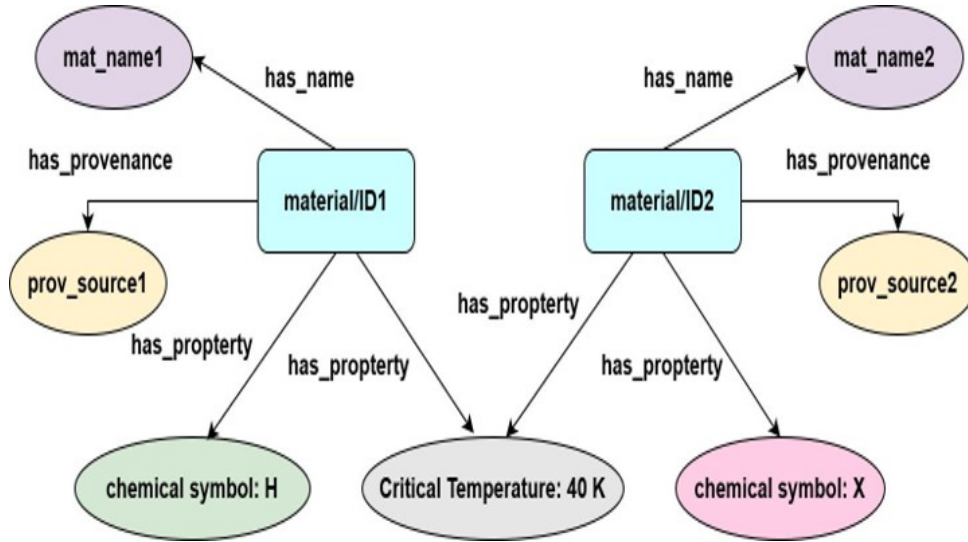


Figure 2.3: Sample RDF Graph for Material Properties

Property Graphs

A Property Graph (PG) is a type of a graph data model where information is organized in the form of nodes and edges. A PG can therefore be formally modeled as $G=(V, E)$, where V stands for the nodes/vertices and E the edges. The nodes represent the different entities of interest while the edges capture the relationships between the various entities. Both the nodes and edges can have attributes to capture more information about them. The nodes in a PG can have one or more labels to describe the type or class of the nodes. The relations between nodes have also have a single type to indicate the nature of the relationship. Both nodes and relationships have attributes/properties in the form of key:value and the relationships have direction.

The PG graph data model is used to capture complex relationships in data hence providing a rich view on how data can be modeled over various datasets as compared to relational data representations. Figure 2.4 shows a PG modeling the relationship between two collaborators who have worked on publishing a book.

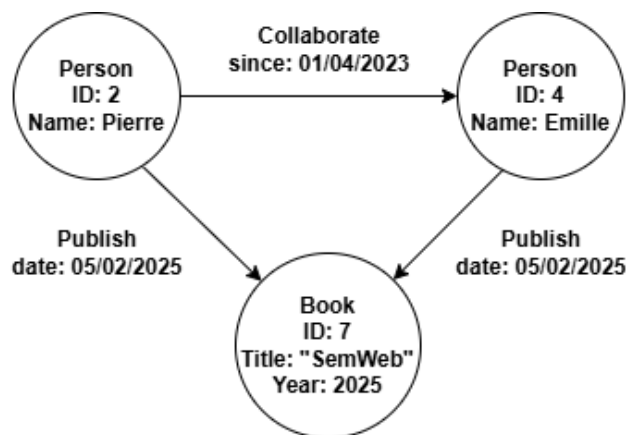


Figure 2.4: Sample Data Modeling with a Property Graph

Both PGs and RDF graphs can be used to model complex relationships captured in data. The semantics of the data representation is provided by standard and controlled vocabularies such as ontologies. However, RDF graphs enable the capturing of semantics on both the entities and the relationships between them. This allows for a formalized way of representing knowledge on various data models, hence enabling machine-readable data and interoperability. The formal semantics and standards in RDF graphs enables standardized data exchange, integration and semantic reasoning on the web of data. This ability is not inherent in PGs.

Knowledge Graph Evaluation

The evaluation of KGs plays a crucial role in validating their quality. The evaluation can be done to assess the structural quality and the ability of the KG in meeting its intended application use case. The major assessment criteria for KGs include the logical consistency, completeness and correctness/applicability to the relevant use cases.

KG consistency refers to the the extent to which the knowledge of a KG does not contradict itself [183, 196]. The main focus by this evaluation metric is on the logical correctness of the KG as opposed to the schema correctness. A KG is consistent if there are no conflicts within the knowledge it represents. Consistency of a KG triple can be assessed by the use of logical reasoners such as Pellet [159] and Hermit [55]. Evaluation of a KG's correctness entails the assessment of the accuracy of the facts and the relationships in the KG. Several methods can be used for instance using gold standard KGs and expert validation. Expert validation may involve checking the correctness of the KG triples through sample expert defined test cases.

KG completeness is a metric that measures the ratio of available information in the KG as compared to the required information that is supposed to be captured by the KG [80]. KG

completeness therefore assesses the possibility of missing crucial information in the KG. This information can be inform of relations between entities or data instances for the relations. KG completeness classified into schema and property completeness [132].

Schema completeness assesses the extent to which classes and properties are represented in the KG. This is done by defining a set of mandatory relations that all classes of entities should have and checking if there are entities without the mandatory relations. This ratio is used to determine the extent of missing information in the KG schema. Proper completeness on the other hand measures the extent of the missing property values of a specific kind of property. This implies that for a particular property, what is the proportion of missing data values. This ratio is used to estimate the amount of missing information hence the KG completeness.

2.2.6 RDF Graph Databases

An RDF graph database/triple store is a type of graph database that stores data in the form of semantic networks. The data is stored in the form of RDF triples (Subject, Predicate and Object) hence the name triple store. This provides a platform for inference to uncover hidden information out of the existing relations in the data network. Examples of RDF graph database include GraphDB¹⁴, AllegroGraph¹⁵ and QLever¹⁶ among others.

RDF graph databases plays a crucial role in managing Linked Open Data datasets. This enables querying of diverse and evolving semantically enriched data from heterogeneous in an interoperable way.

2.2.7 SPARQL Query

SPARQL (SPARQL Protocol and RDF Query Language) Query¹⁷ is the standard and the W3C recommendation query language for querying RDF data [7]. The queries can be executed directly on stored RDF data or indirectly via middleware. We present descriptions of SPARQL query language based on the work presented in [7].

A SPARQL query can be considered to be of the form $head \leftarrow body$, where the head determines the answer format of the query and the body consist of complex RDF graph pattern expression that may include RDF triples with variables, conjunctions, disjunctions, optional parts and constraints over the values of the variables.

Consider a SPARQL query Q and an RDF graph G : the query Q is executed on G by matching the body of Q against G . This will result to variable bindings that are used together with information in the head to retrieve the variable values to construct the answer. Now if we assume the existence of a set of variables V disjoint from U , the graph pattern can be defined as a tuple from $(U \cup V)(U \cup V)(U \cup V)$. Consider a variable $?X \in V$ and $?X \in c$ then $c \in U$. A mapping μ can be considered to be a partial function such that $\mu : U \rightarrow V$ [2, 7]. Let t be the

¹⁴<https://graphdb.ontotext.com/>

¹⁵<https://allegrograph.com/products/allegrograph/>

¹⁶<https://qllever.cs.uni-freiburg.de/wikidata>

¹⁷<https://www.w3.org/TR/2013/REC-sparql11-overview-20130321/>

triple pattern, the $\mu(t)$ can be considered to be triple obtained by replacing variables in t with respect to μ . These concepts forms the basis of a formal definition of a SELECT query.

In this work, we focus on SPARQL queries that retrieve results from RDF data based on the specified matching patterns. These queries contains the *SELECT* clause which implements this task. A SELECT query has three parts: (i) the *PREFIX* (optional), where the short hand names for the namespace URIs are defined, (ii) *SELECT*, which specifies the variables to be selected from the RDF data and lastly (iii) the *triple patterns* to be used during the selection process.

Formally, we adopt the definition of a SPARQL SELECT query from [2] and present it as described below.

Definition 2.2.3 (SPARQL SELECT Query). Consider a set of Variables V and a graph pattern P , a SPARQL SELECT query Q can be defined as a tuple (W, P) , where $W \subseteq \text{var}(P)$.

In definition 2.2.3, $\text{var}(P)$ denotes the set of variables in pattern P while W represents the variables in the SELECT clause of Q .

We present an example of a SPARQL SELECT query based on the QCL RDF graph¹⁸. The query is used for retrieving specific information as illustrated in the example below.

Example 2.2.4. SPARQL SELECT Query: Consider a SPARQL query that seeks to retrieve the possible material compositions of a QCL laser heterostructure with an LO Phonon Design Type.

The SPARQL query expression is shown in listing 2.3 and the query results shown in Figure 2.5.

Listing 2.3: Sample SPARQL Query

```
PREFIX QpOnto:<https://github.com/DeperiasKerre/
qcl_onto/blob/main/qclontology/version-1.0/qclonto.owl#>
SELECT DISTINCT ?mat_composition
WHERE
{
  ?HS QpOnto:hasDesignType QpOnto:LOPhononDepopulation;
    QpOnto:hasMaterials ?HM.
  ?HM QpOnto:matFormula ?mat_composition.
}
```

```
material composition: InGaAs/AlInGaAs
material composition: GaAs/Al0.15Ga0.85As
material composition: GaAs/Al0.3Ga0.7As
```

Figure 2.5: Query Results

For the above example query, the graph pattern is shown in Figure 2.6. This pattern is matched on the RDF graph G in order to retrieve the variable HS (Laser Heterostructure), HM

¹⁸https://github.com/DeperiasKerre/qKG/blob/main/qcl_KG/qcl_KG.rdf

(Heterostructure Materials) and (mat_composition) material composition. All the triples that match this pattern are selected and their variables retrieved. It should be noted that only the variable for material composition specified in the SELECT clause is returned in the query answer.

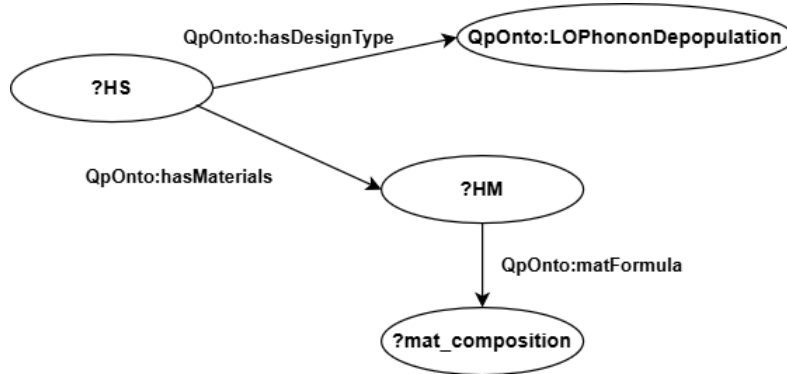


Figure 2.6: Query Graph Pattern

2.2.8 Related Work

Semantic web technologies have been adopted in organizing data and representing knowledge in the materials science domain. This includes the use of Ontologies and Knowledge Graphs in this task. We present the related works in the following section.

Ontologies in the Materials Science Domain

Ontologies have been proposed for knowledge representation in the materials science domain. The Ontologies can be divided into those that represent general materials science knowledge and those that capture knowledge for specific sub-domains. The general materials knowledge Ontologies includes EMMO [54], cheBI [36], freely available dataset of molecular entities e.g atom, molecular ion, etc , the Basic Formal Ontology(BFO) [160], General Formal Ontology(GFO) [70] and the Descriptive Ontology for Linguistic and Cognitive Engineering (DOLCE) [51].

The Ontologies developed for specific sub-domains in materials science are summarized in Table 7.5. Ontologies have also been developed for additive manufacturing in the laser domain. Examples in this case include Laser and Thermal Metamodels [145], Laser Powder Bed Fusion [107] and Metal Additive Manufacturing [146, 56, 149]. Recently, an ontology for the semiconductor domain has also been proposed in [106].

Knowledge Graphs in the Materials Science Domain

Knowledge Graphs have been proposed for exploring materials science data. The semantics of the KGs are provided by several ontologies/vocabularies. The adoption of KGs in materials

Table 2.1: Ontologies for Specific Sub-domains in Material Science

Ontologies	Ontology Metrics	Language	Domain	Intended Application
MatOnto [24]	606 Concepts, 31 relations, 488 instances	OWL	Materials	Materials Discovery
Materials Ontology [8]	78 Concepts, 10 relations, 24 instances	OWL	Crystals	Semantic Querying
MDO [105]	37 Concepts, 64 relations	OWL	Materials Design	Semantic Querying over multiple databases
Polymer Nanocomposite [138]	Being expanded asp per use cases	OWL	Polymer Nanocomposite	Knowledge Representation
NanoParticle Ontology [169]	1904 Concepts, 81 relations	OWL	Nanotechnology	Data Integration, Search
eNanoMapper Ontology [64]	12781 Concepts, 5 relations, 464 instances	OWL	Nanotechnology	Data Integration
MatOWL [198]	(not available)	OWL	Materials	Semantic Querying
MMOY Ontology [199]	2325 Concepts, 9 relations, 1738 instances	OWL	Metals	Knowledge Extraction
Dislocation Ontology [78]	18 Concepts, 16 relations	OWL	Crystalline Materials	Knowledge Representation
MaterialDigital Ontology [3]	13 Concepts, 7 relations	OWL	Material Experiments	Knowledge Representation, Data Curation
MAMBO Ontology [131]	26 Concepts, 33 relations	OWL	Molecular-based Materials	Knowledge Representation
ELSSI-EMD Ontology [9]	35 Concepts, 37 relations, 33 instances	OWL	Molecular-based Materials	Knowledge Representation
Semiconductor Ontology [106]	26 Concepts, 21 relations	OWL	Semiconductors	Knowledge Representation

science has been accelerated by the need for efficient materials data analysis for various reasons such as materials discovery.

A KG for nanocomposite materials dubbed Nanomine, has been proposed in [117]. This KG integrates diverse data from over 1,700 polymer nanocomposite experiments hence providing access to experimental results and their provenance in a linked data format. The Materials Experiment Knowledge Graph is proposed in [163]. This KG captures data and relationships for the general material experiments. Several KGs have also been proposed for general materials science data [123, 125, 176, 203, 200]. Lastly, an approach for accelerated materials discovery enabled by a KG is proposed in [1].

Research Gap and Thesis Positioning

Despite the development of ontologies in the materials science domain, these ontologies cannot be readily used to capture knowledge for the properties in the QCL domain due to several reasons. The general ontologies such as BFO, DOLCE, EMMO and GFO gives general definitions of concepts in materials science domain that are not suitable in representing knowledge for the QCL properties. On the other hand, the specific domain ontologies such as as MatOnto, Materials Ontology, MDO, Nanoparticle Ontology, eNanoMapper, MatOWL, MMOY, Dislocation Ontology, MaterialDigital Ontology are restricted to specific domain concepts that are too limited to capture/represent knowledge for QCL properties and the relationships between them.

The ontologies in the laser domain do not capture knowledge on the QCL design properties such as the design type and layer sequence properties. The relationship between these properties are also not captured. This is also the case with the semiconductor domain ontology where

knowledge on the heterostructure growth sheet has not been formalized [106]. The QCL sequence layer is highly complex compared to the semiconductor lasers in the NIR-Vis domain hence the need to formalize this knowledge. None of these ontologies model the provenance information for QCL properties. There is therefore the need for a formal representation of QCL properties. This will enable formalization of knowledge in the QCL domain hence providing a standardized platform of interacting with QCL data to enable exploration for insights regarding the optimal development of QCL devices with desired working properties.

Despite advancements in the adoption of KGs in accelerated materials discovery, the existing KGs cannot be readily used for capturing and analysis of QCL properties data. The general materials science KG do not capture the knowledge on QCL design and provenance properties. For instance, the heterostructure materials and their relationships with the design types, the working properties and other laser properties such as the working mode. The specific domain KGs are based on ontologies for specific domains of interest. The relationships captured cannot be scaled to capture the relationship between the QCL properties.

A platform for capturing the QCL properties data from various heterogeneous sources is therefore highly desired. A KG with semantics on QCL properties and their relationships can support analysis of these properties in a unified way for insights such as the influence of the design features on the QCL working properties.

2.3 Knowledge Representation and FAIR Data Principles

Knowledge representation plays a critical role in ensuring FAIR access to data. The tools such as Ontologies and KGs allow representation of data in more meaningful ways that allow FAIR access to the data. The concepts in ontologies and KGs are represented with global unique identifiers in the form of URIs. The URIs provide links to resources that comprehensively describe the resources. The concepts are also indexed in controlled vocabularies. This ensures conformance to the FAIR'S Findability principle.

In order to enforce the FAIR Data principle on accessibility, data instances in KGs and Ontologies can be retrieved via links using the common HTTP protocols. The descriptions of the data can still be maintained in global resources referenced by URIs. KGs and Ontologies also enforce the Interoperability principle by adopting formal and broadly applicable languages such as OWL and OWL for Knowledge representation. The use of controlled/widely accepted vocabularies also enhances interoperability. The semantically enriched links between metadata allow for contextualization of the general data being represented.

The FAIR data principle on reusability is enforced by having clear license terms on the usage of the data/concepts within the ontologies or KGs. Provenance information can also be modeled within the ontologies and KGs. Ontologies and KGs can also capture knowledge on domain specific concepts hence providing a standardized way of sharing and processing that within that domain.

2.4 Conclusion

In this chapter, we discuss the concept of Knowledge Representation in the computer science domain and the different technologies used. We also discuss the state of the art for Knowledge Representation in the materials science domain and present the open challenges. This provides a foundation for the next chapters where we propose KR platforms for QCL properties data.

In summary, semantic web technologies and standards have been adopted to allow exchange of machine readable data on the web in various application use cases. Ontologies and KGs are among the most popular tools that have been adopted for knowledge representation in various domains including the materials science domain. Ontologies represent knowledge/facts in a particular context in order to allow reasoning and inference on the represented knowledge. KGs on the other end utilize the standards provided by ontologies to provide an interlinked platform of data instances. The existing Ontologies and KGs in the materials science domain cannot be readily used to capture knowledge on QCL design and performance properties. This is due to the fact that this knowledge has not been formalized. There is therefore no standards/vocabularies for organizing and analyzing QCL properties data in an interoperable way for insights.

Ontologies and Knowledge Graphs can therefore be explored to provide a semantically enriched platform for representing knowledge on QCL properties and their provenance information. This will support FAIR access to QCL data from diverse/heterogeneous sources. The FAIR access to this data will enable analytics in order to uncover hidden relationships among the QCL properties in order to guide the design of QCL devices with optimal working properties.

In the next chapter, we introduce the concept of Information Extraction in the computer science domain and discuss the various methods used for extracting specific entities of interest in structured text. We also present the state of the art for information extraction in the materials science domain and discuss the open research challenges.

Chapter 3

Information Extraction: Definitions, Methods and Related Works

3.1	Introduction	40
3.2	Information Extraction	40
3.2.1	Rule-Based Methods	41
3.2.2	Machine Learning Based Methods	43
3.3	Evaluation of Information Extraction Methods in the Materials Science Domain	49
3.4	Related Works	51
3.4.1	Information Extraction in Materials Science Domain	51
3.4.2	Existing Databases of Materials Science Data	53
3.4.3	Challenges and Thesis Positioning	53
3.5	Conclusion	54

“If I have seen further, it is by standing on the shoulders of Giants.”

– Sir Isaac Newton (1642 – 1727)

3.1 Introduction

In this chapter, we introduce the concepts of Information Extraction with specific focus to the techniques used in materials science information extraction. We also discuss related works for generating materials science databases from scientific text, the existing materials science properties databases, and present open issues that need to be addressed in relation to our domain of interest.

3.2 Information Extraction

Information Extraction is an area of Natural Language Processing (NLP) that entails the process of identifying and extracting specific entities, relationships, events and other information from natural language texts to generate structured output [72]. Information Extraction tasks can be broadly classified into Named Entity Recognition (NER), Relation Extraction, Coreference Resolution, Event Extraction and Open Information Extraction.

NER involves the identification and classification of named entities such as people, organizations, locations, dates within free text. Relation extraction entails the identification and extraction of relationships among entities in text. Event extraction refers to the detection of events and their actors, time and location within free text. Coreference resolution on the other hand involves identifying all expressions that refer to the same entity in a text. Lastly, Open IE entails the extraction of binary relations from text without the need of predefined relations.

IE is an important step in generating structured data from text and has been studied extensively for various domain applications. Potential applications include clinical information extraction [137] and materials science data extraction from text [94]. Other applications include extraction of terminologies and relations for ontology construction in the agricultural domain [85] and the biology domain [130].

One of the common NER processes carried out in material information extraction is the Chemical Name Entity Recognition (CNER). This process entails the identification of chemical and material names of interest in text. This has also been extended to extract other physical properties and other material characteristics such as temperature from scientific texts [165]. The IE methods used extracting structured data from materials science text can be classified into two categories: Rule-based and ML-based methods. We present them in the next subsections.

3.2.1 Rule-Based Methods

Rule-based IE methods are based on handcrafted features which take into consideration the reporting style within the text being analyzed for information extraction. Some of the contextual information used to design the rules include the syntactic and lexical features together with other patterns contained in the text. The methods are helpful in domains where there is no sufficient quality datasets for training ML algorithms for the downstream task of IE. Other advantages of rule-based approaches include readability, maintainability and the possibility of directly transferring domain knowledge into rules [179].

The rule-based IE methods are in most cases developed using grammar rules and dictionaries. Grammar rules capture the patterns in text and are specified in terms of patterns using regular expressions. Whenever the pattern is encountered in text, the particular relationship of interest capturing the pattern is extracted. Dictionaries on the other hand provide an essential resource to provide important information on the semantics and patterns for identifying information in text. They can act as both semantic lexicons and extraction patterns [142].

Semantic lexicons and the extraction patterns work together in order to identify entities of interest in text. The semantic lexicon captures the concepts related to the domain in which the information is to be extracted together with their semantic/meaning categories. This is used to identify and classify the entities and the relationships in the text. Extraction patterns captures predefined rules or patterns that specify how to identify and extract relevant information from text. These patterns can be specified based on a given syntax, semantics or other linguistic features captured in text.

One of the commonly used rule-based toolkit is chemDataExtractor¹, the state of the art rule-based toolkit for NER in the materials science domain [165]. The toolkit was proposed for the automated extraction of chemical entities and their associated properties, measurements, and relationships from scientific documents. Figure 3.1 shows the workflow of the chemDataExtractor toolkit adopted from reference [41]. The toolkit has several steps starting from sentence pre-processing up to the final stage where the data records are extracted.

The toolkit accepts articles in HTML format. The data preprocessing steps include conversion of text from articles into plain text, sentence splitting where the sentences are split into individual units such as words (tokenization) and part of speech (POS) tagging. The preprocessed sentences are then taken through the CNER module in order to identify the chemical entities and their relationships.

The CNER module of chemDataExtractor is based on a set of multiple specialized grammar rules that have been designed to extract more specific types of chemical information. These rules are captured in the grammar logic of the toolkit known as the parser. The *Autosentence Parser* is the default parser (set of grammar rules) for CNER in chemDataExtractor.

The CNER parsing rules in chemDataExtractor are based on the *user model* concept [29]. The *user model* defines a set of property models that are used to specify the information/fields to be parsed from text and the logic/grammar pattern (parser) used to identify the information to be extracted. The user model can be classified into two categories: compound model and

¹<http://www.chemdataextractor2.org/>

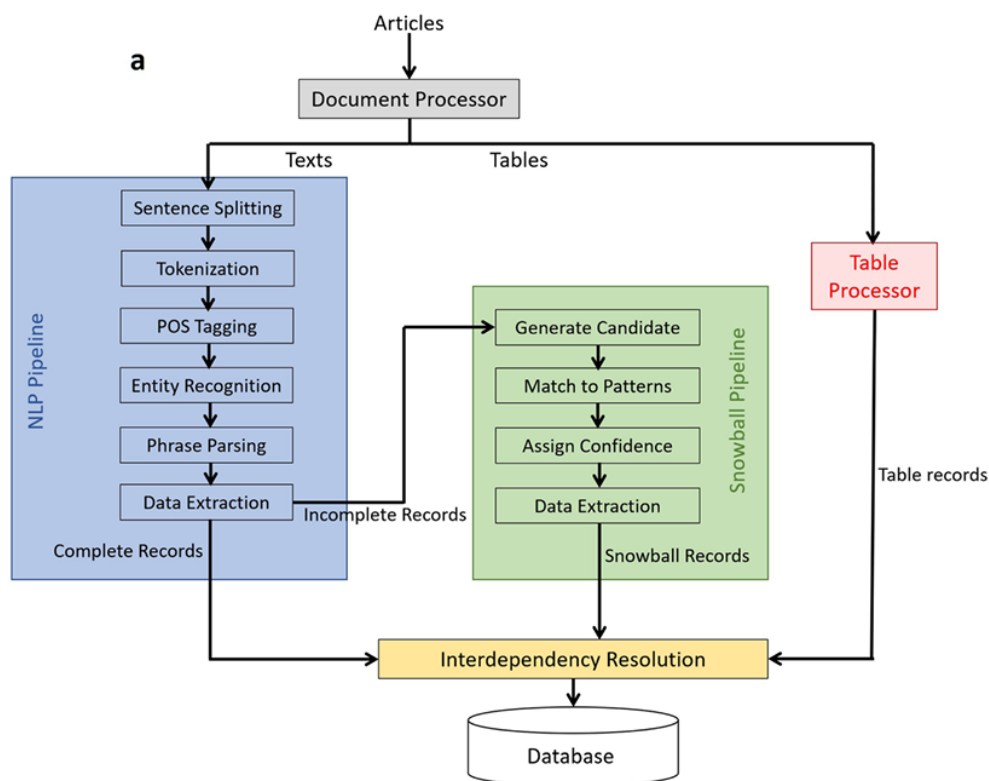


Figure 3.1: The chemDataExtractor pipeline (Image from [41]).

quantity models.

Compound models capture the grammar logic for identifying chemical compounds in text. Quantity models capture the logic for identifying quantitative measurements for properties such as temperature in text. We present an example of the modeling of the grammar logic for identifying a physical quantity such as temperature using the chemDataExtractor toolkit.

Example 3.2.1. Rule Modeling for Property Model: Consider the following text description containing temperature as a property *“The proposed element has a working temperature of 100 K.”*

A property model can be defined to extract the temperature captured in text in the above example. The extraction will entail the extraction of the quantity value and the temperature unit of measurement. This model will fall under the Quantity model as it captures quantitative values. The quantity model in chemDataExtractor defines units and their respective conversions. In cases where the units of interest are not defined, the quantity model can be extended with new units. The temperature model will therefore capture three things: value, units and the parser (grammar pattern) to be used.

In order to specify the text parsing rules, a set of keywords are identified that are used to identify the property of interest. This entails the words mentioned just before or after the property in text. For instance in example 3.2.1, the phrase *“working temperature”*. These key-

words together with the parser elements (in form of regular expression) are used to define the patterns for identifying the properties of interest in text. The predefined parser elements in chemDataExtractor are as presented in Table 5.1.

Table 3.1: The Parser Elements (from [165]).

Elements	Description	Elements	Description
R(Regex)	Match text with regular expression	T (Tag)	Match tags
W(Word)	Match case-sensitive token text	I(IWord)	Match case-insensitive token text
Any	Match any single token	H(Hide)	Ignore the matched tokens
Not	Match only if not followed by some text	FollowedBy	Match only if followed by some text
ZeroOrMore	Match zero or more of the expressions	OneOrMore	Match one or more of the expressions
Optional	Match if it exists	SkipTo	Skips to the next occurrence of text

Based on this, the text matching pattern for extracting temperature in example 3.1 can therefore be expressed as defined below.

Definition 3.2.1 (Working Temperature Pattern). *Working TemperaturePattern* (WP) = $Prefix + Value + Unit$, where *prefix* captures the set of keywords for the particular unit, the value the grammar logic for identifying the value and unit the grammar logic for identifying the units.

The functionality of the grammar parsers of chemDataExtractor is specialized to CNER and generalizes less on the wider materials science domain. The adoption of the toolkit requires further adaptation considering the context of other domains. Possibilities for improvement may involve extensions of the parsing rules and quantity models to cover more quantitative properties. We explore the adaptation of chemDataExtractor for mining QCL properties from text. We extend the toolkit by defining new property models and parsers for mining QCL properties from text. We give more details in the work proposed in Chapter 5.

3.2.2 Machine Learning Based Methods

IE using machine learning methods entails training models to extract information from text. This is done using quality training data from the particular domain of interest. The quality of the information extracted is dependent on the quality of the training data and the training methods applied. The various state of the art ML models adopted for extraction of structured data from text include the traditional Long Short-Term Memory (LSTM) models and the recent adoption of Large Language Models (LLMs). We start by presenting the traditional LSTM models and their variants followed by the recent LLMs in the following subsections:

LSTM Models

LSTM is a type of Recurrent Neural Network (RNN) characterized by the ability to handle long term dependencies captured in sequential data. This enables the model to overcome the vanishing gradient problem commonly exhibited in standard RNNs [53, 73]. Unlike the traditional RNN, it processes the entire sequence of data and not individual points by incorporating feedback from the previous states. This enables it to capture and model the patterns in complex sequential

data such as time series, speech and text. A formal representation of a sequential model as presented in [140] is given as below.

Assume $x = (x_1, x_2, x_3, \dots, x_T)$ represents a sequence of length T and h_t represents the model memory at time step t , the model memory can be updated using the information as presented in equation 3.1 below.

$$h_t = \sigma(W_x x_t + W_h h_{t-1} + b_t) \quad (3.1)$$

where σ is a non-linear function, W_x and W_h the weight matrices and b_t a constant bias.

The memory component in the LSTM model is termed as the ‘gated’ cell. The word gate implies the ability to decide the kind of information to be used/ignored in the memory data. This decision is based on the weight values assigned to this information during the training process. The architecture of an LSTM model is shown in Figure 3.2.

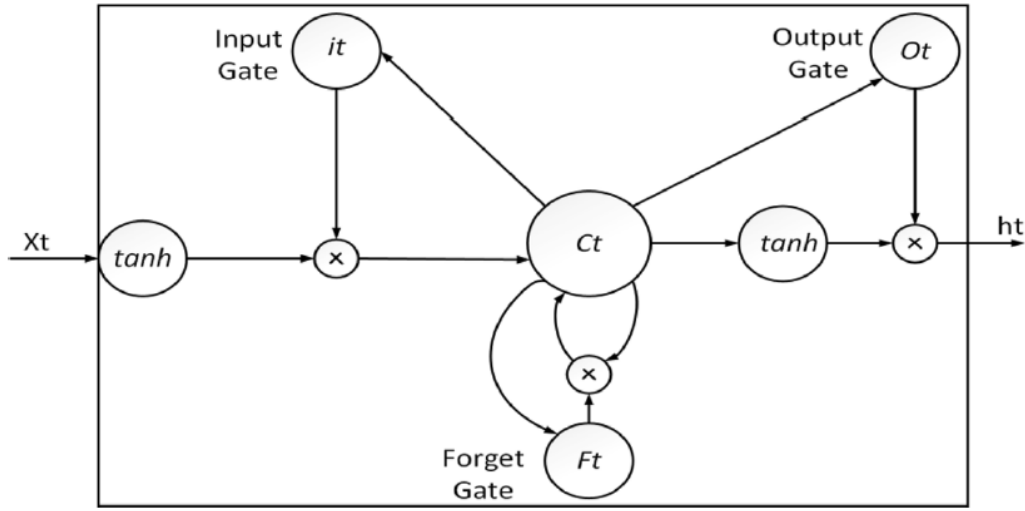


Figure 3.2: The LSTM Model Architecture (Image from [140]).

As shown in Figure 3.2, an LSTM model consist of 3 gates that implement the flow of information through the model. These are the Input gate, the Forget Gate and the Output Gate. We give a brief discussion of them below.

- **Forget Gate:** The role of this gate is to determine which information is to be removed from the LSTM memory. A sigmoid function is applied at this level to implement the decision and this usually based on the values of h_{t-1} and x_t . The output of the Forget Gate f_t is computed as follows:

$$f_t = \sigma(W_{f_h}[h_{t-1}], W_{f_x}[x_t], b_f) \quad (3.2)$$

where b_f is a constant that represents the bias value. The value of f_t fall in the range of 0 to 1, where 0 implies to completely get rid of the learned value, and 1 implies preserving the whole value.

- **Input Gate:** The role of this gate is to determine whether the new information will be added into the LSTM memory or not. It provides a quantitative representation of the importance of the information carried by the input. The input gate consists of two layers: *sigmoid* layer and *tanh* layer. In the sigmoid layer, a sigmoid function is applied to determine which values to be updated. The tanh layer on the other hand creates a vector of new candidate values that will be added into the LSTM memory by assigning some weights. The outputs of the sigmoid and tanh layers are computed as shown in equations 3.3 and 3.4.

$$i_t = \sigma(W_{ih}[h_{t-1}], W_{ix}[x_t], b_i) \quad (3.3)$$

$$c_t' = \tanh(W_{Ch}[h_{t-1}], W_{Cx}[x_t], b_c) \quad (3.4)$$

In equations 3.3 and 3.4, i_t represents whether the value needs to be updated or not, and C represents a vector of new candidate values that will be added into the LSTM memory. The two layers are combined to provide an update for the LSTM memory. The current value in the LSTM memory is forgotten via the forget gate layer by multiplying the old value (c_{t-1}) followed by an addition of the new candidate value $i_t * c_t'$ [140]. The combined memory update is therefore computed as below.

$$c_t = f_t * c_{t-1} + i_t * c_t' \quad (3.5)$$

where f_t represents the results of the forget gate.

- **Output Gate:** This gate performs the following: it first uses a sigmoid layer to determine what part of the LSTM memory contributes to the output, performs a non-linear tanh function to map the values between -1 and 1 and the result is then multiplied by the output of a sigmoid layer. The output is computed by using the following formulas as presented in [140]:

$$o_t = \sigma(W_{oh}[h_{t-1}], W_{ox}[x_t], b_o) \quad (3.6)$$

$$h_t = o_t * \tanh(c_t) \quad (3.7)$$

where o_t represents the output value and h_t its representation value in the range of -1 to 1.

One of the interesting variants of LSTM is the Bidirectional LSTM (BiLSTM) [154]. This is an extension of the LSTM model where two LSTMs are applied to the input data with one taking the input in a forward direction, and the other in a backwards direction. This provides more contextual information to the model in terms of the future and the past hence improving the performance of the model [13]. The general architecture of a BiLSTM model is shown in Figure 3.3.

LSTM models have been adopted in the extraction of structured data from text in various domains. Examples include NER recognition in twitter messages [108], NER for the indonesian language [60] and Chinese NER [110]. Other examples include the use of LSTM models in

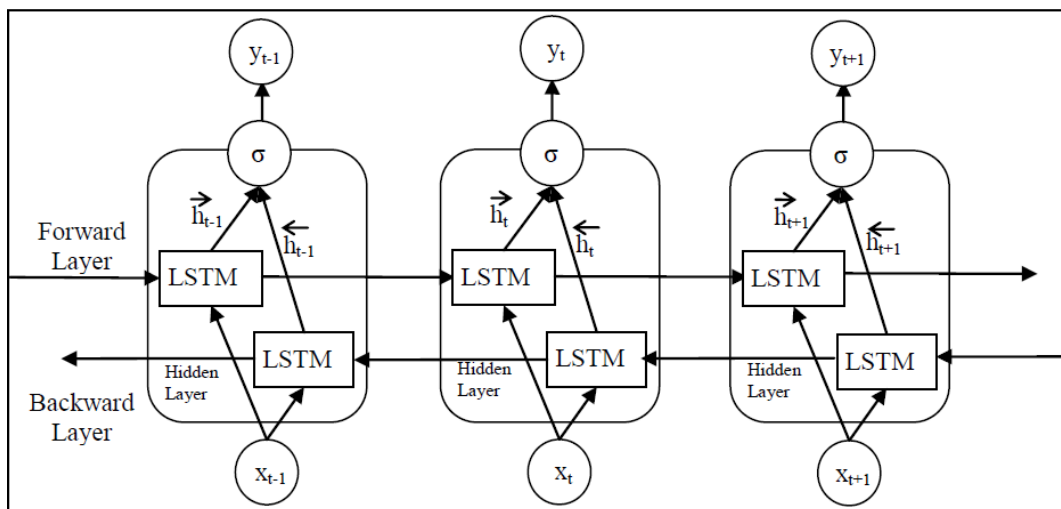


Figure 3.3: General Architecture of a Bidirectional LSTM (Image from [11]).

the biological domain for biological entity recognition [157] and material NER in the materials science domain [148].

The adoption of LSTM models in NER presents their ability in excelling in this task. However, the models cannot be readily utilized in the extraction of structured QCL properties from text. This is due to the fact that many QCL properties entities may be mentioned in text but not all of them are relevant to the corresponding QCL device. This therefore goes beyond mere recognition of the entities hence the need for methods that accurately identifies the entities of interest only.

Language Language Models

Large Language Models (LLMs) are ML models trained on a vast amount of data which enables them to exhibit greater potential in natural language processing tasks. These models are characterized by massive parameter sizes and exceptional learning capabilities. The core principle behind LLM functioning is self-attention mechanism in Transformer architecture² [175]. LLMs have been adopted in the tasks of structured information extraction. Notable examples of such LLMs include the BERT (Bidirectional Encoder Representations from Transformers) model [39].

BERT is a large language model developed by Google with the ability to process text by considering the context of words in a bi-directional manner. BERT model is pretrained on two NLP tasks: masked language modeling (MLM) and next sentence prediction (NSP). MLM entails prediction of the hidden word in a sentence based on the hidden word's context. NSP entails the prediction of whether two given sentences have a logical, sequential connection or whether their relationship is simply random.

Figure 3.4 shows the pretraining process of BERT in which a bi-directional Transformer ar-

²A deep learning architecture for processing sequential data to understand relationships between elements, enabling them to handle long-range dependencies and parallel processing.

chitecture is utilized. $E_1, E_2 \dots E_N$ are the embeddings, $T_1, T_2, \dots T_N$ the token layers while T_{rm} are the individual tokens in the token layers. BERT consist of the following modules: Tokenizer

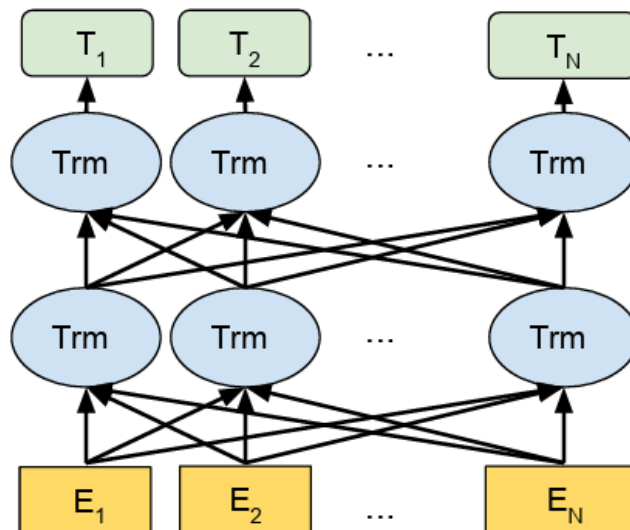


Figure 3.4: Pre-training of BERT (Image from [39]). $E_1, E_2 \dots E_N$ are the embeddings, $T_1, T_2, \dots T_N$ the token layers while T_{rm} are the individual tokens in the token layers.

(for converting a pieces of input text into sequence of integers called tokens), Embedding (for converting the sequence of tokens into an array of real-valued vectors), Encoder (stack of transformer blocks) and Task head (necessary for pre-training by providing probability distributions of the token representation).

Several variants of the BERT model have been proposed for various applications. Notable examples in the scientific domain include SciBERT [15], a pre-trained language model for scientific text and BioBERT, a pre-trained biomedical language representation model for biomedical text mining [101]. The materials science domain has also utilized BERT models for various tasks such as NER, for instance NER for battery materials [77].

Other LLMs include the proprietary GPT models such as GPT-3 [50], instructGPT [129] and GPT-4 [127]. These models are also based on the Transformer architecture. Open source LLMs have also been proposed. Examples of such LLMs include Mistral AI models such as Mistral-7B [81] and Meta Llama models such as Llama 2 [171] and Llama 3 [57]. These models are pre-trained on general knowledge text and can fine-tuned for downstream specific tasks. Several fine-tuning methods have been proposed to enable fine-tuning of LLMs in an efficient manner. Examples of such methods include parameter efficient fine-tuning techniques such as LoRA [76] and QLoRa [37].

A key feature of LLM is in-context learning [23]. This feature enables the models to be trained to generate text based on a given context or a set of examples. Another crucial technique for fine-tuning LLMs is Reinforcement Learning from Human Feedback (RLHF) [26, 205]. Based on the analogy of Reinforcement Learning, this technique involves fine-tuning LLM models with human-generated responses as rewards. This allows the model to learn from its mistakes and improve its performance over time.

Prompt engineering is one of major ways of interacting with LLMs such as GPT and the open source LLMs [188]. This process entails the design of specific prompt texts that guide LLMs in generating desired responses or completing specified tasks. These prompt templates can also be used in the fine-tuning process. The quality of the response generated by the LLM is however dependent on the quality of the prompt design. This implies that for quality LLM responses, quality prompts are required.

One of the biggest challenges for LLMs is hallucinations. This is the generation of content made up and irrelevant by LLMs. Hallucinations are attributed to several factors, for instance the knowledge in the LLM being outdated depending on the task being executed by the LLM. One way of updating this knowledge is by updating the model weights through pretraining which requires a lot of computational resources. To address this limitations, LLMs can be updated with contextual information that is up to date without modification of the model weights. This process is termed as Retrieval Augmented Generation (RAG) [104].

RAG entails combining of a pre-trained retriever with a pre-trained seq2seq model³ (generator) [12]. The process is as summarized in Figure 3.5. In the context of LLMs, the process can be implemented in three steps: Retrieval (involves retrieval of relevant information from external data sources such as databases and documents based on the user's query), Augmentation (augmenting the LLM's input hence providing it with context in relation to the task to be performed) and Generation (generation of a response by the LLM based on the augmented input).

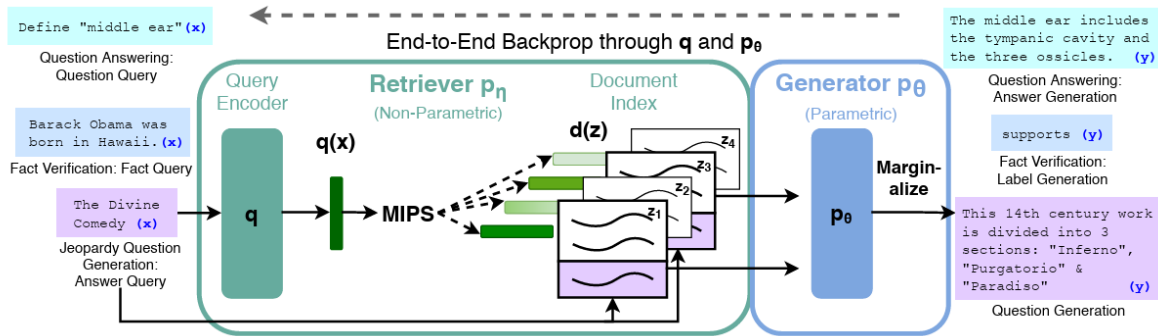


Figure 3.5: RAG Approach (Image from [104]). x denotes the query which is encoded by the process q . The documents provides context that is passed to the generator p that generates contextual information used in getting the response.

The development of RAG approaches has evolved over time and can be divided into three categories: Naive RAG, Advanced RAG, and Modular RAG [52]. These are summarized in Figure 3.6 and we discuss them in the following sections below.

- **Naive RAG:** This represents the earliest RAG methodology. This method adopts the traditional process that consists of three stages: indexing, retrieval, and generation. Indexing entails the cleaning and extraction of raw data captured in various formats such

³Models that are capable of learning and generating a variable-length sequence of tokens such as texts from a variable-length sequence of input data. The input data can be in the form of speech or the same text in another language

as PDF, HTML etc. This raw data is converted into plain text and divided into smaller digestible chunks that are used to provide context to the LLM. The chunks are encoded into a vector representation via an embedding model and stored in a vector database. In the retrieval phase, a user query encoded by using the same embedding model to obtain a vector representation. Similarity scores between the query vector and the vectors in the database are computed and the top K are prioritized for retrieval. The top K vector embeddings in the database and the query are used as the expanded context in the prompt. Lastly in the generation phase, the query and the selected chunks are parsed to the model to generate response.

- **Advanced RAG:** Advanced RAG proposes specific improvements to counter the limitations of Naive RAG. One aims of Advanced RAG is to improve retrieval quality. This is achieved by employing pre-retrieval and post-retrieval strategies. The indexing technique is also refined through techniques such as sliding window approach, fine-grained segmentation, and the incorporation of metadata. In the pre-retrieval strategy, the main aim is to optimize the indexing structure and the original query. Indexing optimization plays a crucial role in enhancing the quality of the content being indexed. Query optimization on the other hand enhances the clarity of the original user question making it more suitable for the retrieval task.
- **Modular RAG:** Modular RAG has better performance than both Naive and Advanced RAG since it enhances adaptability and versatility [52]. This is achieved by the addition of new modules that introduce additional specialized components to enhance retrieval and processing capabilities. An example of such modules is the search module that enables direct searches across various data sources like search engines, databases [185]. Modular RAG also enhances RAG capabilities by introducing new patterns. An example of such patterns is the Rewrite-Retrieve-Read which utilizes the LLM’s capabilities to refine retrieval queries via a rewriting module and a LM-feedback mechanism to update rewriting model [113]. More examples of such patterns are described in [52].

The emergence of LLMs and generative AI in general has opened research opportunities for structured information extraction from text. Various methods have been adopted ranging from prompting and fine-tuning. Examples include NER in the biological domain [83], medical domain [150], chemical domain [191] and lastly general purpose NER [195]. LLMs have also been utilized in the semiconductor domain for various tasks. Examples of these tasks include: semiconductor band gap prediction [194], semiconductor defect classification [109] and semiconductor parameter extraction [186]. These advancements provide opportunities for the adoption of these models for structured information extraction in other domains.

3.3 Evaluation of Information Extraction Methods in the Materials Science Domain

Evaluation of IE workflows in materials science plays a major role in assessing the performance on the IE tasks. In the case of structured data collection, the evaluation is not trivial in most cases. This is due to the fact that there may be multiple synonymous ways of reporting entities [153].

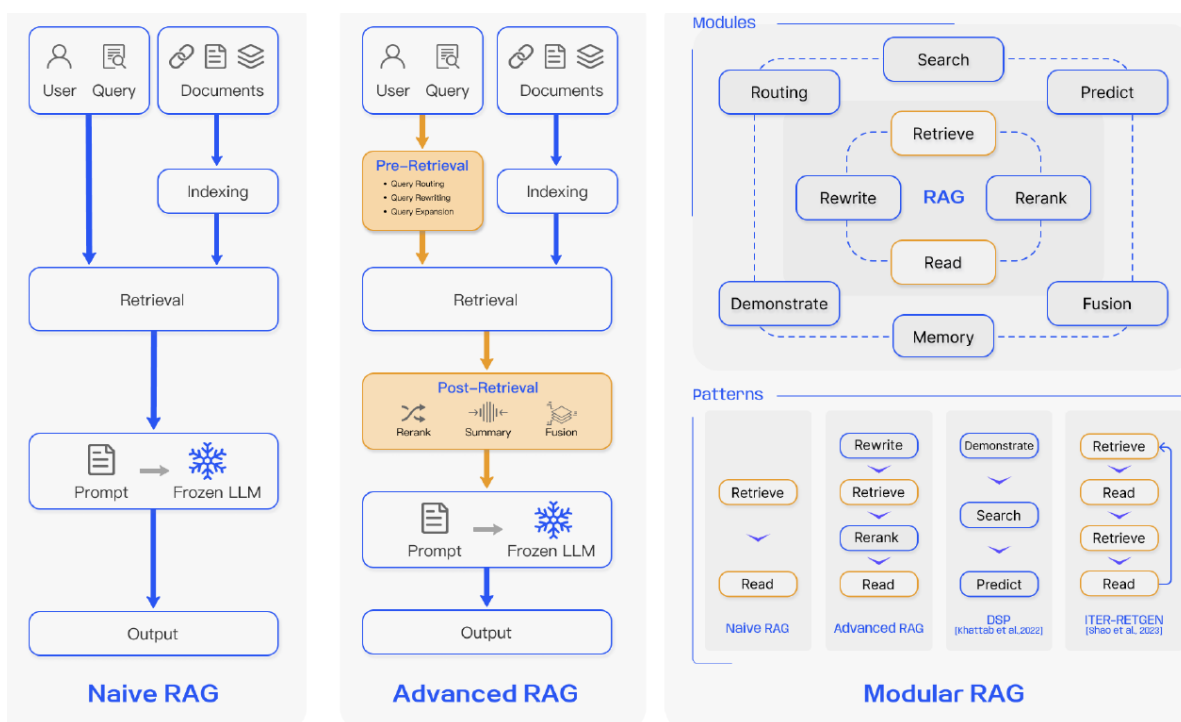


Figure 3.6: RAG Paradigms: Naive RAG, Advanced RAG and Modular RAG (Image from [52]).

As a result of this, automated evaluation becomes a challenge. This is also attributed to the lack of properly defined metrics for structured information extraction [151]. This calls for a careful inspection of the extracted data in order to identify errors at the validation step. Properly defined metrics for structured LLM data extraction helps to develop frameworks for automated evaluation.

The scoring of the IE methods performance for single entity extraction can be thought of as extracting a single value. For instance, if the user is interested in the task of collecting all chemicals in a document, the performance of the IE workflow can be scored by measuring the retrieved entities, the relevant entities and the correct entities out of the retrieved entities [153]. Based on this, two metrics are proposed for scoring an IE method: Precision and Recall. The metrics are computed as shown in the equations below.

$$Precision = \frac{|\{relevant\ entries\} \cap \{retrieved\ entries\}|}{|\{retrieved\ entries\}|} \quad (3.8)$$

$$Recall = \frac{|\{relevant\ entries\} \cap \{retrieved\ entries\}|}{|\{relevant\ entries\}|} \quad (3.9)$$

These definitions of Precision and Recall (as presented in [153]) can be extended to define evaluation metrics for extracting QCL properties in scientific text. In order to define metrics for QCL property extraction from text, we consider a QCL property such as the working temperature which consists of a numerical value and a unit. An extracted working temperature record is considered complete and correct if it contains both the value and the unit. Based on this example, we define the relevant records and retrieved in the definitions given below.

Definition 3.3.1 (Retrieved Entries). This refers to the number of entries extracted, for instance the working temperature entries extracted.

Definition 3.3.2 (Relevant Entries). This refers to the number of all entries captured in the document for a particular property, for instance all the working temperature of the proposed QCL entries in the document.

To illustrate further on Definition 3.3.2, Consider a QCL property mentioned more than once in text. For instance, there may be working temperature of the proposed QCL device and the working temperature of another QCL device referenced to in the text. In this case, the relevant entry for the working temperature will refer to the value that the expert is interested in i.e. the working temperature of the proposed device in the text.

The metrics can be intuitively summarized as follows: Precision estimates the proportion of correct values out of the extracted values while Recall on the other hand estimates the success rate of the extraction process by looking at how many of the values expected are extracted correctly.

The definitions of true positive (TP), false positive (FP), and false negative (FN) are presented in [153]. TP is defined as the correct value extracted. FP refers to extracted values that do not correspond to what is expected. FN corresponds to a value that is in the ground truth/-expected but has not been extracted. True negative (TN) is regarded as value not required to be extracted and it is not extracted by a model. In the case of mining properties of interest from text, such as QCL properties, it is anything that is not a QCL property of interest in the extraction task (including all the other text, figures, symbols etc). This metric cannot be realistically determined as it requires computing the number of all other information in text that is not of interest for the IE task. Based on these error types, Precision and Recall can therefore be computed using the equations below.

$$Precision = \frac{|TP|}{|TP| \cup |FP|} \quad (3.10)$$

$$Recall = \frac{|TP|}{|TP| \cup |FN|} \quad (3.11)$$

3.4 Related Works

3.4.1 Information Extraction in Materials Science Domain

IE methods have been explored in the extraction of materials science data from scientific text. These include both rule-based and ML-based methods. This has led to standard databases of material experiments data. ChemDataExtractor is the state of the art rule-based toolkit for extracting chemical entities from scientific literature [165]. This tool has been extended to other domain-specific use cases. The specific use cases include: semiconductor bandgaps [41], battery materials [77], refractive indices and dielectric constants [201], transition temperatures

of magnetic materials [29], an auto-populated ontology of materials science [114] and thermo-electric materials [158].

Other rule-based tools that have been proposed are LeadMine [111], ChemicalTagger [65] and ChemSpot [144]. Lead mine employs a grammar and a mixture of dictionary based approach for chemical entity recognition. Chemical Tagger is a phrase-based semantic tool for the chemistry domain. The tagging mechanism in ChemicalTagger is based on a modular architecture and uses a combination of OSCAR, domain-specific regex and English taggers. These are structured into tree-based phrases using a specialized grammar. Chemspot is a named entity recognition tool for identifying mentions of chemicals in natural language texts [144]. This tool employs a hybrid approach combining a Conditional Random Field with a dictionary for identifying chemical entities in text.

Machine learning methods have also been widely adopted in the generation of structured data from scientific literature in the materials science domain. An example of ML-methods is tmChem [99]. This tool is based on two independent machine learning models in an ensemble. The generation of datasets of gold nano-particle synthesis procedures, morphologies and size entities is proposed in [31]. The model is based a combination of rule-based (chemDataExtractor) and ML techniques (Recurrent Neural Network). A text-mined dataset of inorganic materials synthesis recipes is also proposed in [93]. The method is based on a bi-directional long-short term memory neural network with a conditional random field layer on top of it (BiLSTM-CRF) for NER. A combination of deep convolutional and recurrent neural networks for named entity recognition are also proposed in [94].

Other ML-Based techniques for extraction of materials science data from text include the use of BERT models. MatBERT is a notable BERT model for material NER in scientific text developed by pretraining of the base BERT model with around 61 million paragraphs containing materials information [172]. NER in the perovskite field is proposed in [197]. The NER pipeline is based on convolutional neural networks and MatBERT. Other examples of BERT-based models for NER in materials sciences include the analysis of optical materials [202] and the extraction of battery materials from scientific text [77]. These models are built on top of the pre-trained base BERT model.

The emergence of generative large language models has also revolutionized the extraction of materials properties from text hence accelerating the generation of structured materials data. The methods adopted for these tasks include simple prompting and in-context learning. Structured information extraction with pretrained LLMs (GPT-3 and Llama-2) is illustrated in [32]. The approach is tested on three tasks in materials chemistry: linking dopants and host materials, cataloging metal-organic frameworks, and general composition/phase/morphology/application information extraction.

Another related work is in the use of LLMs for text parsing in solid-state synthesis for the study of ternary chalcogenides [170]. The model is fine-tuned using an expert-curated dataset and a designed prompt. The in-context learning method is also proposed for assessing the ability of LLMs in processing materials data [25] and extracting materials data from research papers [133]. Lastly, LLMs have also been utilized for NER tasks in the construction of functional materials Knowledge Graph in multidisciplinary materials science [193].

3.4.2 Existing Databases of Materials Science Data

A couple of materials science databases have been developed. The databases contains data mined from text and experimental data collated together in a structured manner. The Open Databases Integration for Materials Design (OPTIMADE) ⁴ is a consortium that advocates to make materials databases interoperable by developing a specification for the common access to these databases [44]. It contains more than 25 databases of experimental materials science data collected from various contributors.

SpringerMaterials ⁵ also provides a platform to access various types of materials data covering multiple material classes, property types, and applications. The materials include Battery materials, Organic-inorganic perovskites, Polymer Thermodynamics among others. MatWeb is also a searchable database of materials properties data. The categories of the materials properties include data sheets of thermoplastic and thermoset polymers such as ABS, nylon, polycarbonate among others. 2D materials databases based on computational methods have also been proposed [63, 204]. The production methods include top-down, bottom-up and other computational approaches such as the density functional theory.

There also exists individual materials properties databases extracted from scientific text using various IE techniques. Examples of these databases include semi-conductor bandgaps [42], battery materials, optical materials [202], inorganic materials synthesis recipes [93], refractive indices and dielectric constants [201], transition temperatures of magnetic materials [29], thermo-electric materials [158] and gold nano-particle synthesis procedures, morphologies and size entities [31]. These databases provides platforms for accelerated discovery of materials properties in the materials science domain. This is by enabling data mining tasks that seek to uncover hidden relations between the materials properties.

3.4.3 Challenges and Thesis Positioning

Despite the great advancements in the adoption of IE methods in generating structured experimental data from text in the materials science domain, there are a couple of open challenges that need to be addressed to successfully generate structured QCL properties data from text. They are as outlined in the sections below.

- Despite the attempts to develop text mining methods for mining properties in the semiconductor domain such as semiconductor band gaps [41], the study of the semiconductor devices and hence their properties at the microscopic level (such as the growth sheet and material composition) has not been explored. There is therefore no existing structured collection of data that can be used in the analysis to understand the device properties and their implication on each other. This limits advancements in scientific discovery of novel semiconductor designs with optimized performance features. We therefore propose initial attempts in understanding these devices by focusing on the QCL and its design and working properties.

⁴<https://www.optimade.org/>

⁵<https://materials.springer.com/>

- The existing rule-based approaches for extracting materials data from text such as chem-DataExtractor are domain-specific and can therefore not be suitable for the extraction of properties from text for other specific domains such as the QCL domain. The domain-specific nature of the QCL domain properties poses difficulties to these methods in identifying the properties of interest in text.
- The existing ML methods need adaptation in order to be fitted to the task of QCL properties extraction from text. This process requires a lot of computing resources. The development of ML-based methods for extracting materials properties data from text also require quality datasets for training and evaluation. To the best of our knowledge, there is no existing dataset for training and evaluation of ML methods for extracting properties from text in the QCL domain.
- Evaluation metrics have been proposed for IE in materials science for both ML and rule-based approaches [41, 153]. There is need to adopt these definitions and propose evaluation metrics for IE in the context of the QCL domain properties. Examples of things to be clarified include the correctness of a property record in the context of the QCL domain properties.
- Despite the existence of databases of materials science data, there is no structured machine readable database for QCL properties. This limits data driven discovery of novel QCL designs. Generation of FAIR QCL data will enable research on novel QCL designs and their impact on performance features.

Therefore, there is need to develop efficient methodologies for extracting QCL properties from scientific texts. The rule-based methods require adaptation in order to capture the context and relationships in the QCL domain. Curation of quality training datasets will support the development and evaluation of ML workflows for QCL properties extraction from text.

The development of both rule-based and ML-based methods for QCL properties extraction from text will accelerate the generation of structured QCL properties data from scientific texts, hence contributing to the existing databases of materials science experimental data that can be used for analysis for various tasks such as predictive material modeling.

3.5 Conclusion

In this chapter, we have seen that information extraction methods play a crucial role in extracting entities of interest from a given piece of text. IE methods (both rule-based and ML-based) have been explored in generating structured experimental data from scientific texts in the materials science domain. The experimental data generated range from general materials to specific domain materials. This has accelerated the curation of materials science data repositories that have been utilized for various purposes such as the prediction of material properties among other data mining tasks.

The existing IE techniques can be explored to develop efficient methodologies for extracting structured QCL properties from text. This structured data can be utilized in developing standard

platforms for analyzing QCL properties data in an interoperable manner for insights regarding the development of novel QC design properties with target optimal working properties.

In the next chapter, we present our first contribution in which we propose an instruction dataset for training and evaluating LLMs for extracting QCL domain properties from scientific texts. The complete dataset generation process and its publication are presented.

A Dataset for Extraction of Quantum Cascade Laser Properties from Scientific Texts

4.1	Introduction	58
4.2	Motivation for the Dataset Generation	58
4.3	Materials and Methods	59
4.3.1	Sentence Sampling	59
4.3.2	Dataset Generation	59
4.3.3	Data Annotation	60
4.4	Value of the Dataset and Valorization	61
4.4.1	Value of the Dataset	61
4.4.2	Dataset Valorization	62
4.4.3	Limitations of the Dataset	62
4.5	Conclusion	62

“A scholar’s positive contribution is measured by the sum of the original data he contributes. Hypotheses come and go, but data remain.”

– Santiago Ramon y Cajal

4.1 Introduction

In this chapter, we address the problem of scarce quality datasets for training and evaluation of ML-based methods for QCL properties extraction from text. We propose an original instruction dataset for training and evaluation of LLMs for QCL property extraction from text [90]. In this work, we envision a structured dataset that can enable advancement of research in the development of efficient methodologies for extracting QCL properties from unstructured text.

We explore the NLP-based data augmentation techniques and domain expert knowledge in order to generate sample sentences which are annotated to provide a standard instruction dataset for QCL properties extraction from text. We detail the dataset generation process, the dataset description and provide details on the volarization. This dataset plays a significant role in the work we propose in Chapter 5.

4.2 Motivation for the Dataset Generation

Existence of structured data for QCL properties could accelerate the analysis of the relationship between these properties. As pointed out in Chapter 3, there is no existing structured data for QCL properties. The main open source of QCL data is in scientific texts. This requires efficient methodologies that can be used to extract this data from text to generate structured data for analysis.

One way is to adopt machine learning (ML) algorithms for developing such methods. QCL properties are domain-specific properties that require adaption of ML methods in order to efficiently extract them from text. This requires quality training data that can be used to train and evaluate these models. To the best of our knowledge, there is no existing dataset for this purpose. There are few publications on QCL devices. Some of the publishers have unfavorable text and data mining policies that do not allow the use of the articles for training and evaluation of AI models. This limits the ability to get enough training data in form of articles hence the need to generate/augment. It is on the basis of this requirement that we propose this work.

4.3 Materials and Methods

4.3.1 Sentence Sampling

Sample sentences for generating the instruction dataset were sourced from sample open access scientific articles that describes proposed QCL devices and their corresponding properties. Review articles or those that describe the applications of these devices are not considered for quality control purposes. QCL experts are also involved at this phase for quality control. We also restricted the sampling to the articles whose publisher permit text and data mining for research purposes. The articles were sourced from two publishers (American Institute of Physics and the Institute of Physics publishers) in line with the text and data mining policies.

A total of 36 original sentences are extracted from 12 articles documenting original QCL devices and their properties [139, 95, 189, 67, 180, 182, 112, 91, 126, 184, 174, 96]. The limited number of sentences is attributed to challenges of few number of articles and the unfavorable text and data mining policies by some publishers. The QCL properties considered are the working temperature, heterostructure materials, power, laser frequency and the laser design types.

4.3.2 Dataset Generation

In this section, we explore the use of NLP techniques for data augmentation in order to generate sentences from the sample sentences extracted from the scientific articles. Large Language Models have been greatly adopted for text data augmentation [46, 33]. Based on this adoption, we use a few shot training strategy to generate an augmented dataset of QCL properties from sample sentences extracted from the scientific articles with the GPT 3.5 instruct model.

Several transformations are applied ranging from rephrasing and generation of new sentences. For the case of rephrasing, we design the LLM prompt such that the meaning of original sentences is not altered and a given number of rephrased sentences is specified for every original sentence. For generating new sentences from original ones, the following transformation is applied: all the numerical and chemical values are replaced with other new values in order to generate new sentences with the same meaning but varied values. Listing 4.1 shows the prompt template for paraphrasing and Listing 4.2 shows the prompt template for generating new sentences based on the value and chemical transformations of the original sentences.

Listing 4.1: Prompt Template for Sentence Paraphrasing

```
{Prompt} {Including 5 examples for each sentences and without  
changing the meaning, please paraphrase the following sentence:}  
Sentence {Sample original Sentence}
```

Listing 4.2: Prompt Template for Sentence Generation

```
Prompt {Please generate 12 sample sentences (Insert different  
numerical/chemical values for the ones mentioned in the text for  
each generated sentence) for the following sentence:}  
Sentence {Sample original Sentence}
```

Once the sentences are generated, we add two columns: Instruction column and the Output column. The instruction column consists of an instruction to the LLM for extracting a particular QCL property while the output column shows the expert annotated column showing the label for the expected output based on the user instruction. Data post-processing steps include screening for the quality of the generated sentences and removing any poor/redundant sentences. The final dataset constitutes a CSV file that consists of 3 columns: Instruction, Input Sentence and the Output with 1300 sample sentences. Figure 4.1 summarizes the dataset generation process. The resources (LLM prompts and original sentences) are publicly shared¹.

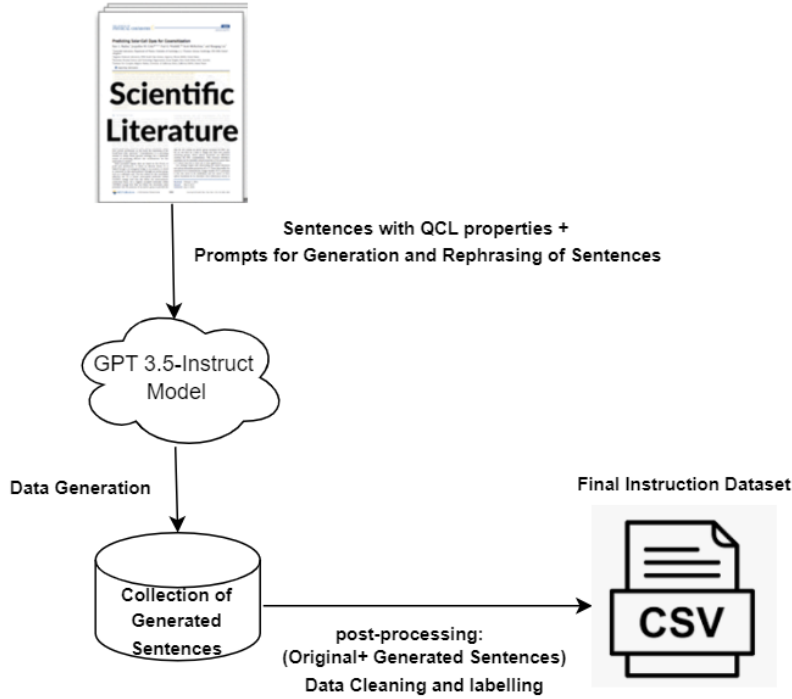


Figure 4.1: Dataset Generation

4.3.3 Data Annotation

The dataset is annotated with the help of QCL experts. The instruction column contains an instruction to the LLM to extract a specified QCL property. The Input column contains the input text from which the QCL property is to be extracted. The last column, the Output column contains an expert assigned label of the extracted property based on the user instruction in the Instruction column and the text input in the Input column. We give an example of an instruction, an input sentence and the corresponding output below.

Example 4.3.1 Consider the following input text from the user: *“At the lasing frequency of 1.5 THz, 30 mW of peak optical power was measured at 7 K, and approximately 8 mW could still be detected at 170 K.”*

¹<https://gite.lirmm.fr/dkerre/qpInstruct>

An instruction to the LLM to extract the lasing frequency of the QCL laser device is given as follows: *From the following sentence, please extract the lasing frequency of the Quantum Cascade Semiconductor laser device. Print none if the value does not exist in the input text.*

The output for this case is therefore 1.5 THz. This is the case with all the other properties. It is important to note that in example 4.3.1, several temperatures exist in the text and it may not be straightforward for an LLM model to determine the correct working temperature. This creates the need to adapt the model to this task by using more examples of this scenario. Table 4.1 shows the summary statistics of the instruction dataset generated.

Table 4.1: Summary Statistics for the Instruction Dataset

QCL Property	Number of Sentences
Working Temperature	386
Lasing Frequency	248
Optical Power	293
Heterostructure Materials	193
Laser Design Type	180
Total	1300

4.4 Value of the Dataset and Valorization

4.4.1 Value of the Dataset

The proposed dataset could accelerate research on QCL device fabrication and research materials science in general. The resources for generating this dataset can also be extended to other domains such as the biological domain. The envisioned benefits of the dataset are as outlined below.

- It provides a background in understanding the relationship between the various QCL properties. This is by enabling training of ML models to generate structured data from text that can be used for analysis. An example of such an analysis could be in the prediction of the laser performance properties based on the design features. The structure and the generation process of the dataset can be used to generate datasets in other domains. Examples may include generating instruction datasets for training ML models for extracting specific properties of interest in the biological domain that can be analyzed to understand relations between various biological entities.
- It provides a benchmark dataset for evaluating large language models' performance for the task of information extraction of properties in the Quantum Cascade Laser domain. This enables advancement in developing efficient LLM-based techniques for QCL property extraction from text. The dataset can also be modified for other tasks such as named entity recognition (NER) for QCL properties.
- The dataset provides a wide range of diverse sentences that capture the Quantum Cascade Laser properties in various ways hence enabling models to learn the various ways

of expressing the properties. This improves the models performance in identifying the domain-specific properties captured in diverse contexts.

- The dataset serves as central resource for researchers in the field of Information Extraction with an interest in developing efficient methodologies for extracting domain-specific properties in materials science domain and even other domains. It can be extended by further augmentation or with additional Quantum Cascade Laser properties to provide a diverse dataset for properties data extraction from text. The format of the dataset can also be adopted in developing information extraction datasets for other properties in materials sciences such as the Physics domain.

4.4.2 Dataset Valorization

The dataset is publicly distributed through the Recherche Data Gouv platform². Recherche Data Gouv is an ecosystem for sharing and opening research data. This platform contains data repositories that enables publication of peer reviewed data in line with the FAIR guidelines. Our dataset is availed to the public for reuse under the CC-BY license and can be accessed through a permanent persistent identifier ³. The data has also been valorized via a description in a published peer reviewed data paper [90].

4.4.3 Limitations of the Dataset

The instruction dataset proposed in this chapter has some limitations. It does not encompass all the QCL properties such as layer sequence and doping characteristics. For the layer sequence, it is possible to extract the materials in the layer sequence but not the details of the sequence of the stack of materials. There is also need to capture scenarios where more than one temperature is given for different values of other properties such as the power. It also has a small context window as it captures properties at sentence level. The short sentence descriptions may also not fully represent the real nature of descriptions captured in scientific articles documenting QCL working and design properties.

4.5 Conclusion

In this chapter, we proposed an instruction dataset for extracting QCL properties from scientific texts. The dataset acts as a central resource for adapting LLMs to the domain-specific task of QCL properties extraction from text. It allows models to learn the complex relationships within the domain specific data hence enabling them to efficiently identify the QCL properties mentioned in text in various contexts.

The dataset is however limited to the following QCL properties: the working temperature, power, frequency, design type and the heterostructure materials. It was challenging to generate

²<https://recherche.data.gouv.fr>

³<https://doi.org/10.57745/U3U7XR>

more sentences due to the high cost of using GPT and the required labeling and post-processing efforts. The quality of the generated sentences may not fully represent the nature of descriptions given in scientific articles detailing the QCL properties. We argue that the data is generated with examples from real articles (not pure text generation), involving paraphrasing and reversing values. This implies that the generated sentences may to some extent reflect the variability exhibited in real QCL text. Researchers can use this dataset to augment and generate larger instruction datasets for QCL properties extraction from text. The dataset format can also be adopted for the generation of other datasets in the semiconductor domain or materials science in general. Lastly, the dataset generation process and the dataset format can also be adopted to generate instruction datasets for other domains. An example may be the generation of an instruction dataset for extraction biological entities from text.

In the next chapter , we present our second contribution where we propose two methods for extracting QCL properties from scientific texts. These methods are based on ML and rule-based techniques.

Extraction of Quantum Cascade Laser Properties From Scientific Texts

5.1	Introduction	66
5.2	Rule Based Approach	66
5.2.1	Introduction	66
5.2.2	Methodology	67
5.2.3	Experiments and Results	72
5.3	Machine Learning Approach	75
5.3.1	Introduction	75
5.3.2	Methodology	75
5.3.3	Experiments and Results	79
5.4	Conclusion	82

“The important thing in science is not so much to obtain new facts as to discover new ways of thinking about them”

– William Lawrence Bragg (1890 – 1971)

5.1 Introduction

In this chapter, we address the problem of QCL property extraction from scientific texts. As explained in Chapter 3, we focus on open IE so as to generate structured machine readable QCL properties data that can be used for a wide range of tasks. This contributes to the existing databases of materials properties datasets. We propose two approaches for mining QCL properties from scientific texts. One approach is rule-based and the other one is based on ML techniques. With these methods, we envision efficient methodologies for generating structured QCL properties data from scientific texts for designing Knowledge Representation platforms for structured exploration of QCL properties data. We implement the approaches, define the evaluation metrics and test them on scientific texts data containing QCL properties.

5.2 Rule Based Approach

5.2.1 Introduction

The proposed rule-based approach entails the initial attempts in extracting and manipulating QCL properties data as described in [88]. In this method, we explore the possibility of extracting QCL properties data from scientific texts in situations where there is limited or no training data for ML models. This approach is based on an extension/adaptation of the chemDataExtractor pipeline, a state of the art rule-based pipeline used for mining properties in materials chemistry domain [165].

The chemDataExtractor pipeline has several ways of extracting materials properties which includes: (i) Rule-based approach (using grammar logic in the form of regular expressions), (ii) automatic parsing and (iii) the modified snowball algorithm. We adopt the rule-based approach in which we define new syntactic rules (inform of regular expressions) that extend the pipeline to enable it to mine properties in the QCL domain from text. We present some preliminary definitions of concepts in the section that follows.

Definitions of Preliminary Concepts

We begin by a giving a summary definitions of based on the user model concept proposed in [29] and as detailed in Chapter 3. The user model concept entails a collection of defined property models that are used for mining different categories of information from text. A property model specifies two items: the information to be extracted and the rules to extract them. This information is captured in a property parser which contains the keywords to be used in identifying the information to be extracted and the fields to be populated. A property model

can fall under the following categories based on the information it intends to capture: Quantity model and a Compound model. The definitions of the Property Model, Parser, Quantity Model and Compound Model are given below.

Definition 5.2.1 (Property Model). Property Model: *A Property Model is an IE model that specifies the information to be retrieved and the grammar rules to be used in identifying the patterns in text that contains this information.* [29].

Examples of information could be value, units and chemical names/ compound formulae.

Definition 5.2.2 (Quantity Model). A Quantity Model is a property model that defines physical quantities such as time, electric charge, volume, density etc. [29].

Quantity model captures materials properties that are represented by measurable numerical values. An example of a quantity model is the Temperature Model in chemDataExtractor that defines the units/unit conversions for identifying temperature units in text.

Definition 5.2.3 (Compound Model). A Compound Model is a property model that defines chemical names and their corresponding chemical name labels and roles [29].

The compound model is used to retrieve information on chemical formulae of the various materials combinations together with their respective ratio of combination.

Definition 5.2.4 (Parser). Refers to the formal representation of the grammar logic used for identifying the particular properties of interest that needs to be extracted from text [29].

The parser captures information such as the keywords used for identifying the properties and the sequence of phrases to be used in capturing the properties. The grammar logic is expressed in the form of regular expressions to capture the patterns of interest in text.

The previous definition forms a basis on which the concepts that we implement in the methodology of the rule-based approach for extracting QCL properties from text. This is detailed in the subsection that follows.

5.2.2 Methodology

The QCL properties information extraction pipeline is developed in a series of steps. These are detailed below.

Document Acquisition and Preprocessing

This is the first stage in the information extraction pipeline. It involves acquiring articles documenting the various QCL devices with their corresponding properties. The documents can be either be retrieved in the HTML or PDF formats. The retrieval can be automated i.e by web scrapping or it can be done manually over few selected papers depending on the requirements. It

is important to note that in our case we use manually downloaded articles in the HTML format as they requires less processing efforts as compared to those in the PDF format.

The articles in HTML format have a hierarchical structure with several semantic markup tags, for instance, the `<head>` tag. The `chemDataExtractor` toolkit has the processing capability of using these tags to identify key information in the articles for instance the title, abstract, paragraphs and individual sentences. The reader package in the `ChemDataExtractor` preprocessing unit is used to convert the documents' content into plain text and stored in a `Document` object of the toolkit. The plain text is utilized for further processing tasks.

Natural Language Processing

In this step, we perform state of the art NLP preprocessing steps in order to prepare the plain text data for the information extraction task. These are implemented using the data preprocessing unit of the `chemDataExtractor` toolkit. The preprocessing steps carried out includes: Sentence splitting, Tokenization and Part-of-Speech-Tagging.

QCL Properties Extraction

In this phase, we define property models and the grammar logic (in form of property parsers) to be used in extracting the QCL properties of interest. We define the property models based on the type of information to be extracted. The property models for the QCL working temperature, power, frequency and the layer sequence are defined as quantity models as the information to be extracted from these properties consists of quantitative values. The QCL heterostructure on the other hand is defined as a compound model as it retrieves combinations of material names.

The `chemDataExtractor` toolkit has predefined models for power and temperature which handles the units and their conversions. The default power model does not contain a conversion module for power unit in milliwatts (mW). We extend the power model in order to capture this as the QCL optical power is in most cases expressed in milliwatts. For the working temperature, all the unit conversions are covered by the default Temperature Model and it is adopted as is. For the QCL frequency and barrier thickness, we define the Frequency Model and the Barrier Thickness Model from scratch to handle their unit conversions.

The next step after handling the unit representation is to define property models that will utilize the property parsers in order to extract the properties of interest. The property models specify the fields to be populated during property extraction. These include the grammar logic in form of a property parser and the fields to be populated based on the grammar logic. For the property models that retrieve quantity values, we specify two fields to be populated i.e a value and a unit and the parser (grammar logic) to be used. These are extracted based on the grammar logic captured in the property parser. For the QCL properties, we define five new property models to be used in extracting these properties.

The power, frequency, working temperature and barrier thickness property models fall under the quantity models and inherits the their respective parent models for unit handing. This

implies that the working temperature model inherits the Temperature Model, optical power model inherits the power model, lasing frequency model inherits the Frequency Model and the barrier thickness property model inherits the Barrier Thickness Model. For the heterostructure, it inherits the Base model which is the superclass of all property models. For heterostructure, there is no need for unit handling as the information to be extracted does not contain units. The property model for the QCL heterostructure therefore contains the parser and the heterostructure materials field.

The final stage is the definition of property parsers that captures the grammar logic used in identifying specific QCL properties in text. These are utilized by the respective property models in order to populate the required fields. This implies that a frequency property parser is utilized by the lasing frequency property model in order to populate the fields for frequency values and units from the text. The same applies to the other property models. Extraction of QCL properties in the rule-based approach entails extraction of a relation that is either in the form of (i) a specifier expression/keyword and a chemical name or (ii) a specifier expression, a value and a unit. We give examples of QCL property relationships in text. This forms the basis on which the property parsers (grammar logic) for mining QCL properties from text are defined. We present the examples below.

Example 5.2.1 Consider the following description of a QCL device from a sample scientific article: “*A dual lasing channel Terahertz Quantum Cascade laser (THz QCL) based on GaAs/Al_{0.17}Ga_{0.83}As material system is demonstrated. The device shows the lowest reported threshold current density (550A/cm² at 50K) of GaAs/Al_xGa_{1-x}As material system based scattering-assisted (SA) structures and operates up to a maximum lasing temperature of 144K.*” (text extracted from [187]).

From the sample text description in example 5.2.1, the grammatical format of the working temperature can be expressed in the following form: **prefix + value + unit**, where prefix is the keyword for identifying the property i.e “maximum lasing temperature of”, value i.e 144 and the unit K. From example 5.2.1, it can be noted that several temperatures are mentioned in text. However, only the working temperature of interest. This poses challenges in identifying the particular temperature of interest. The other difficulty is with the identification of the material in the layer sequence captured in text.

Example 5.2.2 Consider the following description of a QCL device from a sample scientific article: “*A 2.1 THz quantum cascade laser (QCL) based on a scattering assisted injection and resonant-phonon depopulation design scheme is demonstrated. The QCL is based on a four-well period implemented in the GaAs/Al_{0.15}Ga_{0.85}As material system.*” (text extracted from [92]).

In example 5.2.2, the grammar logic for the QCL heterostructure can be formally expressed as follows: **prefix + materials formulae** where the prefix is the keyword/phrase “implemented in the” and the materials formulae “GaAs/Al_{0.15}Ga_{0.85}As”.

The chemDataExtractor has the default AutoSentence parser that utilizes multiple specialized grammar rules designed for the extraction of more specific types of chemical information. The Autosentence parser performs well on the chemical domain but fails to generalize well on other domains, for instance the physics domain. Some of the properties in the QCL domain such as the barrier thickness has special characters (for instance commas and forward slashes) that

cannot be completely captured by the Autosentence parser. Based on experimental analysis, it also fails in cases where the units are put in bracket after the value. We therefore define five parsers for every QCL property that are used with the respective predefined property models.

The grammar logic in property parsers is defined by combining keyword(s) for QCL properties of interest with parser elements predefined in ChemDataExtractor in form of tokens. This forms an expression that will be used to extract a QCL property when encountered in text. The predefined parser elements in chemDataExtractor are shown in Table 5.1.

Table 5.1: The Parser Elements (elements extracted from [165]).

Elements	Description	Elements	Description
R(Regex)	Match text with regular expression	T (Tag)	Match tags
W(Word)	Match case-sensitive token text	I(IWord)	Match case-insensitive token text
Any	Match any single token	H(Hide)	Ignore the matched tokens
Not	Match only if not followed by some text	FollowedBy	Match only if followed by some text
ZeroOrMore	Match zero or more of the expressions	OneOrMore	Match one or more of the expressions
Optional	Match if it exists	SkipTo	Skips to the next occurrence of text

For the keywords, we consider several keywords used in relation with the QCL properties considering the reporting style in the QCL domain scientific literature. These keywords usually occur with a possible combinations of expressions that can be used to uniquely identify the QCL properties in text. We summarise the possible keywords and units used for each QCL property in Table 5.2.

Table 5.2: Quantum Cascade Laser Properties, Units and the Keywords.

Target Property	Keyword/Sentence	Unit
Working Temperature	heat-sink, Tmax, Maximum Temperature, Working Temperature	K
Optical Power	Optical power, Output Power, Peak Power	W
Hetero-structure (Material)	Growth, Grown in, Wafer, implemented in the, MBE, Laser-structure	N/A
Frequency	Laser Frequency, Lasing at, output Frequency	THz
Barrier Thickness	Layer thicknesses	Å

Based on the keywords and parser elements, we present rule definitions for extracting the working temperature and the heterostructure material in the following examples:

Example 5.2.3 Consider the following text description capturing QCL properties: “A dual lasing channel Terahertz Quantum Cascade laser (THz QCL) based on GaAs/Al_{0.17}Ga_{0.83}As material system is demonstrated. The device shows the lowest reported threshold current density (550A/cm² at 50K) of GaAs/Al_xGa_{1-x}As material system based scattering-assisted (SA) structures and operates up to a maximum lasing temperature of 144K.” (text extracted from [187]).

Given the text in example 5.2.3, the rule for extracting the working temperature can be defined in the form of prefix + value + unit. We define regular expressions to match this relationship using the keyword and parser elements below.

```
prefix = (I('maximum') + I('lasing') + I('temperature') + I('of')),
units = (Optional(W('°')) + Optional(R('^?[CFK]\.?$', 'units'))),
value = R('^d+(\.\d+)?$', 'value'), and therefore
working temperature = prefix + value + units.
```

The parsers for the laser frequency, power and barrier thickness are defined in the same way as they both consist of values and units.

Example 5.2.4 Consider the following text description capturing the QCL heterostructure materials : “A 2.1 THz quantum cascade laser (QCL) based on a scattering assisted injection and resonant-phonon depopulation design scheme is demonstrated. The QCL is based on a four-well period implemented in the GaAs/Al0.15Ga0.85As material system.” (text extracted from [92]).

For the case of the QCL heterostructure, only two fields are defined i.e the prefix (to capture the keywords for the property) and the materials formulae field. Using the keywords and parser elements, the rule can therefore be expressed as: prefix + material. For the case in example 5.2.4, the relationship can be defined as illustrated below.

```
prefix = (I('implemented')+I('in')),
material
=(R('G[aAs]')+W('/')+R('A[l]')+R('^d+(\.d+)?$')+R('G[a]')+R('^d+(\.d+)?$')
+R('A[s]')), and therefore
heterostructure material = prefix + material .
```

Formally, the process of QCL property extraction from text using the property models and parsers is as summarized in Algorithm 1.

Algorithm 1: Mining the QCL Properties from Text

Input: D-Union{Ei}, input document object.

Output: : R-Union{Ri}

```
1  $S \leftarrow \text{Union}\{S_i\}$  /* Prefix for the various keywords describing qcl properties
   */
2  $M \leftarrow \text{PropertyModel}$  /* specifies the fields to be captured for a particular
   property. */
3  $P \leftarrow \text{Parser}$  /* grammar logic. */
4 Set  $D.\text{model} \leftarrow M$  and  $\text{parser} \leftarrow P$  /* Defining the parser and property model.
   */
5 for each document element  $E_i$  in  $D$  do
6   scan(Ts) if  $T_i \leftarrow S_i \subseteq S$  then
7     match  $R_i$ . /*  $R_i$  is an individual record in  $R$  (set of records
       retrieved). */
8   else
9     skip to the next  $E_i$  and repeat step 6 and 7.
10  end if
11 end for
12 repeat line 5-11 until all  $R_i$ s are merged.
13 return record  $R$  with the matched property relationship(s).
```

Consider a document object D containing several QCL properties of interest. The document object D consists of several document elements $E_1 \dots E_n$ which consists of titles, paragraphs and sentences. We defined prefix S to capture the possible expressions S_i for a particular property. A

particular property may be expressed by several combinations of expressions. These individual combinations are grouped together to form a set S of possible expressions. The expressions contains keywords/a combination of the keywords used in context of a particular QCL property of interest as illustrated in Table 5.2. R is a set of the property records which may consist of several of the individual record elements R_i . R_i captures the extracted property information which includes values, units and materials. M and P consists of the defined property models and the parsers respectively which work together in order to extract the QCL properties based on the defined relationship rules.

Before initiating the the extraction process, all the parsers have to be set to their respective property models. Once this is done, subsequent steps involves iteratively searching for the matching token expressions for the prefix, property values, units and names which are merged into complete records. Figure 5.1 shows a sample heterostructure material tree structure(upper) and extracted record output(lower) for the following text description: *We present two different terahertz quantum cascade laser designs based on GaAs/Al_{0.3}Ga_{0.7}As heterostructures that feature a depopulation mechanism of two longitudinal-optical phonon scattering events* (text extracted from [190]).

```
b'<material>GaAs / Al0.15Ga0.85As</material>'
[{'QclMaterialDesign': {'heterostructure': 'GaAs / Al0.3Ga0.7As'}}]
```

Figure 5.1: Sample Extracted Heterostructure Material Record (example data extracted from [190]).

5.2.3 Experiments and Results

In this section, we present an experimental evaluation of the rule-based approach for mining QCL properties from text. We also provide the implementation of the rule-based approach and define the metrics for evaluation.

Source Code

The source code for the rule-based approach is available at our GitHub repository: <https://github.com/DeperiasKerre/qclProperties>

Evaluation Metrics

We adopt the expert validation approach for the evaluation of the text mining pipeline. This entails comparison of the model's output with the expert extracted and annotated records. For the evaluation metrics, we use the Precision and the Recall scores. Definitions of these metrics for the task IE are presented in [41, 153] and detailed in Chapter 3. We provide definitions for the terms 'correct' and 'incorrect' for this context by the definitions below.

Definition 5.2.5 (Correct). Implies that the property record extracted can be validated by a human expert when reading the corresponding sentence containing the property. It should also match with the the expert annotated property record. For property records involving units, they are only considered complete and correct if the respective units are also included.

Definition 5.2.6 (Incorrect). Suggests that the value extracted does not correspond to the correct/expected value as compared to the expert annotated property record.

Test Dataset

The test dataset comprises of text data extracted from 43 open access articles capturing QCL devices and their respective properties. The data is annotated with the assistance of an expert which involves identifying all the QCL property records contained in the text. A label (QCL properties records contained in text) is determined by an expert. This is done for each text description hence having several pairs of text with a corresponding label of QCL properties contained in it. The main aim is to assess whether the output of the information extraction model (the properties extracted from a sample test text) will match the expected properties as indicated in the corresponding label for that piece of text. This consists a total of 192 QCL property records that are manually identified. The distribution of records is as follows: Optical power (33), Working Temperature (32), Lasing Frequency (44), Hetero-structure (66) and Barrier thickness (15 records). Some of the QCL properties have multiple mentions in text.

Baseline

We compare the performance of the proposed rules with the default rule-basing capabilities (the AutoSentence Parser) of the chemDataExtractor toolkit. The performance evaluation of the pipeline on the sequence layer thickness is done separately as the AutoSentence parser is not completely able to extract this property due to the special characters it contains. We compare the outputs of the pipeline with the expert annotated records in order to determine the metrics.

Results and Discussions

We present a comparison of performance results between the defined rules and those of the default Autosentence parser. The analysis is done per QCL property as the various properties have varying levels of difficulty in the extraction process. The results are summarized in Table 5.3 (Default Autosentence parser in chemDataExtractor) and Table 5.4 (the defined QCL property parsers). The comparison does not include the results for the barrier thickness since the Autosentence parser is not able to extract this property. For this property, we present the results for the defined parser at the end.

For the case of optical power, the default autosentence parser attains a precision of 68.75% and a recall of 33.33%. This implies that around 68.75% of the extracted power records correspond to

Table 5.3: Performance Evaluation Metrics for the Default Autosentence Parser.

Property	Records Extracted	TP	FP	FN	Precision(%)	Recall(%)
Optical Power	16	11	5	22	68.75	33.33
Working Temperature	128	31	97	1	24.22	91.17
Lasing Frequency	16	10	6	34	62.50	22.73
Hetero-structure	140	50	90	16	35.71	75.76
Total	300	102	198	73	47.80	55.75

Table 5.4: Performance Evaluation Metrics for the Defined Parsers.

Property	Records Extracted	TP	FP	FN	Precision(%)	Recall(%)
Optical Power	25	24	3	9	96.00	72.73
Working Temperature	32	25	7	7	78.13	78.13
Lasing Frequency	37	29	8	15	78.34	65.91
Hetero-structure	60	59	1	7	98.33	89.39
Total	154	137	19	38	87.70	76.55

the annotated power records. The lower recall of 33.33% is attributed to a large number of power records being not extracted due to the varying expression styles for this property in text which presents generalization challenges for the Autosentence parser. The performance is however not competitive due to its inability to adapt to the context for the QCL power. The defined optical power parser on the other hand extracts power records that corresponds to 96% of the annotated power records. The parser also achieves a relatively higher recall of 72.73% hence indicating that a smaller number of power properties not extracted. The defined optical power parser takes into consideration the various contexts in which the power values are expressed hence achieving a better performance than the default autosentence parser.

The default autosentence parser has a higher false positive rate (hence a lower precision of 24.22%) for the working temperature and a higher recall of 91.17%. The defined working temperature parser on the other hand extracts the least number of incorrect records and misses the least number of correct records hence attaining a higher precision and recall of 78.13% and 78.13% respectively. This is attributed to the fact that as more temperatures are mentioned in literature, the more difficult it is to retrieve the temperature of interest.

Despite the increase in performance with the defined frequency parser for the lasing frequency, this property poses challenges for the extraction process for both the default autosentence parser and the defined frequency parser. This is because of the many possibilities of expressing this property in text hence making it difficult for the rules to completely capture all the possible scenarios.

For the laser heterostructure materials, the default autosentence parser achieves a higher false positive rate. This results to a precision of 35.71% and a recall of 75.76%. The higher false positive rate is contributed to by records having compound-like names but are not necessarily QCL heterostructure materials. The defined parser on the other hand achieves a precision of 98.33% and recall of 89.39% which are attributed to the more specialized rule combination for the QCL heterostructure materials.

Lastly for the QCL barrier thickness, we present the performance of the defined barrier thickness parser. The records not extracted records consists of a combination of values with units in different positions and not after the reading. An example of such scenario is when a unit is mentioned before the value for instance: *"The barrier thickness of the QCL device (in*

nm) is 3/4.5/10/6.7/8.5". This entails an initial attempt in extracting properties with special characters.

In summary, there is an improved average performance in the QCL property extraction from text with the defined property parsers. The defined parsers take into consideration the textual relationships for identifying the specific properties. However, it should be noted that the rules cannot be deemed complete for capturing all the possible ways of expressing the QCL properties in text. This is indicated by the cases in which the rules fail to extract the properties of interest when the patterns in text don't match with those specified in the rules. This also creates the need to update the rules whenever a new expression is encountered. The rules are helpful in cases where there is no or its not possible to produce quality data for training machine learning models for extracting QCL properties from text.

5.3 Machine Learning Approach

5.3.1 Introduction

In this section, we present a method for extracting QCL properties from text using a machine learning based approach. This method addresses the limitations posed by the rule-based approach as discussed in section 5.2. We explore the use of large language models in developing the proposed method using an instruction dataset that we present in Chapter 4.

5.3.2 Methodology

In this method, we explore the use of large language models (LLMs) in developing an efficient method for mining QCL properties from text. As presented in the related works in Chapter 3, LLMs have recently posted remarkable state of the art performance in extraction of materials science properties from scientific literature [25, 133]. LLMs are trained on a lot of general knowledge data and often perform poorly when utilized for domain specific tasks. This implies that they cannot be efficiently used for mining QCL properties from text without adaptation.

We hypothesize that exposing the LLMs to updated labeled data of QCL properties improves the models' ability in recognizing the domain specific properties and hence improving the general performance in extracting these properties from text. In-context learning can be adopted to improve the learning ability of the model. But the normal static prompt engineering task entails specification of few shot examples in every prompt. This limits the efficiency in generating quality prompts with the best examples for the model input context. Therefore, we explore Instruction tuning with RAG approach for mining QCL properties from text [52, 104].

In this method, we adopt a hybrid technique of few short learning where the training examples are automatically generated from an instruction dataset via the RAG process as opposed to the normal prompting where static examples have to be specified in every prompt [43, 192]. The dynamic prompting strategy improves the efficiency of quality prompt generation. The training examples consists of the top-K best examples retrieved from the training dataset. This approach

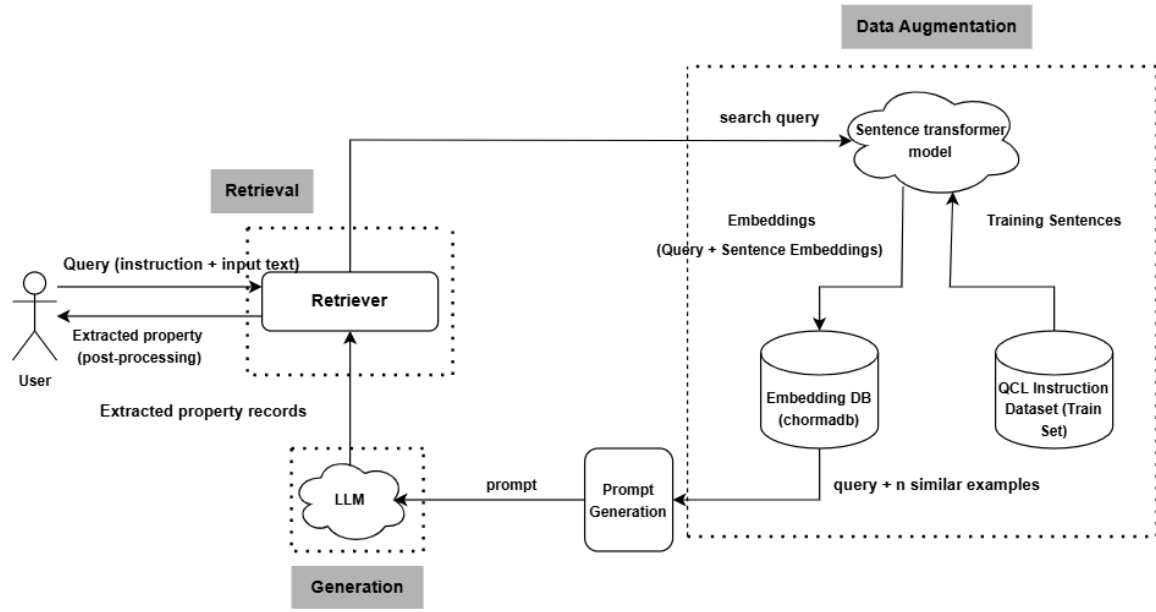


Figure 5.2: The RAG Pipeline for Mining QCL Properties

is computationally less expensive as compared to LLM training/fine-tuning and requires less effort in specifying the training examples for each prompt. It therefore combines the benefits of RAG (less computational resources) and Instruction tuning (improved performance with quality instruction dataset). The approach development steps are given below.

Dataset Curation

One of the main challenges in developing and evaluating ML algorithms in mining QCL properties from text is the lack of quality training datasets. We therefore adopt the dataset presented in Chapter 4 for this workflow. The dataset captures the following QCL properties: Power, Working Temperature, Frequency, Design Type and the QCL Heterostructure materials.

Information Extraction

The proposed method for QCL properties extraction from text based on RAG has three steps: Retrieval, Data Augmentation and Data Generation. These steps are summarized in Figure 5.2 and we describe them in the following sections.

In the **Retrieval phase**, a query in form of an input sentence is submitted to the retriever module by a user. The query contains the QCL property of interest and an instruction to the large language model for extracting the QCL property. The retriever then forwards the query to the data augmentation module which will fetch responses with the highest similarity to the query.

The **Data Augmentation** module entails a collection document embedding consisting of

training examples for the model. This is used to provide the context for the model during the data extraction process. In this case, the context is provided by the instruction dataset described subsection 5.3.2. We use 80% of the dataset for this purpose. The data augmentation module enables the retrieval of the most relevant examples to be used in training the model during the QCL property extraction phase. This is achieved by computing the similarity score between the query embeddings and the embeddings of the training examples stored in a vector database. This implies that for a particular user query, a given number K of the best examples will be retrieved from the vector database based on the similarity score.

The individual sentence embeddings for the query and the training samples are computed using a cased pre-trained sentence transformer model [162]. In our case, we adopt the all-mpnet-base-v2³ version of the sentence transformer model in our approach. The similarity scores between the query and the training embedding are determined using the cosine similarity metric [18]. Consider two sentence vector embeddings A and B . The cosine similarity metric is determined as shown below.

$$\text{CosineSimilarity}(A, B) = \frac{A \cdot B}{\|A\| \|B\|} \quad (5.1)$$

where $A \cdot B$ is the dot product of the two vector embeddings and $\|A\|, \|B\|$ the length (magnitude) of the vector embeddings. The values of this metric range between -1 and 1. If the value of the cosine score is close to 1, it means that the angle value between the two vectors is small hence the sentences are similar while if its close to 0, the angle value is larger implying that the sentence vectors are dissimilar. This score is preferred as it is independent of the magnitude and it is relatively quick and efficient to compute.

The retrieved training examples are combined with the query and used to generate a prompt that is passed to the LLM in order to extract the QCL properties. In order to achieve this, we define a prompt template that captures the user query and the training examples which are passed to the LLM during the generation phase. Listing 5.1 shows the defined prompt template and Figure 5.3 shows an example of a generated prompt.

Listing 5.1: RAG Prompt Template

```
Problem Definition {Extraction of Quantum Cascade Laser Properties
from text entails extracting properties from a given text
description. This should be done without providing any other
additional information or explanations. The output format should
correspond to the one in the example sentences. Example sentences
containing an instruction, input text and the extracted properties
are given below:}
```

```
Example Sentences {Example sentences containing qcl properties, LLM
instruction and the corresponding extracted properties.}
```

```
Instruction {An instruction to the LLM for extracting QCL property
and the input text containing the properties.}
```

³<https://huggingface.co/sentence-transformers/all-mpnet-base-v2>

```
[INST]
Problem Definition: Extraction of quantum cascade laser properties from text entails extracting properties from a given text description. This should be done without providing any other additional information or explanations. The output format should correspond to the one in the example sentences. Example sentences containing an instruction, input text and the extracted properties are given below:

Example Sentences: ['From the following sentence, please extract the design type of the Quantum Cascade Semiconductor laser device. Print none if the value does not exist in the input text. Input Text: Given the above-normal operating temperatures and duty cycles, we assert that utilizing direct LO-phonon-based depopulation proves to be a robust method in the realization of long-wavelength THz quantum cascade lasers.\t\t\n\nExtracted Properties:\n\nLO-phonon', 'From the following sentence, please extract the design type of the Quantum Cascade Semiconductor laser device. Print none if the value does not exist in the input text. Input Text: Considering the relatively high temperatures at which they operate and the frequency of their cycles, we propose that leveraging direct LO-phonon-based depopulation is a dependable means of obtaining quantum cascade lasers that emit long-wavelength THz frequencies.\t\t\n\nExtracted Properties:\n\nLO-phonon', 'From the following sentence, please extract the design type of the Quantum Cascade Semiconductor laser device. Print none if the value does not exist in the input text. Input Text: Our experimental results show that by increasing the operating temperature to 100 degrees Celsius and doubling the duty cycle to 0.8, direct LO-phonon-based depopulation becomes an effective technique for generating quantum cascade lasers operating in the long-wavelength THz range.\n\nExtracted Properties:\n\nLO-phonon']

Instruction: Please extract the design type of the quantum cascade laser device. Please print none if the value is not in text and do not give any explanations. In the output, just include only the extracted property. Input Text: We report the development of a quantum cascade laser, at 1587.2 nm, corresponding to 3.44 THz or 14.2 meV photon energy. The GaAs/Al0.15Ga0.85As laser structure utilizes longitudinal-optical LO-phonon scattering for electron depopulation. Laser action is obtained in pulsed mode at temperatures up to 65 K, and at 50% duty cycle up to 29 K. Operating at 5 K in pulsed mode, the threshold current density is 840 A/cm2, and the peak power is approximately 2.5 mW. Based on the relatively high operating temperatures and duty cycles, we propose that direct LO-phonon-based depopulation is a robust method for achieving quantum cascade lasers at long-wavelength THz frequencies.

[/INST]
```

Figure 5.3: Sample Regenerated Prompt by GPT-4 Turbo Model

Lastly in the **Generation phase**, the generated prompt is passed to the LLM model for extracting the QCL properties based on the user input sentence and the retrieved examples. The whole extraction process is summarized in Algorithm 2.

Algorithm 2: The QCL Properties Extraction Steps

Input: Set of input sentences S , Prompt P , Instruction I , base model M , an embedded instruction dataset to provide the context C .

Output: QCL property record R .

- 1 Set the number of examples to be retrieved (K) to a specific value. /* Number of examples to be retrieved from the context documents based on the user query. */
 - 2 Enter an instruction I and an input sentence S .
 - 3 Set the query $Q = (Instruction(I) \cup InputSentences(S))$.
 - 4 Convert the queries to a vector embedding E_q .
 - 5 Pass query E_q to the retriever.
 - 6 Fetch K examples from C based on E_q .
 - 7 Set $E_q = (E_q \cup K \text{ examples})$.
 - 8 Set $Q = E_q$ (decoded to plain text).
 - 9 Update the prompt P with Q .
 - 10 Pass P to the base model M .
 - 11 **return** R
 - 12 **Repeat** steps 2-9 until all the property records of interest are extracted.
 - 13 Post-process R . /* Entails removal of incomplete records and any unnecessary responses. */
-

Based on Algorithm 2, we can formally frame the QCL property extraction from text in a conversation like manner as below.

Consider a set of input sentences S , where $S = \{S_1, S_2, S_3 \dots S_n\}$ and a prompt P that contains an instruction I and S . The context $C = \{C_1, C_2, C_3 \dots C_k\}$ is set to constitute the best k examples from the training set, where k depends on the number of desired training examples to be retrieved from the training embeddings. The prompt P is passed to the LLM model to generate the required QCL property R from the input text.

5.3.3 Experiments and Results

In this section we present experimental evaluation results for the machine learning based approach for mining QCL properties from text. We also provide the implementation of this approach.

Source Code

The source code for the machine learning based approach for QCL property extraction from text can be found at our GitHub repository: <https://github.com/DeperiasKerre/qKG>

Evaluation Metrics

For the evaluation metrics, we adopt the metrics defined in section 5.2.3 in order to evaluate the ML based approach for QCL properties extraction from text. We also examine the execution times of the models in extracting QCL properties from text.

Test Dataset

We evaluate the IE method on a test dataset obtained from the instruction dataset presented in section 5.3.2. This constitutes 10 % of the dataset containing a total of 130 sentences containing the various QCL properties and their annotated labels. The statistics of the test data are summarized in table 5.5.

Table 5.5: Test Data Statistics

Quantum Cascade Laser Property	Number of Sentences
Laser Optical Power	26
Working Temperature	34
Laser Design Type	27
Laser Frequency	29
Laser Heterostructure	14
Total	130

Baselines

We test the proposed RAG-based approach for the QCL properties extraction from text on the following large language models: GPT 4-Turbo and the Mistral-7b-Instruct model. We do not perform tests on GPT-4 due to the context window limitations. We compare the performance of these models on QCL property extracting from text with the proposed RAG-based instruction tuning approach and with the simple query approach (zero shot). We present the analysis per QCL property as the different properties have varying styles of presentation in text hence varying levels of difficulty during the extraction process.

Results and Discussions

We present the results for the ML based approach for QCL property extraction from text. The results are both for the individual QCL properties and the average performance of the models for all the QCL properties. The results are as follows: Table 5.6 for the laser working temperature, Table 5.7 laser power, Table 5.8 lasing frequency, Table 5.9 the laser design type, Table 5.10 the laser heterostructure materials and lastly Table 5.11 the average performance for all the QCL properties.

Table 5.6: Performance Metrics for the Laser Working Temperature.

Model	Precision	Recall
gpt 4-turbo (RAG + Instruction Tuning)	100%	100%
gpt 4-turbo (zero shot)	90.91%	96.77%
mistral-7b-instructv0.3 (RAG + Instruction Tuning)	85.30%	100%
mistral-7b-instructv0.3 (zero shot)	85.29%	100%

Table 5.7: Performance Metrics for Laser Output Power.

Model	Precision	Recall
gpt 4-turbo (RAG + Instruction Tuning)	100%	100%
gpt 4-turbo (zero shot)	100%	100%
mistral-7b-instructv0.3 (RAG + Instruction Tuning)	100 %	100%
mistral-7b-instructv0.3 (zero shot)	100%	96.15%

Table 5.8: Performance Metrics for the Laser Frequency.

Model	Precision	Recall
gpt 4-turbo (RAG + Instruction Tuning)	100%	100%
gpt 4-turbo (zero shot)	92.86%	96.30%
mistral-7b-instructv0.3 (RAG + Instruction Tuning)	96.43%	96.43%
mistral-7b-instructv0.3 (zero shot)	92.59%	92.59%

Generally, the models attain higher precision for the QCL power, working temperature and lasing frequency properties for both zero shot and the RAG-based instruction tuning approach. This is due to the fact these properties are in the general knowledge and not specific to a particular domain hence higher chances of them being captured in the models' knowledge. There is also an improvement in precision with the RAG-based approach for the QCL working temperature and lasing frequency (Tables 5.6 and 5.8). Temperature and frequency properties are characterized by having more than one value in text (for instance temperature based on the working mode) hence the models struggle to identify both of them in the case of simple queries hence leaving

Table 5.9: Performance Metrics for the Laser Design Type.

Model	Precision	Recall
gpt 4-turbo (RAG + Instruction Tuning)	100%	33.33%
gpt 4-turbo (zero shot)	92.31%	46.15%
mistral-7b-instructv0.3 (RAG + Instruction Tuning)	100%	92.60%
mistral-7b-instructv0.3 (zero shot)	33.33%	29.41%

Table 5.10: Performance Metrics for the Laser Heterostructure Materials.

Model	Precision	Recall
gpt 4-turbo (RAG + Instruction Tuning)	85.71%	100%
gpt 4-turbo (zero shot)	91.67%	84.62%
mistral-7b-instructv0.3 (RAG + Instruction Tuning)	85.71%	85.71%
mistral-7b-instructv0.3 (zero shot)	92.86%	100%

Table 5.11: Average Performance of the Models for all the Properties

Model	Precision	Recall
gpt 4-turbo (RAG + Instruction Tuning)	97.14%	86.67%
gpt 4-turbo (zero shot)	93.55%	83.59%
mistral-7b-instructv0.3 (RAG + Instruction Tuning)	93.49 %	92.01%
mistral-7b-instructv0.3 (zero shot)	80.82%	80.69%

them out during the extraction process. This is also the case when frequency values are presented in ranges. Provision of examples capturing such scenarios improves the model performance on these properties. This explains the improved precision with the RAG-based approach.

For the laser Design type, the models perform well in cases where the term ‘design type’ is explicitly expressed in the text description and in most instances leaving out this property when the this word is not mentioned. This is attributed to the fact this a domain specific context hence posing a high level of difficulty for the model to extract it. For all the models, there is an improvement in both precision and recall with the RAG-approach as the models’ ability to identify the properties is improved with access to sample training examples. The best performance is posted by the Mistral-Instruct model as shown in Table 5.9. GPT-4 Turbo excels at correctly identifying the design type names. However, there is a higher proportion of design types not extracted in cases where it is not explicitly mentioned in text hence a lower recall rate.

The QCL heterostructure materials property is also a domain specific concept. There is however no improvement with the exposure of the model to the training examples. This is attributed to the fact that despite it being a domain specific concept, material composition knowledge is well developed in the general knowledge. This increases the chances of the models having acquired this knowledge during the training process. The best performance is attained by the Mistral-7b-Instruct model as shown in Table 5.10.

The model execution times results are presented in Figure 5.4. The Mistral-7B-instruct model has the highest execution time in zero shot training approach. This is attributed to the excessive generated responses as a result of imprecise responses by the model. This same model together with RAG and instruction tuning has a lesser execution time (-197 seconds) due to short and precise responses generated by the model. The model output also corresponds to the output format in the instruction dataset. With the GPT 4-Turbo model, there is a slightly higher execution time (+23 seconds) with the RAG and instruction tuning due to the comparison of the model input and the training example embeddings. This is however advantageous with regard



Figure 5.4: Model Execution Time

to the better extraction performance achieved by the model.

In summary, there is an improvement in the average performance of the LLMs in extracting QCL properties from text with the RAG-based instruction tuning approach/inclusion of domain-specific instructions and labels (+12.67% precision and +11.32% recall for Mistral-7b Instruct, +3.58% precision and +3.08% for GPT-Turbo) as shown in Table 5.11. The best performance is attained by the GPT-4-Turbo model with the RAG-based approach. Exposure of the models to labeled instruction dataset of QCL properties improves the models' ability in identifying these properties and efficiently extracting them. With this method, it is also possible to fine-tune the models' output format hence minimizing unnecessary explanations. This allows a better performance with efficient execution times.

5.4 Conclusion

In this chapter, we propose two approaches for mining QCL properties from scientific texts. These are based on rules and ML techniques. Based on the experimental results in Sections 5.2.3 and 5.3.3, the rule-based approach performs well in simple scenario (where properties are explicitly mentioned in text) but experience challenges when the text description does not match with the defined rules hence it fails when there are text description variations. This implies that models are limited in terms of scalability as compared to the ML models. The ML based approach can be able to generalize on a couple of scenarios with better performance. It however needs quality data to provide context to the model during the text mining process. In terms of Interpretability, the rule-based approach is more interpretable as is based on rules-captured in text expressions while the ML-based has limited interpretability. In terms of resources required,

the rule-based approach requires minimal resources while the ML-based approach require more annotation efforts for the dataset. In general, allowing LLMs to access structured labeled data improves their performance in identifying and extracting the QCL properties.

The rule-based approach can be used in situations where there is no quality training data for training ML algorithms for QCL property extraction from text. Both the rule-based and ML methods provide an opportunity for extracting structured data for QCL properties from text. This can be used to create structured collections/databases of QCL properties data that can be used to develop KR platforms to explore the relationships between the QCL properties.

In the next chapter, we present our third contribution which entails an Ontology for the QCL properties. This allows for a standardized access and analysis of QCL properties in an interoperable manner. The ontology development process and its evaluation is detailed in this chapter.

Formal Representation of the Quantum Cascade Laser Properties: An Ontological Approach

6.1	Introduction	86
6.2	Ontology Modeling	86
6.2.1	Ontology Engineering Methodology	86
6.2.2	Requirements Analysis	87
6.2.3	Modeling the Quantum Cascade Laser Properties	89
6.3	Ontology Evaluation	95
6.3.1	Source Code	96
6.3.2	Evaluation Metrics	96
6.3.3	Results and Discussions	98
6.3.4	The QCL Ontology Model and the FAIR Data Principles	102
6.4	Ontology Limitations	103
6.5	Conclusion	103

“Having knowledge but lacking the power to express it clearly is no better than never having any ideas at all”

– Pericles, *Brainy Quote*

6.1 Introduction

In this chapter, we address the problem of formal representation of the QCL properties and the relationships among them. This is in order to allow for a standardized representation and analysis of QCL properties data. We propose a semantically enriched model for formalizing the knowledge on QCL properties in form of an Ontology model [89].

In this work, we envision a platform that provides formal semantics for the QCL properties and the relationships among them hence enabling standardized access and analysis of QCL properties data from various sources in a unified and interoperable manner. We explore the extension and combination of concepts in existing Ontologies and Vocabularies in the materials science domain in order to define a unified platform for formalizing the QCL properties, the relationships between them and their provenance information.

6.2 Ontology Modeling

In this section, we detail the QCL Ontology development methodology and the description of the various Ontology concepts.

6.2.1 Ontology Engineering Methodology

In developing the QCL Ontology model, we adopt the the Neon ontology engineering methodology [164]. As elaborated in chapter 2, this methodology is characterized by a set of scenarios that correspond to the common activities within the ontology engineering life cycle. The activities range from the early stages of ontology engineering such as requirements analysis to advanced stages such as the reuse of existing ontologies, restructuring and adaptation of ontological resources to specific use cases.

The Neon methodology has a total of 9 scenarios for building ontology networks which enables the use of both ontological¹ and non-ontological resources² in the ontology development life cycle . The scenarios adopted are dependent on the use case for development of a particular ontology network. In most cases, this implies that not all scenarios may be adopted in the development life cycle (for instance scenario ii is not adopted in this work). The Neon methodology is suitable for development of ontology networks as it captures most of the crucial steps taken in developing Ontologies [105].

¹Ontological resources are ontologies adopted as a resource in the development of other ontologies

²Refers to other resources (not ontologies) used in the ontology development life cycle.

In this work, we intend to specify the user requirements for the proposed ontology model and explore the use of the existing ontologies and vocabularies to develop the ontology model. We therefore adopt the following scenarios from the Neon methodology: scenario i (From Specification to Implementation), Scenario iii (Reusing Ontological resources), Scenario iv (Reusing and re-engineering ontological resources) and Scenario viii (Restructuring ontological resources). We give the description of these scenarios and how we adopt them for our use case. We adopt the scenarios as described below.

- **Scenario i. From Specification to Implementation:** This scenario entails development of an ontology from scratch. The core ontology development activities in this scenario includes the specification of the requirements that the ontology should fulfill, by means of the ontology requirements specification activity. We use this scenario to specify the scope, purpose, the implementation language, the target group and the set of requirements to be fulfilled by the proposed ontology in the form of competency questions.
- **Scenario iii. Reusing Ontological Resources:** In this scenario, ontology resources are used in the ontology development process. Examples include using ontologies as a whole, ontology modules, and/or ontology statements. We adopt this scenario for re-using concepts from other ontologies in our use case. Examples in our use case include using terms from other ontologies to capture information on units.
- **Scenario iv. Reusing and Re-engineering ontological Resources:** Ontological resources are both reused and re-engineered to meet specific needs. This is suitable in cases where existing ontological resources are not readily usable and therefore needs modification.
- **Scenario viii. Restructuring ontological resources:** In this phase, ontological resources are pruned, modularized, extended before being used in the ontology network being developed.

We adopt the Web Ontology Language (OWL2)³ for representing the ontology model and the RDF/XML syntax for serialization of the Ontology. The use of OWL Description Language as the representation language of the ontology is suitable for this task as it enables reasoning and consistency checks on the developed Ontology using the available standard reasoners. We also adopt the metadata terms from the Dublin Core Metadata Initiative (DCMI)⁴ to represent the metadata of the proposed ontology. Based on the adopted scenarios, the ontology models is therefore developed as detailed in the following sections:

6.2.2 Requirements Analysis

This is the first step in the ontology development life cycle. In this step, we specify the ontology requirements in relation to scenario i of the Neon ontology engineering methodology. We present the user requirements in form of use cases (UCs), the competency questions and other additional domain restrictions to be captured by the proposed ontology model. The user requirements are developed based on discussions with domain experts in the QCL domain and we present them below.

³<https://www.w3.org/TR/owl2-overview/>

⁴<http://purl.org/dc/terms/>

- **Use Case 1:** The Ontology will be used to represent knowledge (definition of concepts and relations) on the QCL design properties (heterostructure materials/design types, layer sequencing) and the optoelectronic/working properties (optical power, working temperature, lasing frequency etc).
- **Use Case 2:** The Ontology will be used for representing the various working modes at which the working properties are attained based on the various design properties.
- **Use Case 3:** The Ontology model will be used to maintain provenance information about the QCL properties (both the design and working properties). This will enable tracking of data on QCL development with information such as the authors, year in which the design was proposed and permanent links to the resources such as the DOI.

Based on the Ontology use cases, we propose a list of 10 competency questions (CQs) for the QCL properties ontology. The CQs presented below.

- CQ1: What are the material composition and sequence layer of a heterostructure with a particular design type?
- CQ2: For a particular design type, what are the possible layer sequences and material composition?
- CQ3: What is the material composition of a particular hetero-structure with a particular sequence layer?
- CQ4: What are the resultant performance characteristics of a QCL laser working in a particular working mode?
- CQ5: For a particular heterostructure (as described by the sequence and material composition), what are the resultant laser performance characteristics?
- CQ6: For a particular laser performance characteristics, what are possible design properties (i.e layer sequence and material composition)?
- CQ7: What is the sequence layer of a heterostructure with a particular material composition?
- CQ8: Who are the authors of the particular laser device having certain properties?
- CQ9: When was the laser device proposed?
- CQ10: Where is the information published?

QCL data captured in scientific articles is characterized by ambiguity and conflicting information. In order to handle this, property constraints are enforced on relations between properties. The ontology is also aligned with existing standard vocabularies in the semantic web. Provenance modeling is also used to track conflicting information for review. The following additional restrictions are specified for the proposed ontology:

- A hetero-structure corresponds to one particular design type.

- A property can relate to the working mode of the laser.
- A property corresponds to only one laser working mode.
- A property can also relate to the hetero-structure.
- A Layer sequence corresponds to one material combination formula.

6.2.3 Modeling the Quantum Cascade Laser Properties

The proposed QCL ontology is developed in a modular way based on the category of information to be represented. The information captured is divided into several categories: **Properties** (which represent knowledge on the QCL optoelectronic/working properties and the relations among them), **Design Information** (which represent knowledge on the QCL heterostructure materials and the design types), **Working Modes** (to represent the nature of laser beam emission i.e either in pulses or continuously) and **Provenance Information** (which represent all details regarding the provenance information for the QCL properties data, for instance links to the sources of this information).

We provide some definitions for the upper level concepts and relationships in the ontology. The **QpOnto:QuantumCascadeLaser** concept provides a formal definition for the QCL device. The **QpOnto:LaserHeterostructure** concepts defines the laser design properties in form of the heterostructure materials, layer sequence and the design types. The **Property** concept from the Materials Design Ontology is used to represent information on the QCL optoelectronic properties. The **QpOnto:LaserWorkingMode** mode represents knowledge on the various QCL working modes while the **AcademicArticle** concept from the BIBO ontology is used to represent provenance information for the various QCL properties. The prefix **QpOnto** denotes the QCL ontology prefix.

The relationships between these concepts can be represented by description logic axioms denoted by the symbol $\sqsubseteq$. These are expressed by the following expressions: A *quantum cascade laser* has a *laser heterostructure* (U1), a *Property* (U2), a *working mode* (U3) and it is attributed to an *AcademicArticle* (U4) which gives link to resources documenting the laser. The description logic axioms for the upper concepts are expressed by the equations below.

$$(U1) \text{ QuantumCascadeLaser } \sqsubseteq = 1 \text{ hasHeterostructure.LaserHeterostructure} \quad (6.1)$$

$$(U2) \text{ QuantumCascadeLaser } \sqsubseteq \forall \text{ hasProperty.Property} \quad (6.2)$$

$$(U3) \text{ QuantumCascadeLaser } \sqsubseteq = 1 \text{ hasWorkingMode.LaserWorkingMode} \quad (6.3)$$

$$(U4) \text{ QuantumCascadeLaser } \sqsubseteq \text{ wasAttributedTo.AcademicArticle} \quad (6.4)$$

Figure 6.1 shows the representation of the upper level concepts.

The development of the ontology classes and relations for the various category of information captured in the ontology model is done as illustrated in the sections described below.

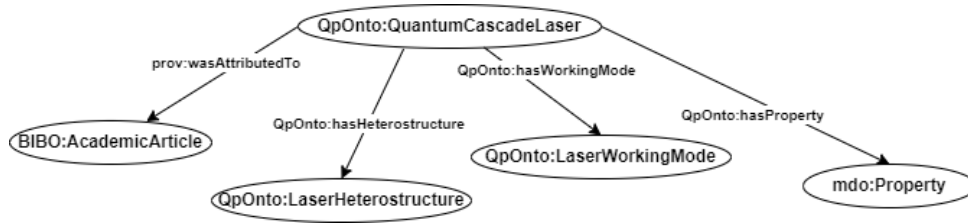


Figure 6.1: Upper Concepts in the Ontology.

Laser Optoelectronic/Working Properties

In order to represent knowledge on the QCL working properties, we adopt terms that define physical properties from the MDO ontology [105] and those terms that describe quantity values from the QUDT ontology [62]. We adopt the concept **Property** from the MDO ontology to provide descriptions for the working properties such as the optical power, frequency and working temperature. *Property* is viewed as a quantifiable aspect of a material system and it is defined as a sub-concept of *Quantity* from the QUDT ontology. *Quantity* represents the measurement of an observable property and it has the *QuantityValue* and *QuantityKind*. *QuantityValue* captures the numerical value represented and its corresponding unit. The unit information is represented by the *Unit* class in the QUDT ontology. *QuantityKind* represents knowledge on any observable property that can be measured and quantified numerically. Examples of quantity kinds include the temperature, power etc. Descriptions on these concepts can be found in the QUDT ontology⁵.

The concept *Property* is defined as a subclass of *Quantity* so as to reuse the relations and classes for quantity values, kinds and units. We can denote the description logic axioms for relations for *Property* with the symbol $\sqsubseteq$ and expression them as follows: A *Property* is a subclass of *Quantity* (P1), related to the *laser heterostructure* (P2) and corresponds to a particular *laser working mode* (P3).

The description logic axioms for the optoelectronic properties are presented in the equations below.

$$(P1) \text{ Property} \sqsubseteq \text{Quantity} \quad (6.5)$$

$$(P2) \text{ Property} \sqsubseteq \forall \text{ relatedToHeterostructure.LaserHeterostructure} \quad (6.6)$$

$$(P3) \text{ Property} \sqsubseteq = 1 \text{ correspondsToWorkingMode.LaserWorkingMode} \quad (6.7)$$

The concepts are diagrammatically illustrated in Figure 6.2 respectively.

Laser Design/Heterostructure Properties

The QCL design/heterostructure properties captures information about the heterostructure materials combination used in developing the laser device. The information also includes the details on the layer sequencing patterns used. We adopt the term *Material* from the EMMO ontology and the CheBi ontology in order to represent knowledge about the heterostructure materials [54, 36]. We also define the following terms to represent knowledge about the QCL design properties:

⁵<https://qudt.org/>

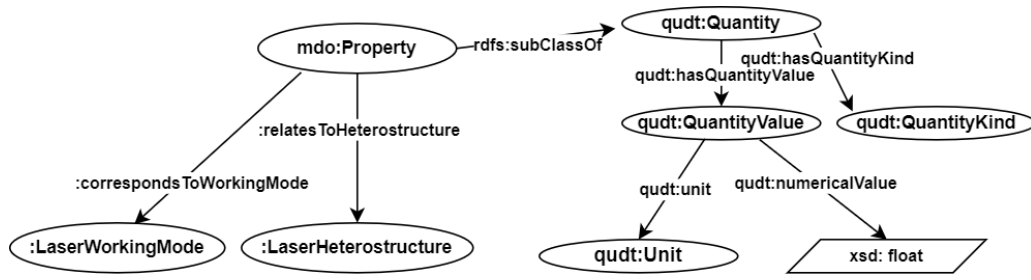


Figure 6.2: Ontology Section for Quantum Cascade Laser Optoelectronic Characteristics.

`QpOnto:LaserHeterostructure`, `QpOnto:DesignType`, `QpOnto:HeterostructureMaterials` and `QpOnto:LayerSequence`. We provide formal definitions for these concepts below.

Definition 6.2.1 (*LaserHeterostructure*). A layered structural material composed of different semiconductor materials.

This concept represents the laser layer design comprising of the various semiconductor materials.

Definition 6.2.2 (*DesignType*). The design type of the laser refers to the geometrical arrangement of materials in the laser design.

Sample instances of this class include the LO phonon design and the bound to continuum design types.

Definition 6.2.3 (*HeterostructureMaterials*). Heterostructure Materials refers to the materials forming the heterostructure layer.

This concept gives more information on the different materials used and their respective proportions of combination.

Definition 6.2.4 (*LayerSequence*). The layer sequence represents the thickness of materials composing the heterostructure.

This concept constitutes the materials combinations thickness of measurement in Nanometers or Angstrom.

We capture the relationship between these concepts in form of description logic axiom denoted by the symbol D . A particular *laser heterostructure* has a *design type* (D1), *heterostructure materials* (D2), the *layer sequence* (D3) and relates to a Material (D4). The *heterostructure materials* are composed of *atoms* (D5) and has a *matFormula* in a string which captures the chemical elements and their ratio of combination. The *layer sequence* is based on the *heterostructure materials* (D6), has a unit (D7) and has the *sequence* in a string. The description logic axioms for the laser heterostructure properties are expressed in the following equations:

$$(D1) \text{ LaserHeterostructure } \sqsubseteq = 1 \text{ hasDesignType.DesignType} \quad (6.8)$$

$$(D2) \text{ LaserHeterostructure } \sqsubseteq = 1 \text{ hasMaterials.HeterostructureMaterials} \quad (6.9)$$

$$(D3) \text{ LaserHeterostructure } \sqsubseteq = 1 \text{ hasLayerSequence.LayerSequence} \quad (6.10)$$

$$(D4) \text{ LaserHeterostructure } \sqsubseteq \text{ relatesToMaterial.Material} \quad (6.11)$$

$$(D5) \text{ HeterostructureMaterials } \sqsubseteq \forall \text{ composedOf.Atoms} \quad (6.12)$$

$$(D6) \text{ LayerSequence } \sqsubseteq \forall \text{ basedOnMaterials.HeterostructureMaterials} \quad (6.13)$$

$$(D7) \text{ LayerSequence } \sqsubseteq = 1 \text{ unit.Unit} \quad (6.14)$$

Figure 6.3 summarizes the concepts and relations for the QCL heterostructure properties.

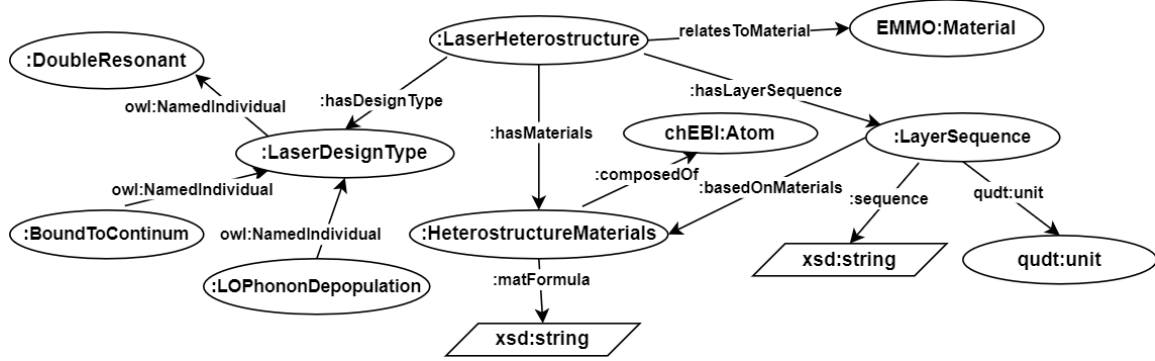


Figure 6.3: Heterostructure Design Properties Concepts and Relations

A visualization of the heterostructure class from the ontology model using the WebVOWL platform⁶ is shown in Figure 6.4.

Laser Working Modes

For the laser working mode, we define the term `QpOnto:Laser WorkingMode` to represent the knowledge. The term is defined below.

Definition 6.2.5 (LaserWorkingMode). This refers to the nature in which the QCL laser beams are emitted. This can either be in a continuous mode or in pulses (i.e in pulse mode).

Some of the QCL working properties such as the working temperature vary depending on the working mode of the QCL laser device. Examples of instances for the laser working mode are shown in Figure 6.5.

Provenance Information

Provenance information is captured in the proposed ontology for the purpose of tracking the QCL properties to their sources. We consider sources that have a permanent/persistent identifier (in this case scientific articles). We adopt the terms `Agent` and `AcademicArticle` from the

⁶<https://service.tib.eu/webvowl/>

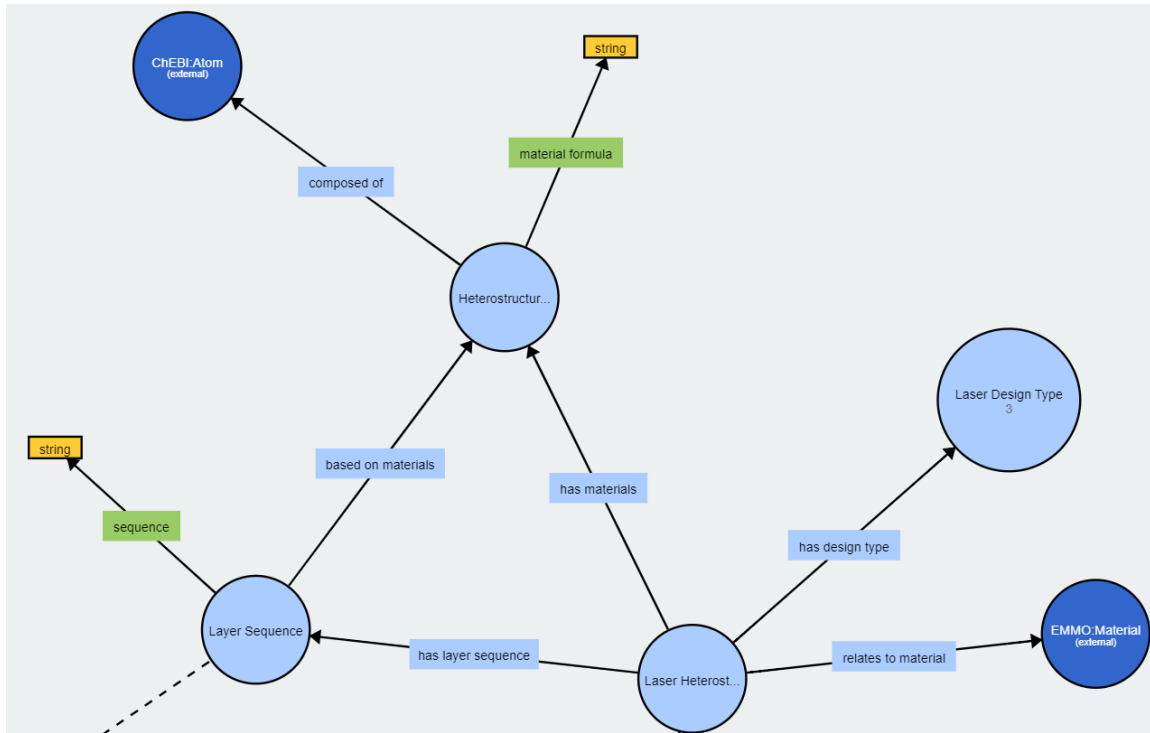


Figure 6.4: A Visualization of the QCL Heterostructure Class from the Ontology.

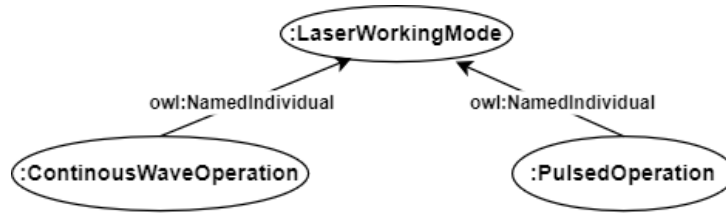


Figure 6.5: Concepts for the Quantum Cascade Laser Working Mode.

provenance ontology [100] and the BIBO⁷ ontology respectively in order to represent knowledge on provenance information for the QCL properties.

An *Agent* from the provenance ontology refers to an entity that is responsible for the existence of another entity or the activity of another agent. An *AcademicArticle* from the BIBO ontology refers to a peer reviewed article. We use this term to refer to the articles documenting the QCL properties. An *AcademicArticle* is set to be a sub-concept of an *Agent* in order to create the intuition that academic articles are ‘responsible sources’ of the various QCL properties documented in text. We denote the description logic axiom for this relation as P1.

We also define some terms for data properties (relations) that are associated with an *AcademicArticle* for its unique identification. The definitions are as outlined below.

Definition 6.2.6 (PublisherName). An academic article has a publisher name in a string.

⁷[4](#)

Definition 6.2.7 (URL). An academic article has a URL with the URL datatype.

Definition 6.2.8 (DOI). An academic article has a DOI in a string.

Definition 6.2.9 (PublicationTitle). An academic article has a publication title in a string.

Definition 6.2.10 (AuthorName). An academic article has an author name in a string.

Definition 6.2.11 (PublicationYear). An academic article has a publication year in an integer.

The description logic axiom for provenance information is expressed in the equation below.

$$(PR1) \text{AcademicArticle} \sqsubseteq \text{Agent} \quad (6.15)$$

Figure 6.6 details the concepts for representing the provenance information.

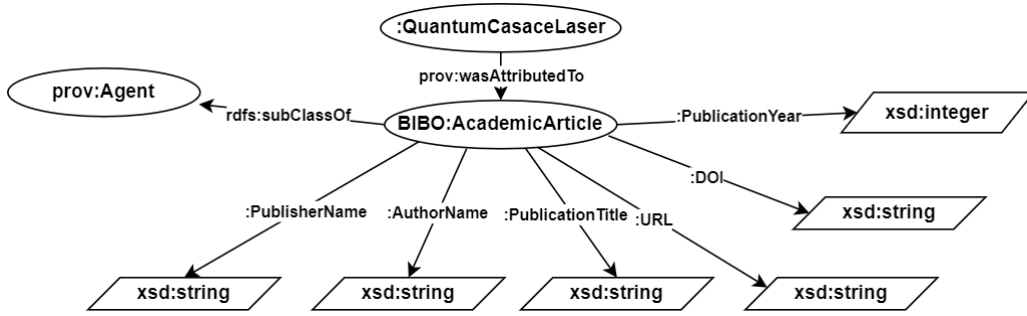


Figure 6.6: Concepts for Provenance Information.

In summary, the proposed ontology model has a total of 15 concepts, 23 relations and 11 instances. Figure 6.7 shows the ontology concepts and relations.

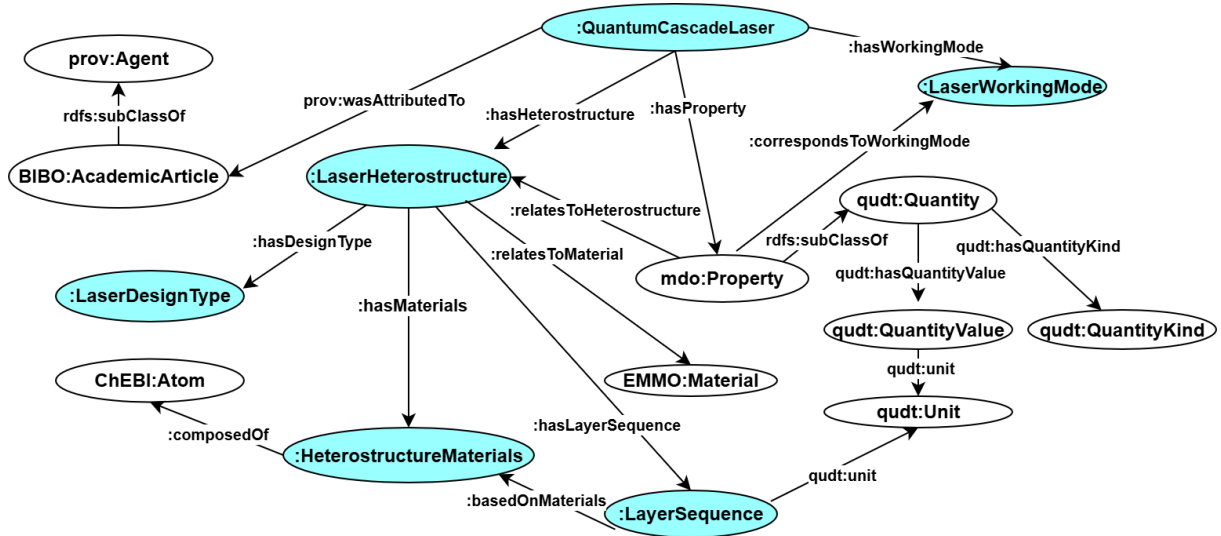


Figure 6.7: Concepts in the QCL Ontology.

A Neo4j⁸ visualization of the ontology concepts is detailed in Figure 6.8.

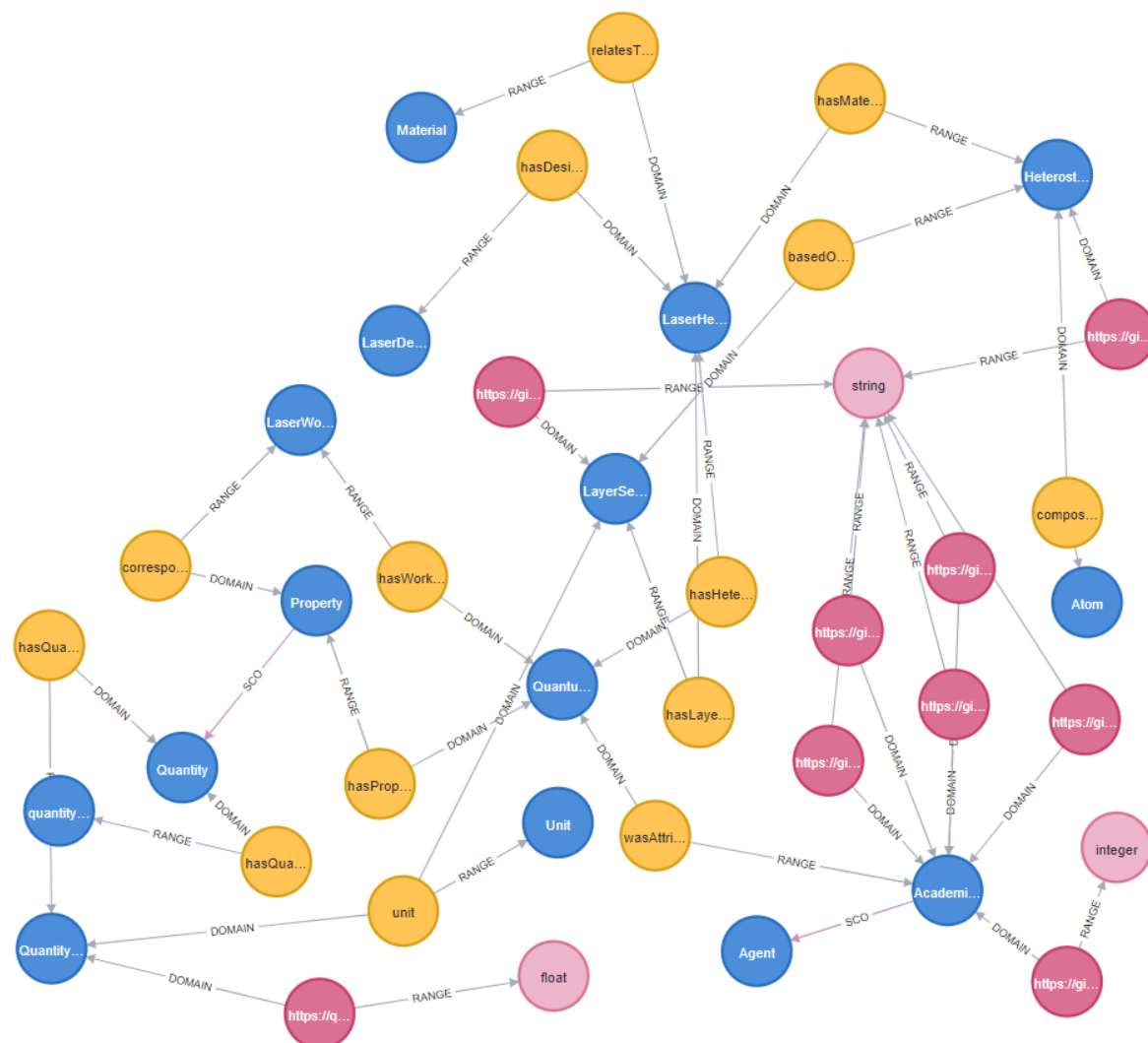


Figure 6.8: A Neo4j Visualization of the Ontology model.

6.3 Ontology Evaluation

In this section, we present an experimental evaluation of the proposed ontology model and discuss how the ontology addresses some of the FAIR data principles for the QCL domain.

⁸Neo4j is graph database that stores data as a network of connected nodes.

6.3.1 Source Code

The implementation of the proposed ontology is publicly available at our GitHub repository: https://github.com/DeperiasKerre/qcl_Onto

6.3.2 Evaluation Metrics

For the evaluation of the proposed Ontology, we focus on three metrics: (i) Consistency, (ii) Formative Evaluation (Evaluating the ability of the Ontology in meeting the domain requirements) and (iii) Richness of the ontology based on quality evaluation metrics. We explain these metrics below.

Consistency

The consistency of an ontology entails the logical correctness of the definitions in its concepts. More formally, ontological consistency can be defined as below.

Definition 6.3.1 (Ontology Consistency). A given Ontology definition is said to be consistent if there is no contradiction in the interpretation of the formal definition with respect to the outside world. [61].

Therefore, there should be no contradictory conclusions derived from the meaning of all the definitions and axioms in the defined ontology, and the ontologies included in this ontology. The following ontology consistency checking tools are used in verification of the ontology consistency: HermiT Reasoner [55] and Pellet Reasoner [159] both embedded in the Protege Software⁹ and the Pitfall Scanner [135].

Formative Evaluation

In formative evaluation, we assess the ability of the developed Ontology in meeting the user requirements for the QCL domain. This is done by carrying out Ontology Validation [141]. Ontology Validation can be defined as below.

Definition 6.3.2 (Ontology Validation). Ontology validation involves evaluating the ontology using sample test cases in order to verify its applicability to the intended problem. The validation is deemed successful if the ontology passes the test cases [141].

This evaluation is important in verifying the ability of the ontology in capturing capturing the concepts in the QCL domain hence being able to organize QCL properties data from scientific texts for structured exploration.

In order to carry out ontology validation, we first generate a KG instance from the ontology by mapping sample QCL properties data (for 15 QCL devices) onto the Ontology schema. The data

⁹<https://protege.stanford.edu/>

is generated from scientific texts adopted from [96, 92, 189, 95, 139, 4, 182, 126, 38, 174, 47, 14, 178, 190, 20] based on the method presented in section 5.2. Some of the laser properties (such as design type, working mode) and metadata (such as the URL) are added by the manually. We then define a set of test cases in form of competency questions (CQs) based on the user requirements and competency questions presented in the requirements analysis in section 6.2.2. Table 6.1 shows the 7 classes of CQs that we define for the test cases.

Table 6.1: Competence Questions (CQs)

ID	Question Text
CQ1	What is the material composition/sequence layer of a heterostructure with a design type X ?
CQ2	For a particular heterostructure X, what are the possible design type?
CQ3	What is the layer sequence of a heterostructure with a material composition X ?
CQ4	What is the performance property X of a laser working in mode Y ?
CQ5	What is the performance property X of a laser having a heterostructure with material composition Y or layer sequence Z?
CQ6	For a particular performance property X, What are the corresponding heterostructure designs?
CQ7	What is the DOI and/ the URL of the scientific article documenting a laser with performance property W or with heterostructure X or working mode Y or design type Z.

W, X, Y and Z are basically placeholders for any specific value for the QCL design and working properties. We run several possibilities for each of the test cases on the KG instance in order to assess the ability in passing the test cases by successfully answering the queries.

Quality Metric

We adopt a schema metric for evaluating the quality of the Ontology. The metrics are presented in [167]. The schema metrics address the design of the ontology. In this study, we use the *Inheritance Richness*(*IR*) assessment metric. This metric describes the distribution of information across the across different levels of the ontology inheritance tree. The IR metric for a class C_i is defined as the *average number of subclasses per class in the a given ontology subtree* which is determined as $|H^C(C_1, C_i)|$. Therefore, the IR for the entire ontology schema is given by the equation below.

$$IR = \frac{\sum_{C_i \in C'} |H^C(C_1, C_i)|}{|C|} \quad (6.16)$$

In this case, H is the number of inheritance relationships and C is the number of classes in the ontology. Lower IR levels implies that each class has a relatively large number of subclasses hence the more detailed an ontology is to cover relationships in the domain requirements and vice versa. We use this metric to analyze the detail level of the ontology in capturing the domain requirements.

6.3.3 Results and Discussions

Ontology Consistency

For the case of consistency checks, we observe that no inconsistencies are identified by the HermiT and Pellet logical reasoners. There are also no critical pitfalls detected by the Pitfall Scanner tool. This validates the logical correctness in the ontology schema as shown by the logical reasoners. We also encounter no pitfalls in the ontology design process as verified by the ontology pitfall scanner tool. This validates adherence to best practices in the ontology design process.

Ontology Validation

We define a set of 12 CQs for the CQ classes in Table 6.1 with help of QCL domain experts. The CQs entails questions that QCL domain experts could need answers to during the QCL design fabrication process. The CQs for each of the CQ classes are as follows: CQ1 (2 competency questions), CQ2 (1 competency question), CQ3 (2 competency questions), CQ4 (2 competency questions) and CQ5 (3 competency questions). The CQs for each of the classes and the respective ontology classes and relations that answers them are shown in Table 6.2.

The CQs are translated into the their respective SPARQL queries which are executed on the KG instance [58]. We present examples where the QCL expert wants to know the possible working temperatures of QCLs working in pulsed mode and the possible materials combinations for QCLs with lasing frequency greater than 3 THz:

Example 6.3.1. Competency Question 4.2: *What are the operating temperatures of semiconductor laser devices working in a pulse mode operation?*

The SPARQL query for this CQ can be expressed as illustrated in Listing 6.1 and the query result in Figure 6.9.

Listing 6.1: SPARQL Query for Competency Question 4.2

```
PREFIX QpOnto:<https://github.com/DeperiasKerre/qcl_Onto/blob/main/qclontology/version-1.0/qclonto.owl#>
PREFIX qudt:<https://qudt.org/schema/qudt/>
SELECT ?value ?unit
WHERE
{
  ?wt QpOnto:correspondsToWorkingMode QpOnto:PulsedOperation;
  <https://qudt.org/schema/qudt#hasQuantityValue> ?qv.
  ?qv qudt:numericValue ?value;
  qudt:hasUnit ?unit.
}
```

Example 6.3.2. Competency Question 6.1: *What is the possible material composition for heterostructures with a lasing frequency greater than 3 THz?*

Table 6.2: Competence Questions (CQs)

CQ ID	Competency Question	Classes	Properties
CQ1.1	What is the possible heterostructure material composition(s) of a semi-conductor laser with a bound to continuum design type?	QuantumCascadeLaser, LaserHeterosrtucture, LaserDesignType, HeterostructureMaterials	hasHeterostructure, hasDesignType, hasMaterials, matFormula
CQ1.2	What is the possible heterostructure layer sequence(s) of a semi-conductor laser with an LO Phonon design type?	QuantumCascadeLaser, LaserHeterostructure, LaserDesignType, LayerSequence	hasHeterostructure, hasDesignType, hasLayerSequence, sequence
CQ2.1	What is the design type for a heterostructure with material GaAs/Al0.15Ga0.85As?	QuantumCascadeLaser, LaserHeterostructure, LaserDesignType, HeterostructureMaterials	hasHeterostructure, hasDesignType, hasMaterials, matFormula
CQ3.1	What is the possible layer sequence(s) of a heterostructure with material composition GaAs/Al0.15Ga0.85As?	HeterostructureMaterials, LaserSequence, Unit	basedOnMaterials, sequence, matFormula, unit
CQ4.1	What are the operating temperatures of semiconductor laser devices working in a continuous wave operation?	WorkingTemperature, LaserWorkingMode, QuantityValue, Unit	correspondsToWorkingMode, hasQuantityValue, unit, numericalValue
CQ4.2	What are the operating temperatures of semiconductor laser devices working in a pulse mode operation?	WorkingTemperature, LaserWorkingMode, QuantityValue, Unit	correspondsToWorkingMode, hasQuantityValue, unit, numericalValue
CQ5.1	What is the lasing frequency of a semiconductor laser with a material composition GaAs/Al0.15Ga0.85As?	LaserHeterostructure, HeterostructureMaterials, Property, QuantityValue, Unit	hasMaterials, relatesToHeterostructure, hasQuantityValue, unit, numericalValue, matFormula
CQ5.2	What is the output power of a semiconductor laser with a a layer sequence of 54/78/24/64/38/148/24/94 Å?	LaserHeterostructure, LayerSequence, Property, QuantityValue, Unit	hasLayerSequence, relatesToHeterostructure, hasQuantityValue, unit, numericalValue, sequence
CQ6.1	What is the possible material composition for heterostructures with a lasing frequency greater than 3 THz?	LaserHeterostructure, HeterostructureMaterials, Property, Unit, QuantityValue	hasMaterials, hasQuantityValue, relatesToHeterostructure, unit, numericalValue, matFormula
CQ7.1	What are the DOIs and URLs of scientific articles documenting semi-conductor laser devices with a working temperature greater than 40K at continuous wave mode?	AcademicArticle, QuantumCascadeLaser, Property, LaserWorkingMode, Unit, QuantityValue	wasAttributedTo, hasProperty, correspondsToWorkingMode, unit, DOI, URL
CQ7.2	What are the DOIs and URLs of scientific articles documenting semi-conductor lasers with a material composition of GaAs/Al0.15Ga0.85As?	AcademicArticle, QuantumCascadeLaser, LaserHeterostructure, HeterostructureMaterials	wasAttributedTo, hasHeterostructure, hasMaterials, matFormula, DOI, URL
CQ7.3	What are the DOIs and URLs of scientific articles documenting semi-conductor lasers with An LO Phonon Design Type?	LaserDesignType, QuantumCascadeLaser, LaserHeterostructure, LaserDesignType, AcademicArticle	hasHeterostructure, wasAttributedTo, hasDesignType, URL, DOI

temperature value: 110 unit: <https://qudt.org/vocab/unit/K>
 temperature value: 144 unit: <https://qudt.org/vocab/unit/K>
 temperature value: 65 unit: <https://qudt.org/vocab/unit/K>
 temperature value: 85 unit: <https://qudt.org/vocab/unit/K>
 temperature value: 138 unit: <https://qudt.org/vocab/unit/K>

Figure 6.9: Result for the Query for Competency Question 4.2

The corresponding SPARQL query for this CQ is expressed as shown in Listing 6.2. The result obtained after executing this query is shown in Figure 6.10.

Listing 6.2: SPARQL Query for Competency Question 6.1

```
PREFIX QpOnto:<https://github.com/DeperiasKerre/qcl_Onto/blob/main/qclontology/version-1.0/qclonto.owl#>
PREFIX qudt:<https://qudt.org/schema/qudt/>
SELECT DISTINCT ?mat_composition
WHERE
{
  ?lf <https://qudt.org/schema/qudt#hasQuantityValue> ?fv;
  QpOnto:relatesToHeterostructure ?HS.
  ?fv qudt:numericValue ?value;
  qudt:hasUnit ?unit.
  ?HS QpOnto:hasMaterials ?HM.
  ?HM QpOnto:matFormula ?mat_composition.
  FILTER(?unit=<https://qudt.org/vocab/unit/TeraHZ>&&?value>3).
}
```

```
material composition: GaAs/Al0.15Ga0.85As
material composition: GaAs/Al0.25Ga0.75As
material composition: In0.53Ga0.47As/GaAs0.51Sb0.49
```

Figure 6.10: Result for the Query for Competency Question 6.1

In order to obtain a lasing frequency greater than 3 THz, there are 3 options for materials composition. These are explained below.

- *GaAs/Al_{0.15}Ga_{0.85}As*: The materials composition may consist of three elements. These are Ga (Gallium Atom), Al (Aluminium Atom) and As (Arsenic Atom). The combination will consist of two materials as follows: material 1 (*GaAs* - 1 atom of Ga and 1 atom of As), material 2 (*Al_{0.15}Ga_{0.85}As* -for 1 atom of Al, 15 % of Ga and 85 % of As).
- *GaAs/Al_{0.25}Ga_{0.75}As*: The same material combinations as the previous ones with material 2 having 25 % of concentration Gallium Atom and 75 % concentration of Arsenic Atom for every 1 Atom of Aluminium.
- *In_{0.53}Ga_{0.47}As/GaAs_{0.51}Sb_{0.49}*: he materials composition may consist of four elements. These are Ga (Gallium Atom), In (Indium Atom), As (Arsenic Atom) and Sb (Antimony Atom). The combination will consist of two materials as follows: material 1 (*In_{0.53}Ga_{0.47}As* – for 1 atom of In, 53 % of Ga and 47 % of As), material 2 (*GaAs_{0.51}Sb_{0.49}* -for 1 atom of Ga, 51 % of As and 49 % of Sb).

We also present an example that demonstrates access to provenance information for particular QCL properties:

Example 6.3.3. Competency Question 7.1: *What are the DOIs and URLs of scientific articles documenting semi-conductor laser devices with a working temperature greater than 40 K in continuous wave mode?*

The SPARQL query for CQ 7.1 is represented in Listing 6.3 and the retrieved results shown in Figure 6.11.

Listing 6.3: SPARQL Query for Competency Question 7.1

```
PREFIX QpOnto:<https://github.com/DeperiasKerre/qcl_Onto/blob/main/
qclontology/version-1.0/qclonto.owl#>
PREFIX qudt:<https://qudt.org/schema/qudt/>
PREFIX prov:<http://www.w3.org/ns/prov#>
SELECT ?doi ?url
WHERE
{
  ?qcl QpOnto:hasProperty ?wt;
  prov:wasAttributedTo ?article.
  ?wt <https://qudt.org/schema/qudt#hasQuantityValue> ?qv;
  QpOnto:correspondsToWorkingMode QpOnto:ContinuousWaveOperation.
  ?qv qudt:numericValue ?value;
  qudt:hasUnit ?unit.
  ?article QpOnto:DOI ?doi;
  QpOnto:URL ?url.
  FILTER(?unit=<https://qudt.org/vocab/unit/K>&&?value>40).
}
```

```
DOI: 10.1063/1.2189671 URL: https://www.doi.org/10.1063/1.2189671
DOI: 10.1364/OE.16.003242 URL: https://www.doi.org/10.1364/OE.16.003242
DOI: 10.1063/1.2737129 URL: https://www.doi.org/10.1063/1.2737129
DOI: 10.1063/1.2216112 URL: https://www.doi.org/10.1063/1.2216112
DOI: 10.1063/1.5110305 URL: https://www.doi.org/10.1063/1.5110305
DOI: 10.1063/1.3114418 URL: https://www.doi.org/10.1063/1.3114418
DOI: 10.1063/1.4862177 URL: https://www.doi.org/10.1063/1.4862177
DOI: 10.1088/1367-2630/11/12/125022 URL: https://www.doi.org/10.1088/1367-2630/11/12/125022
DOI: 10.1088/0256-307X/30/6/064201 URL: https://www.doi.org/10.1088/0256-307X/30/6/064201
DOI: 10.1088/1367-2630/18/12/123004 URL: https://www.doi.org/10.1088/1367-2630/18/12/123004
DOI: 10.1063/1.4766915 URL: https://www.doi.org/10.1063/1.4766915
DOI: 10.1088/1367-2630/17/2/023050 URL: https://www.doi.org/10.1088/1367-2630/17/2/023050
DOI: 10.1364/OE.20.003866 URL: https://www.doi.org/10.1364/OE.20.003866
```

Figure 6.11: Result for the Query for Competency Question 7.1

The full results for all the SPARQL queries can be found at the GitHub repository: https://github.com/DeperiasKerre/qcl_Onto.

All the 12 queries executed were successfully answered. We compare the query responses with the expected responses to determine the precision. All the 150 responses retrieved by the queries match the expected responses hence resulting to a precision of 1. This demonstrates the ability of the ontology schema in successfully modeling relationships in the various QCL design and working properties. In addition, as seen in the queries for CQs 7.1 to 7.3, it is possible to maintain provenance information for the QCL properties hence providing links to the sources.

Metric Based Evaluation

The proposed Ontology model attains an Inheritance Richness metric of 0.133. This implies each class has a relatively large number of subclasses capturing knowledge about a particular

class. This is a reflection of the level of detail of the ontology model in capturing the domain requirements. Despite having general information in the inheritance sub-tree, there is enough distribution of information (in form of relations) that enables to it to capture well the relations between the QCL properties.

6.3.4 The QCL Ontology Model and the FAIR Data Principles

We present the role of the proposed ontology in addressing the FAIR Data principles. The QCL ontology addresses the FAIR data principles in two ways described below.

Meeting the FAIR Data Standards in Ontology Representation

The QCL ontology meets the FAIR Data principles in representation. We give descriptions for this below.

- **Metadata Completion:** The ontology metadata is completed using a checklist for completing vocabulary metadata¹⁰. This information is used to represent the ontology Authors, version and license using the Dublin Core metadata terms. This enables to achieve FAIR's Findability (F2 and F3) to make it easier to find and reuse the ontology model.
- **Indexing and Availability:** The QCL Ontology is published under the CC-BY 4.0 license. The ontology source code is also publicly available at the GitHub repository: https://github.com/DeperiasKerre/qcl_Onto. This addresses the FAIR'S Reusability (R1.1) principle and for replication purposes.

Enabling FAIR Access to QCL Properties Data

The QCL ontology also allows FAIR access to QCL data in several ways. This is as outlined below.

- **Interoperability:** The ontology is implemented using OWL 2. This enables logical reasoning on QCL data by both humans and machines that utilize logical reasoners. This enables achieving FAIR'S Interoperability (I1). Based on this, automatic inferences can be made on the QCL data gathered from heterogeneous scientific texts sources. This aspect also enables the community to have standards for accessing, sharing and analyze QCL properties data irrespective of the scientific texts source. This will have a positive impact on the accelerated discovery of novel QCL designs with optimal performance features.
- **Findability and Reusability:** The QCL ontology provides rich formal description on the QCL properties and also provides provenance for this information hence enabling them to be traced to the source. This achieves FAIR'S Findability (F2) and Reusability (R1.2) in providing access to QCL properties data. With this aspect, it is possible to maintain and

¹⁰<https://w3id.org/widoco/bestPractices>

trace the source of data on QCL properties over a given period of time hence enabling the analysis of development trends. The standard formalization enables the data to be reused for various purposes due to the shared vocabularies that are utilized to analyze the data.

6.4 Ontology Limitations

The proposed ontology does not fully focus on all the aspects of the semiconductor domain. It is limited to the QCL design and working properties. Some of the properties not covered include the semi-conductor doping formalizations. This can be extended in future to provide a deeper view on this knowledge.

6.5 Conclusion

In this chapter, we address the issue of formal representation of QCL properties. We propose an Ontology that formalizes the Knowledge for QCL design and working properties and the relationships among them. The ontology also captures knowledge on the provenance information for these properties in order to allow for their traceability to the sources.

We validate the logical consistency of the Ontology using logical reasoners. We also validate the ability of the Ontology in capturing the domain requirements by demonstrating its success in the test cases and the distribution of information for capturing the relations in the QCL domain. We demonstrate the role of the ontology in providing a FAIR access to data for the QCL properties domain hence enabling interoperability among the community in sharing and accessing QCL data for insights for optimized device fabrication. The concepts in the proposed ontology can also be extended/re-engineered to represent knowledge on concepts in the semiconductor domain.

However, the proposed ontology does not capture all the properties for the laser design such as the sequence layer doping and barrier thickness properties. Despite this limitation, the ontology provides a foundational knowledge representation platform that can be extended to capture more QCL properties (either working or design properties). The concepts in the ontology can also be reused/extended to represent knowledge on other related semiconductor properties.

In the next chapter, we present our fourth and last contribution, a KG prototype for semantic modeling of QCL properties in scientific texts. We detail the KG generation process and the validation process.

Semantic Enrichment of the Quantum Cascade Laser Properties

7.1	Introduction	106
7.2	Knowledge Graph Generation	106
7.2.1	Data Extraction from Scientific Texts	106
7.2.2	Knowledge Graph Generation	107
7.3	Knowledge Graph Evaluation	110
7.3.1	Source Code	111
7.3.2	Evaluation Metrics	111
7.3.3	Evaluation Results	112
7.4	Conclusion	117

“To know that we know what we know, and to know that we do not know what we do not know, that is true knowledge”

– Nicolaus Copernicus, *Good Reads*

7.1 Introduction

In this chapter, we address the problem of the QCL properties semantic enrichment in text. We envision a platform that can be used to capture and organize QCL properties data in text together with their semantic enrichment for meaningful exploration. We propose a Knowledge Graph (KG) that captures the QCL properties in text and also provides links to provenance and references for these properties. We focus on the following QCL properties: Optical power, Lasing frequency, Working Temperature, Heterostructure Materials and the Design type.

7.2 Knowledge Graph Generation

The QCL KG is generated in two steps. The first step entails generating QCL properties data from scientific texts and organizing it in a structured format. This is then followed by the KG schema definition and data mapping in order to generate the KG. We detail these in the following subsections.

7.2.1 Data Extraction from Scientific Texts

The first step in the KG generation process is to generate structured data for the QCL properties. We implement the data extraction process by adopting the machine learning based approach presented in Chapter 4. We use the GPT-4 Turbo model for the data extraction step. This is due to the model’s improved efficiency in generating responses, its large context window [128] and its relatively best average performance as presented in the experimental results in section 5.3.3.

First, we manually retrieve plain text from abstracts of scientific articles documenting QCL properties and organize them into a structured CSV file. The data extraction algorithm is then executed on the file in order to extract the QCL properties from its contents. This step generates multiples files each corresponding to a particular QCL property. These are merged to combine all the properties from every abstract (i.e for every QCL device). This is followed by data post-processing in order to remove incomplete records and update the missing values.

The next step involves adding the provenance data to all the QCL properties. This includes adding columns for the DOI and URL for every abstract capturing the particular QCL properties. We perform these steps on 36 abstracts documenting QCL devices together with the corresponding design and working properties. The limited data is attributed to the text and data mining

limitations for some publishers as detailed in Chapter 4. We generate QCL properties data that we utilize in the subsequent steps for KG generation.

7.2.2 Knowledge Graph Generation

At this step, we perform two tasks: we define a KG schema that will be used to organize the QCL data and we automatically map the data to generate the KG.

Knowledge Graph Schema Definition

In order to model a schema to organize QCL properties data, we consider a KG that is defined as below.

Definition 7.2.1 (Knowledge Graph). $KG = \{t_1, t_2, t_3 \dots t_n\}$ and t refers to the various triples in the KG and n the number of triples in the KG. A triple $t = \{s, p, o\}$ where s is the subject, p the predicate and o the object. These are uniquely identified by Uniform Resource Identifiers (URIs).

In our case, s and o are the various QCL property entities and p the relations between the various QCL properties. We provide an example of this modeling below.

Example 7.2.1 Consider the following text description capturing the QCL heterostructure materials : “A 2.1 THz quantum cascade laser (QCL) based on a scattering assisted injection and resonant-phonon depopulation design scheme is demonstrated. The QCL is based on a four-well period implemented in the GaAs/Al_{0.15}Ga_{0.85}As material system.” (text extracted from [92]).

In this example, we have the following entities: *frequency with a value of 2.1 THz*, the *laser design type (resonant phonon)* and the *heterostructure materials (GaAs/Al_{0.15}Ga_{0.85}As)*. From this text description, the relationships between the QCL properties are not explicitly mentioned. In fact, even the entities names for instance ‘design type’ are also not explicitly mentioned. The entity names and relationships can then be inferred based on the expert knowledge captured in the QCL properties Ontology proposed in Chapter 6.

Based on the QCL ontology, semantic modeling for the QCL properties can be implemented by the following triples; *Lasing Frequency (S) has a value (P-data property) 2.1 (O). In this case the object is a literal of float datatype. Another triple can be Heterostructure Material (S) has a material formula (P-data property) GaAs/Al_{0.15}Ga_{0.85}As (O-Literal of string datatype.)*. This implies that predefined knowledge in the QCL ontology can be used to infer the relationships between the various QCL properties.

We define a KG schema based on the QCL Ontology¹¹ presented in chapter 6 and other vocabularies such as BIBO and Schema.org¹². The schema also captures concepts from the QUDT ontology [62] to represent the QCL property numerical values and units and the provenance ontology [100] to represent knowledge that provides links to QCL properties’ sources.

¹¹https://github.com/DeperiasKerre/qcl_onto/blob/main/qclontology/version-1.0/qclonto.owl

¹²<https://schema.org/>

with their literal values through the data properties defined in the KG schema. The object relations between the data class instances are also implemented as per the Schema. The units, design types and the relevant QCL working modes are also enriched with the relevant URIs. The redundant triples are also examined and eliminated from the generated Knowledge Graph. The final RDF file is then serialized into Turtle and RDF/XML formats.

QCL data in scientific texts is characterized by ambiguity and conflicting facts. Ambiguity is dealt by Ontology property constraint enforcement, aligning the KG schema with standard vocabularies, units are standardized, abbreviations and synonyms are resolved using a controlled vocabulary. For conflicting data, the KG retains provenance metadata. Contradictions are not overwritten; instead, they are flagged so that experts or downstream algorithms can decide on resolution strategies.

The generated KG contains a total of 2979 triples and provides a semantic representation for information on QCL properties, the relations between them, the references and their provenance details. Despite the fact that the KG captures few devices, it captures the relevant relationships as the representation is usually standard. It provides a proof of concept on how QCL properties data can be semantically enriched using standard vocabularies and explored for insights.

We present a sample modeling of the relation between the lasing frequency, the heterostructure and its provenance information. The visualization of the sample heterostructure and frequency instances is shown in Figure 7.2.

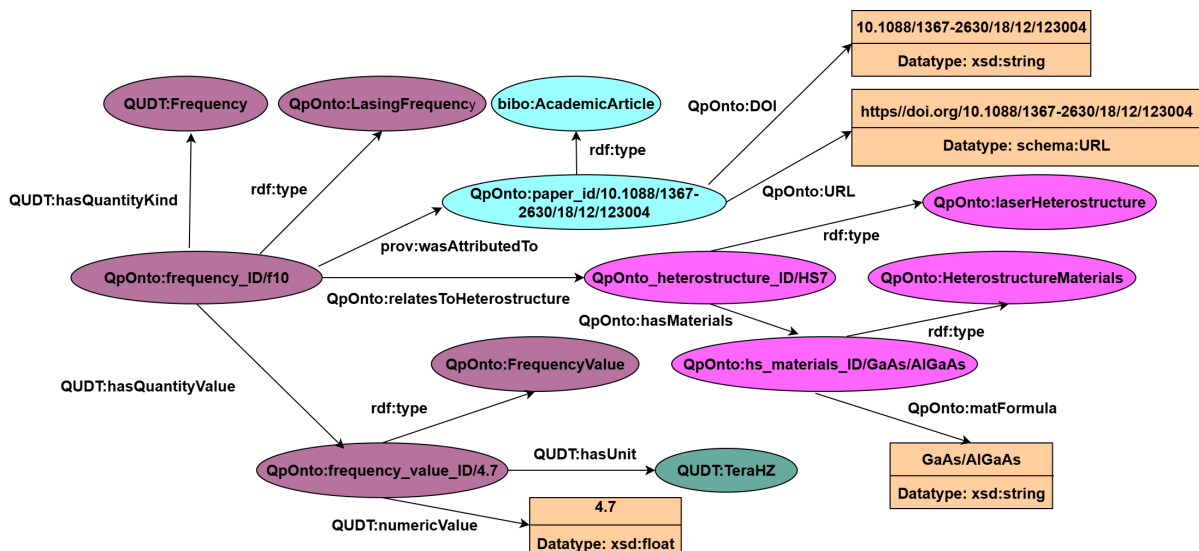


Figure 7.2: Modeling the Relationship Between Frequency and the Heterostructure.

Figure 7.3 shows a visualization of a sample heterostructure instance in the generated KG using the RDF Visualizer tool ¹⁴.

The whole QCL KG generation process starting from QCL properties extraction from scien-

¹⁴<https://www.ldf.fi/service/rdf-grapher>

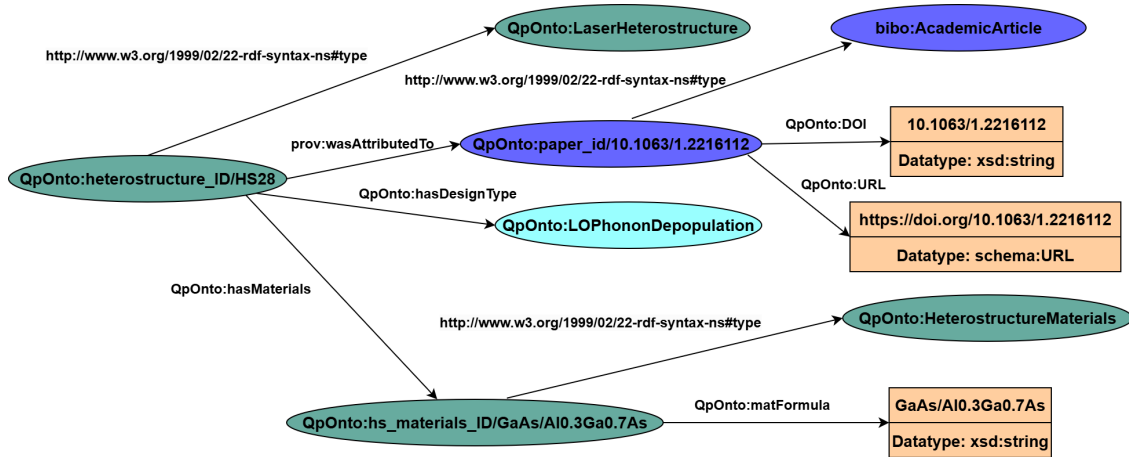


Figure 7.3: A Visualization of an Instance of the Heterostructure

tific texts to the final stages of data mapping and post-processing is summarized as shown in Figure 7.4.

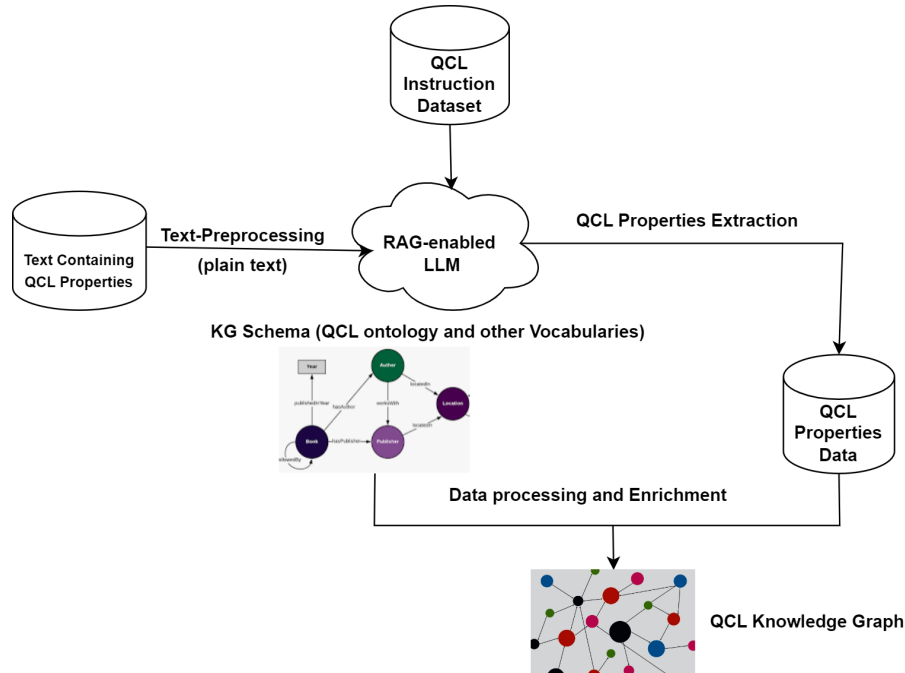


Figure 7.4: The Knowledge Graph Generation Workflow

7.3 Knowledge Graph Evaluation

In this section, we present the experimental evaluation of the generated KG for the QCL properties domain. We also provide the source code for the implementation of the KG and the serialized

files of the KG in both Turtle and RDF/XML formats.

7.3.1 Source Code

The source code for the KG generation, the KG files (serialized in Turtle and RDF/XML formats) and the competency questions results can be accessed through our GitHub repository: <https://github.com/DeperiasKerre/qKG>

7.3.2 Evaluation Metrics

In order to evaluate the generated KG, we focus on three strategies: (i) We check and validate the logical consistency of the generated triples, (ii) we evaluate the KG correctness based on the requirements of the QCL properties domain and (iii) the KG completeness to assess the ratio of missing information in the KG. The definitions of these evaluation metrics are given in Chapter 2. For case of the logical consistency of the KG triples, we use Pellet [159] and Hermit [55] reasoners for validation. The aim of the domain validation is to verify the ability of the generated KG in answering questions on the relationships between the various QCL properties.

Experimental Setup

(i) **Knowledge Graph Correctness:** A set of expert test cases in form of competency questions (CQs) are prepared with the help of QCL experts in order to assess the quality of the knowledge represented by the KG.

The CQs constitutes inquiries that QCL domain experts would need answers to in the process of designing QCL devices with desired working properties. The answers to these questions should allow access to structured information about the relations between the QCL properties. We define 7 classes of CQs that encompass the various types of information i.e the design properties, working properties, provenance information (sources of information for the properties) and references on which the properties are based. Table 7.2 presents the classes of the CQs.

The variables X, Y and Z are place holders for specific QCL properties. For each CQ class in Table 7.2, we define several possibilities of CQs, translate them to SPARQL queries and execute them on the KG to obtain the query results.

(ii) **Knowledge Graph Completeness:** We assess two quality metrics: *schema completeness* and *property completeness* in order to establish the proportion of missing information in the KG. For the *schema completeness*, we define a set of mandatory relations for a class in order to determine the missing facts in a particular class in the KG [97]. The mandatory properties are determined by considering the following conditions:

- QCL properties class instances (design/working properties) should be linked to the relevant provenance information (should contain links to papers describing them).

Table 7.2: Classes of Competency Question

CQ ID	Question Text
CQ1	What are the possible material compositions of a QCL laser heterostructure with a design type X ?
CQ2	What is the working property X of a QCL laser working in mode Y ?
CQ3	What is the performance property X of a QCL laser having a heterostructure with material composition Y?
CQ4	For a particular performance property X, what are the corresponding laser heterostructure designs?
CQ5	For a particular performance property X, what are the corresponding heterostructure material compositions?
CQ6	What are the DOIs and/ the URLs of the scientific articles documenting a laser with performance property W or with heterostructure materials X or working mode Y or design type Z.
CQ7	What are the DOIs and URLs of the articles being referenced by a QCL device with property W or with heterostructure materials X or working mode Y or design type Z?

- The working temperatures should have a corresponding working mode.
- The optoelectronic properties should be linked to units, have quantity kind, quantity value.
- The optoelectronic characteristics should be related to corresponding design features.

Based on these conditions, the mandatory relations are *wasAttributedTo*, *correspondsToWorkingMode*, *hasQuantityValue*, *hasQuantityKind*, *hasUnit* and *relatesToHeterostructure*. For each of these mandatory attributes, we determine the the ratio of instances that actually have these relations in the KG.

For the *property completeness*, we consider the data properties and check whether there is any missing information for these properties. We argue that all scientific articles in the KG should have DOIs, all heterostructure materials should have a material formula, and all quantity values for the optoelectronic properties should have numeric values. We determine the ratio of the values as compared to the data properties for the instances in the KG in order to determine the extent of missing information.

7.3.3 Evaluation Results

KG Consistency

The consistency of the generated KG is validated by lack of inconsistencies in the KG triples. This is as validated by the results obtained by running of the logical Pellet and HermiT logical reasoners on the KG data file.

KG Correctness

A set of 20 competency questions are evaluated on the KG in order to assess the quality of the knowledge captured in the KG in relation to the domain requirements. These are presented in Table 7.4. Each CQ ID is preceded by the ID of the CQ class in which it belongs, for instance

Table 7.3: The Competency Questions

Query Number	Query Text
1.1	What are the possible material compositions of a QCL laser heterostructure with an LO Phonon Design Type ?
1.2	What are the possible material compositions of a QCL laser heterostructure with a Resonant Phonon Design Type?
1.3	What are the possible material compositions of a QCL laser heterostructure with a Bound to Continuum design type?
2.1	What are the working temperatures for a QCL laser operating in the continuous wave mode?
2.2	What are the working temperatures for a QCL laser operating in the pulsed mode?
3.1	What are the possible power values for a QCL laser with a heterostructure with material composition GaAs/Al _{0.15} Ga _{0.85} As?
3.2	What are the possible frequency values for a QCL laser with a heterostructure with material composition In _{0.53} Ga _{0.47} As/GaAs _{0.51} Sb _{0.49} ?
3.3	Query 3.3: What are the possible working temperature values for a QCL laser with a heterostructure with material composition GaAs/Al _{0.3} Ga _{0.7} As?
4.1	Query 4.1: What are the possible heterostructure designs for a QCL device with a working temperature greater than 100 K in Pulsed Mode?
4.2	What are the possible heterostructure designs for a QCL device with an optical power less than 50 mW?
5.1	What are the possible heterostructure material compositions for a QCL device with a working temperature less than 85 K in the continuous wave mode?
5.2	What are the possible heterostructure material compositions for a QCL device with a lasing frequency greater than 1.5 THz?
6.1	What are the DOIs and URLs of scientific articles documenting QCL laser devices with an optical power greater than 10mW?
6.2	What are the DOIs and URLs of scientific articles documenting QCL laser devices with a working temperature greater than 100 K in pulse mode?
6.3	What are the DOIs and URLs of scientific articles documenting a QCL laser with a material composition of GaAs/Al _{0.25} Ga _{0.75} As?
6.4	What are the DOIs and URLs of scientific articles documenting QCL lasers with bound to continuum design type?
6.5	What are the DOIs and URLs of articles documenting QCL lasers with a heterostructure of material composition GaAs/Al _{0.15} Ga _{0.85} As, LO phonon design type and working temperatures greater than 70 K in pulse mode operation?
7.1	What are the DOIs and URLs of the articles being referenced by a QCL device with a working temperature greater than 225 K in the continuous wave mode?
7.2	What are the DOIs and URLs of the articles being referenced by a QCL device with an optical power less than 1 mW?
7.3	What are the DOIs and URLs of the articles being referenced by a QCL device with a lasing frequency greater than 2.5 THz and an LO Phonon design type?

CQs 1.1, 1.2 and 1.3 belong to CQ class 1. The queries range from simple to complex queries based on the relationship between QCL properties and their provenance information.

We present examples of three scenarios where a QCL expert is in need of particular information on QCL properties during the device fabrication process. We illustrate how this information can be retrieved from the KG with example CQs in Table 7.4.

Example 7.2.2: *Consider a scenario where a QCL expert is interested in the possible heterostructure materials composition of a QCL device with a certain working property, for instance, a lasing frequency value greater than 1.5 THz.*

This information can be retrieved from the KG by executing CQ 5.2 in Table 7.4. The corresponding SPARQL query is as shown in Listing 7.1 and the obtained results are shown in Figure 7.5.

Listing 7.1: SPARQL Query for CQ 5.2

```
PREFIX QpOnto:<https://github.com/DeperiasKerre/qcl_Onto/blob/main/qclontology/version-1.0/qclonto.owl#>
PREFIX qudt:<https://qudt.org/schema/qudt/>
PREFIX qudt_qv: <https://qudt.org/schema/qudt/#>
SELECT DISTINCT ?material_composition
WHERE
{
  ?lf qudt_qv:hasQuantityValue ?fv;
  QpOnto:relatesToHeterostructure ?HS.
  ?fv qudt:numericValue ?value;
  qudt:hasUnit ?unit.
  ?HS QpOnto:hasMaterials ?HM.
  ?HM QpOnto:matFormula ?material_composition.
  FILTER(?unit=<https://qudt.org/vocab/unit/TeraHZ>&&?value>1.5).
}
```

```
material composition: GaAs/AlGaAs
material composition: InAs/AlAs0.16Sb0.84
material composition: AlInGaAs
material composition: GaAs/Al0.1Ga0.9As
material composition: Al0.03Ga0.97As
material composition: GaAs/Al0.15Ga0.85As
material composition: GaAs/AlAs
material composition: In0.53Ga0.47As/GaAs0.51Sb0.49
material composition: GaAs/Al0.25Ga0.75As
material composition: GaAs/Al0.14Ga0.86As
```

Figure 7.5: SPARQL Query Results for CQ 5.2

Example 7.2.3: *Consider a scenario where a QCL expert is interested in knowing/accessing information on the possible heterostructure materials composition for QCL devices with working temperature, less than a given value x , for instance 85 K in continuous mode.*

This information can be retrieved from the KG by executing CQ 5.1 on the KG. Listing 7.2 shows the corresponding SPARQL query for CQ 5.1 and Figure 7.6 shows the results.

Listing 7.2: SPARQL Query for CQ 5.1

```
PREFIX QpOnto:<https://github.com/DeperiasKerre/qcl_Onto/blob/main/qclontology/version-1.0/qclonto.owl#>
PREFIX qudt:<https://qudt.org/schema/qudt/>
PREFIX qudt_qv: <https://qudt.org/schema/qudt/#>
SELECT DISTINCT ?material_composition
WHERE
{

```

```

?wt qudt_qv:hasQuantityValue ?tv;
QpOnto:correspondsToWorkingMode QpOnto:ContinuousWaveOperation;
QpOnto:relatesToHeterostructure ?HS.
?tv qudt:numericValue ?value;
qudt:hasUnit ?unit.
?HS QpOnto:hasMaterials ?HM.
?HM QpOnto:matFormula ?material_composition.
FILTER(?unit=<https://qudt.org/vocab/unit/K>&&?value<85).
}

```

```

material composition: GaAs/Al0.1Ga0.9As
material composition: GaAs/Al0.14Ga0.86As

```

Figure 7.6: Results for CQ 5.1

Example 7.2.4: *Lastly for provenance information, consider a scenario where a QCL expert is interested in the DOIs and URLs for articles documenting QCL devices with a certain property, for instance a working temperature greater than 100 K.*

This inquiry can be expressed by CQ 6.2. The corresponding SPARQL query can be expressed as in Listing 7.3. The results are shown in Figure 7.7. All the 20 queries are successfully answered by the generated KG. The results for all the queries can be found at the GitHub repository ¹⁵.

Listing 7.3: SPARQL Query for CQ 6.2

```

PREFIX QpOnto:<https://github.com/DeperiasKerre/qcl_Onto/blob/main/qclontology/version-1.0/qclonto.owl#>
PREFIX qudt:<https://qudt.org/schema/qudt/>
PREFIX qudt_qv:<https://qudt.org/schema/qudt/#>
PREFIX prov:<http://www.w3.org/ns/prov#>
SELECT ?doi ?url
WHERE
{
?wt prov:wasAttributedTo ?article;
qudt_qv:hasQuantityValue ?tv;
QpOnto:correspondsToWorkingMode QpOnto:PulsedOperation.
?tv qudt:numericValue ?value;
qudt:hasUnit ?unit.
?article QpOnto:DOI ?doi;
QpOnto:URL ?url.
FILTER(?unit=<https://qudt.org/vocab/unit/K>&&?value>100).
}

```

The query results demonstrates the ability of the generated KG in capturing QCL properties data and semantically enriching them. This provides a platform that enables a quicker analysis of the QCL properties from the various heterogeneous sources of scientific texts. This is as

¹⁵<https://github.com/DeperiasKerre/qKG/tree/main/Results/KG>

DOI: 10.1063/1.2189671 , URL: <https://doi.org/10.1063/1.2189671>
 DOI: 10.1063/1.2189671 , URL: <https://doi.org/10.1063/1.2189671>
 DOI: 10.1088/0268-1242/20/7/013 , URL: <https://doi.org/10.1088/0268-1242/20/7/013>
 DOI: 10.1063/1.3068496 , URL: <https://doi.org/10.1063/1.3068496>
 DOI: 10.1063/1.2136222 , URL: <https://doi.org/10.1063/1.2136222>
 DOI: 10.1063/1.2216112 , URL: <https://doi.org/10.1063/1.2216112>

Figure 7.7: Results for CQ 6.2

opposed to the traditional first principle computational analysis of the various properties that is time consuming, error prone and requires domain expertise. This enables an accelerated understanding of the implications of the various QCL properties on each other without manually accessing the properties in text. The insights obtained may be used to inform decisions regarding the design of QCL devices with desired/optimal properties. With the KG, it is also possible to track the sources of QCL properties via links to provenance information. The QCL experts may also be able to access the references on which some properties are based on.

KG Completeness

The results for schema completeness and property completeness are presented in Tables 7.4 and 7.5 respectively.

Table 7.4: Results for Schema Completeness Analysis

Mandatory Relation	Number of Instances in the KG (Expected to have Relation)	Instances with the Relation	Ratio
wasAttributedTo	100	100	1
correspondsToWorkingMode	36	36	1
hasQuantityValue	79	79	1
hasQuantityKind	79	79	1
hasUnit	69	69	1

Table 7.5: Results for Property Completeness Analysis

Property	Number of Instances in the KG (Expected to have Data)	Number of Data Values in KG	Ratio
DOI	474	474	1
matFormula	14	14	1
numericValue	69	69	1

As illustrated in 7.4, all the mandatory relations are adequately captured in the KG schema with no information missing from the individual KG classes. This implies that all the class instances are adequately associated with other instances for the mandatory relations hence validating the KG schema quality. As indicated in Table 7.5, all data values for property instances are adequately captured in the KG. This implies that all articles in the KG have DOIs, all

heterostructure materials have a corresponding material formula and all the quantity values in the KG have a corresponding numerical values. This validates the property completeness of the generated KG.

7.4 Conclusion

In this chapter, we address the issue of semantic enrichment of QCL properties in scientific texts. The main aim is to provide a platform that integrates QCL properties data from different sources to allow a unified, structured and quicker analysis while providing links to the sources. We propose a KG for the QCL design and working properties, which contains a total of 2979 triples for 36 QCL devices. The KG also captures references and the provenance information for these properties. The KG generation workflow involves first generating structured QCL properties data from scientific texts (based on a RAG approach with GPT-4 Turbo model) and performing data mapping and enrichment in order to generate the KG. We validate the consistency of the generated KG using logical reasoners. We also validate the ability of the generated KG in capturing the domain requirements using expert defined test cases in form of competency questions. The completeness of the KG is also verified to assess the extent of missing information in the proposed KG.

The proposed KG prototype lays a foundation for data driven discovery of novel QCL designs with optimized working properties. It also provides insights for domain KG generation from text and KR for specific domain data. From the domain perspective, it provides a platform for analysis of QCL data in a structured manner and maintains provenance information hence enabling tracking development of QCLs overtime.

However, the proposed KG is limited on properties for 36 QCL devices. With the proposed KG generation workflow, the KG can be extended with more QCL data as more QCL devices are proposed in scientific articles. Some other QCL properties such as current density can also be included in the extended KG schema. With the proposed KG and the generation workflow, we hope that it will provide a platform for quicker analysis of QCL properties hence transform the way QCL experts and researchers interact with scientific texts containing QCL properties to provide more opportunities for accelerated QCL device discovery.

The next chapter is the last chapter of this thesis. We present the summary of our contributions, the significance and limitations of the study. We also present the research perspectives that describe possible future research directions for our study.

Conclusion and Perspectives

8.1	Summary of the Thesis	120
8.1.1	Comprehensive Literature Review of Knowledge Representation and In-formation Extraction in Materials Science	121
8.1.2	Generation of Datasets for Extraction of QCL properties from Text	122
8.1.3	Extraction of Quantum Cascade Laser Properties from Text	122
8.1.4	Formal Representation of Quantum Cascade Laser Properties	122
8.1.5	Semantic Enrichment of Quantum Cascade Laser Properties in Text	123
8.2	Significance and Limitations	123
8.3	Perspectives	124
8.3.1	Further Improvement of Efficiency in Mining QCL properties from Text	124
8.3.2	Extension of Knowledge Representation for QCL Properties	124
8.3.3	Learning Methods for QCL Properties Prediction	124

“Research is what I’m doing when I don’t know what I’m doing”

– Wernher von Braun (1912 – 1977)

8.1 Summary of the Thesis

This research study addresses the *development of standardized platforms for accessing and analyzing QCL properties data captured in various heterogeneous scientific texts in a FAIR manner*. This enables interoperable access and analysis of QCL properties data in order to uncover the relationships between the QCL properties. The insights obtained from this analysis can be used for various purposes such as making decisions on optimized QCL device fabrication.

The main sources of QCL properties data is in the form of scientific texts which are usually unstructured. The properties are also captured in various presentation styles within the texts. This presents challenges in their identification and extraction. Therefore, the objective of this thesis was to focus on the research directions outlined below.

- RD1-development of **robust datasets** and **methods for extraction of structured QCL properties data** from scientific texts.
- RD2-the development of **robust KR platforms** for **formalizing the knowledge** about QCL properties and for the **semantic enrichment** of the QCL properties in text.

The two research directions are further subdivided into 4 individual research questions as outlined below.

- *RQ 1: How can Natural Language Processing (NLP) techniques be utilized together with QCL expert knowledge to curate an original dataset for extraction of QCL properties from scientific texts?*
- *RQ 2: How can information extraction techniques be utilized for the task of generating structured QCL properties data from scientific texts?*
- *RQ 3: How can knowledge representation systems such as Ontologies be utilized for representing knowledge on QCL design and working properties?*
- *RQ 4: How can knowledge representation systems such as Knowledge Graphs be utilized for semantic enrichment of QCL properties in scientific texts?*

In line with the research directions, we explore each of the research questions in order to generate standardized platforms for representing and analyzing QCL properties data in a FAIR manner. Therefore, we focus on the tasks of extracting QCL properties from text and proposing KR platforms for these properties. The main contributions of this thesis in relation to the research directions are represented in Figure 8.1.

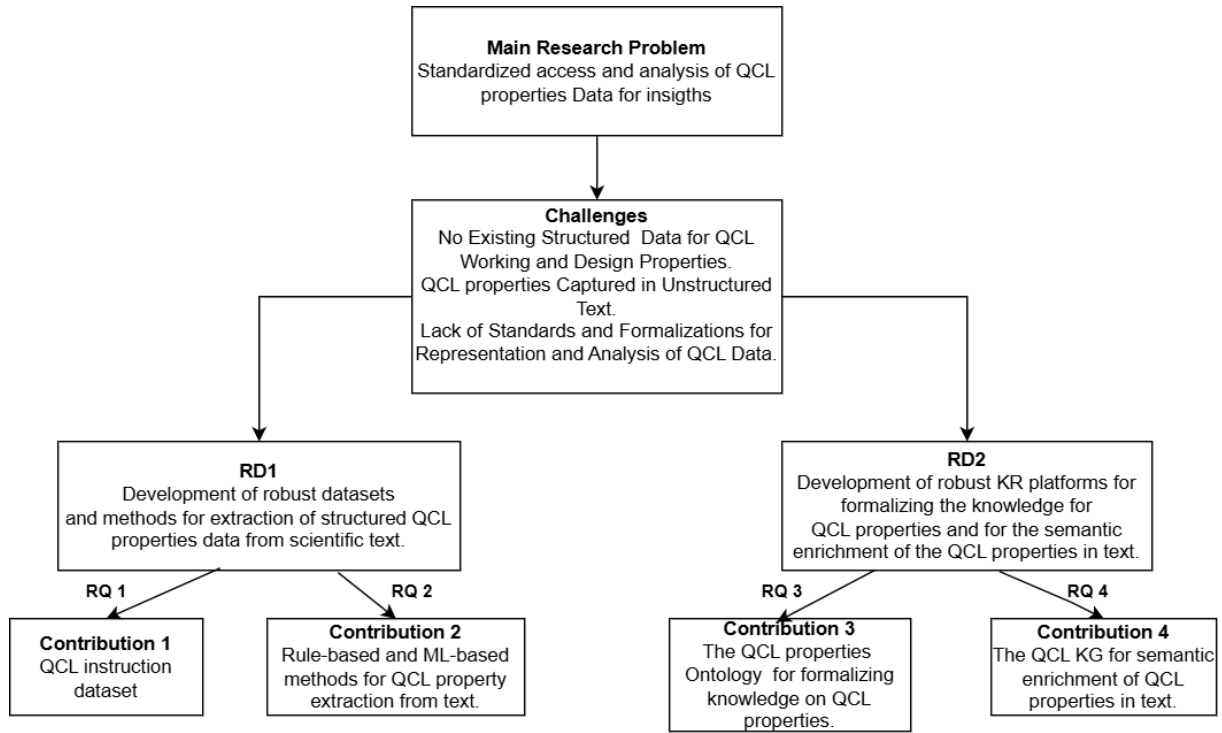


Figure 8.1: Summary of the Contributions

In the following subsections, we give a summary of the specific contributions made by each chapter of this thesis. In Section 8.1.1, we give a summary of the contributions from the survey of Knowledge Representation and Information Extraction in materials science. In Section 8.1.2, we discuss how we addressed research question 1 by developing a novel instruction dataset that enables development of LLM-based techniques for extracting QCL properties from text. In Section 8.1.3, we summarize how we addressed research question 2 by proposing two methods for extracting QCL properties from scientific texts. In Section 8.1.4, we summarize how we addressed research question 3 by proposing a formal representation of QCL properties. Lastly in Section 8.1.5, we summarize how we addressed research question 4 by proposing a model for semantic enrichment of QCL properties in text in the form of a Knowledge Graph.

8.1.1 Comprehensive Literature Review of Knowledge Representation and Information Extraction in Materials Science

In Chapter 2 and 3, we provide a comprehensive review of literature on IE and Knowledge representation in the materials science domain. We established that there is need for adaptation of IE techniques in order to improve the efficiency in mining QCL properties. We also discovered that the existing knowledge representation systems cannot be readily adopted to represent knowledge on QCL properties. This is due to lack of formalization of the QCL concepts of interest. This calls for development of standards and shared vocabularies for representing, accessing and analyzing QCL properties data in an interoperable manner. The insights obtained from the standardized analysis can be utilized for the development of QCL devices with optimal performance. We also

reviewed existing datasets and established the need for structured QCL properties data that can be used for data mining tasks that seek to understand the relation between the various QCL properties.

8.1.2 Generation of Datasets for Extraction of QCL properties from Text

In chapter 4, we tackle the problem of lack of quality training datasets for developing ML-based techniques for QCL property extraction from text. We explore NLP techniques and expert Knowledge in the QCL domain to propose a novel instruction dataset that can be used in the development of LLM-based techniques for mining QCL properties data from text. The dataset provides a valuable benchmark for advancing research in the field of IE for the QCL domain. We utilize this dataset to adapt LLMs to the domain-specific task of QCL properties extraction from text.

8.1.3 Extraction of Quantum Cascade Laser Properties from Text

In chapter 5, we propose two methods for extracting QCL properties from scientific texts. In order to understand the nature of text in the QCL domain, we start by developing a rule-based approach for QCL properties extraction from text. This method is suitable in cases where there is no quality training data for training and evaluating machine learning models for QCL property extraction from text. In the second step, we propose an LLM-based approach for mining QCL properties from scientific texts. We employ a hybrid few shot learning strategy where the few shot examples consist of the best examples generated automatically from the training set of the instruction dataset by the RAG process.

Based on experimental evaluation, we establish that the rule-based approach is limited when there are changes in the text that are not captured by the rules. The LLM-based approach attains more flexibility and greater performance but it is highly sensitive to the quality of the data provided through the RAG process. These two methods constitutes initial attempts in generating structured QCL properties data from unstructured scientific texts. The experimental data can be used for a wide range of data mining tasks.

8.1.4 Formal Representation of Quantum Cascade Laser Properties

In order to formalize knowledge on QCL properties and provide standards for accessing and analyzing QCL properties in an interoperable manner, we propose a formal representation of properties in the QCL domain in form of an Ontology in chapter 6. The ontology provides formal descriptions for the various QCL properties, the relations between them and their provenance information. We validate the ability of the ontology in capturing the domain requirements. We also show that it is possible to capture the knowledge in QCL properties and provide links to the sources of this data.

8.1.5 Semantic Enrichment of Quantum Cascade Laser Properties in Text

Lastly, we propose a Knowledge Graph for exploring QCL properties in chapter 7. The KG generation workflow is based on the LLM-based approach for mining QCL properties from text. The semantics of the KG are provided by the QCL ontology that we have proposed together with terms from other Ontologies. We validate the logical consistency of the generated KG and its ability to capture and provide semantics for the QCL properties data. We show examples scenarios where the KG provides a unified and semantically enriched access to QCL properties data that could be used in informing decisions regarding design of QCL devices with optimal/target performance.

8.2 Significance and Limitations

The work of this thesis in the field of KR and IE with the application in the QCL domain holds significant importance as it addresses the challenges posed by the process of providing structured and interoperable platforms for analyzing QCL properties from heterogeneous sources of scientific texts. By proposing IE methods for extracting QCL properties from text and generating KR platforms for their analysis, the study enhances the process of seamless analysis and exploration of the QCL properties and the relationships between them. The projected impact of this study can be visualized in two aspects: Academia and Industry.

In academia, the study enriches the body of knowledge on IE particularly the methods for extraction of physical quantities from scientific texts. The study also provides a benchmark dataset to further research in this direction. The formalization of the knowledge on QCL properties holds potential in enabling the community to have a shared vocabulary for analyzing QCL properties. The formalization could also be extended to generate formal representations of other semiconductor properties. Lastly, the study also provides novel insights regarding the KG generation from text methods.

In industry, the study provides formal representations for structured QCL properties data. This allows massive analysis and other data mining tasks regarding the implications of design features on the laser performance. The proposed ontology and KG schema can be adopted by QCL fabrication teams and be used to analyze massive amounts of experimental data. The insights obtained from this analysis can be used in the optimal design of QCL devices.

While this work presents the mentioned contributions, we also acknowledge its limitations. The proposed rule-based for extracting QCL properties from text approach is limited to the QCL domain context and the nature of descriptions in QCL domain text. The RAG-based approach is directly dependent on the quality of the data provided in the context during input. Not all QCL properties are also captured by these methods. The proposed ontology and KG do not also capture all the QCL properties. The evaluation of the QCL KG is also based on a limited number of QCL devices. This may be extended in future for scalability analysis.

8.3 Perspectives

With regard to the contributions made by this work in the field of information extraction and knowledge representation, we present the following research perspectives which can be considered to be future research directions.

8.3.1 Further Improvement of Efficiency in Mining QCL properties from Text

In chapter 5, we describe two methods for mining QCL properties data from scientific texts. We observe that the rule-based approach is limited on varying textual representations. The LLM-based method presents good performance but not for all the properties. It is also dependent on the quality of the training data.

Interesting research directions may involve more exploration of LLM-based techniques for QCL property extraction from text. This may include generation of higher quality datasets for training and evaluation purposes. It may also be interesting to perform an analysis on the various learning methods such as fine-tuning and compare this with the proposed RAG approach. The scope of the properties to be handled could also be extended to capture other QCL properties such as current density among others. Other interesting research directions may also entail extension of rule based approaches to adapt with variability in text, hybrid methods (combine rule-based and machine learning techniques for less annotation efforts and relatively higher performance) [82].

8.3.2 Extension of Knowledge Representation for QCL Properties

In chapters 6 and 7, we have proposed KR platforms for semantically enriching and organizing QCL properties data for structured exploration. The list of properties captured does not include all the QCL properties. Future works on this platform may involve exploring ontology learning techniques to extend the Ontology with more concepts for the QCL properties, for instance the barrier thickness, doping properties and the detailed layer sequence (growth sheet), which material, which thickness and emerging QCL designs. The generated KG may also be augmented with more QCL properties data as more devices are proposed in scientific literature.

Recently, there has been a lot of research interest at the intersection of LLMs and KGs. One of the notable research directions is the LLM-enabled KG generation from text [119]. This area can be explored more in the context of KG generation tasks for QCL properties in text in order to capture more information on QCL properties captured in text.

8.3.3 Learning Methods for QCL Properties Prediction

Lastly, we envision a possibility of developing learning methods for predicting various QCL properties, for instance, the working properties based on the QCL design properties. This could be possible through the analysis of data captured by the QCL properties Knowledge Graph

proposed in chapter 7. KG-based learning methods such as link prediction and node classification methods have been proposed for material properties prediction [177].

In order to achieve the task of predicting QCL properties, future works may involve exploration of graph machine learning methods (for instance graph embeddings and graph neural networks) to develop algorithms that utilize knowledge in the proposed KG for the task of QCL properties prediction. This may also include prediction of relations between the QCL properties. Sample tasks towards this direction may include node classification and link prediction for the QCL knowledge graph.

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Appendix A

Links to Resources

A.1 Source Code Links

A.1.1 Instruction Dataset Source Code

Source Code Link: <https://gite.lirmm.fr/dkerre/qpInstruct>

A.1.2 Rule-Based IE Source Code

Source Code Link: <https://github.com/DeperiasKerre/qclProperties>

A.1.3 RAG-Based IE Source Code

Source Code Link: <https://github.com/DeperiasKerre/qKG>

A.1.4 The QCL Ontology Source Code

Source Code Link: https://github.com/DeperiasKerre/qcl_Onto/tree/main/qclontology/version-1.0

A.1.5 The QCL KG Generation Source Code

Source Code Link: <https://github.com/DeperiasKerre/qKG>

A.2 Dataset Links

A.2.1 Instruction Dataset Link

Dataset Link: <https://doi.org/10.57745/U3U7XR>